THE CRYSTALLINE WORLDSHEET: A STRING THEORETICAL FRAMEWORK BASED ON $\mathbb{F}_1$, FOR THE DE SITTER OBSERVER PROBLEM

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ABSTRACT. Holography and the Riemann hypothesis are missing the same kind of object. The pair correlation of the zeros of ζ follows GUE statistics; so does the boundary side of low-dimensional holography, where the bulk of JT gravity is recovered only as an average over an ensemble of theories. An ensemble is what one writes for want of an individual—the Hermitian operator of Hilbert–Pólya in one case, a single microstate in the other—and an average is a fact about the description, not about what exists. Based on this reading, and on other known connections between the primes and black holes, and between the odd zeta values and string and particle amplitudes, we propose that the obstruction in de Sitter is *epistemological*—a circumstance of mathematical circularity—rather than *ontological*—a matter of vacuum selection.

At this intersection the obstruction takes one form: how to describe the object without referring to the object, which the $\mathbb{F}_1$ program seeks to circumvent in arithmetic and the Hilbert–Pólya problem poses for the operator. We trace it to the non-paradoxical self-reference of binary language, in the precise sense of the Lawvere fixed-point theorem and the Yanofsky diagonal $g(t) = \alpha(f(t,t))$, as one of the most studied cases of self-referential circularity; and we analyse the strategies for confronting this circularity and their relation to the $\mathbb{F}_1$ program for proving the Riemann hypothesis—particularly Yuri Manin's proposed use of the same noncommutative tori that M-theory uses, but to approach that program.

From our own use of this torus we construct a string cocone, from which we later derive previously proposed criteria for holography in de Sitter, as well as the operator we propose as the microstate.

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Date: **Received:** July 13, 2026; **Last modified:** September 1, 2026.

2020 *Mathematics Subject Classification.* Primary 81T30, 14G99, 11M26; Secondary 68V20.

Key words and phrases. Holography; de Sitter; M-theory; string theory; field with one element ($\mathbb{F}_1$); λ-rings; golden ratio; moduli spaces; formal verification.

Zenodo All Versions DOI: 10.5281/zenodo.21343602.

1. Introduction

1.1. The problem: the observer in de Sitter.

1.1.1. *The participatory observer.* The problem of holography in de Sitter space is, as Witten frames it in *Quantum Gravity in De Sitter Space*, that General Relativity "cannot be quantized—unless it is embedded in a more precise theory that determines the value of the cosmological constant" [1]: a theory in which the observer is not introduced from outside but "described by the theory" [2]. In a cosmology with positive cosmological constant there is no asymptotic region to which an external observer may retreat; the observer is part of the system it measures and must therefore model the system that contains it. This is what makes the de Sitter case structurally different from the anti-de Sitter case, which has been the most successful approach to finding a UV complete description of quantum gravity, yet.

1.2. **Proposed solutions and their open issues.** In this section we review some of the most important proposed solutions to the problem, together with the issues that each one leaves unresolved.

1.2.1. *Strominger: dS/CFT and non-unitarity.* In 2001 Strominger proposed the dS/CFT correspondence [3]: quantum gravity in a D-dimensional de Sitter space is equivalent to a Euclidean conformal field theory living at future (or past) infinity, conceptualised as a $(D-1)$-dimensional sphere. The difficulty is that the resulting theory is generically non-unitary: the boundary conformal weights

$$h_{\pm} = \frac{D-1}{2} \pm \sqrt{\left(\frac{D-1}{2}\right)^2 - m^2\ell^2} \tag{1.1}$$

become complex once $m^2\ell^2 > (\frac{D-1}{2})^2$, so probabilities do not sum to one and the dual theory is mathematically elusive.

1.2.2. *Susskind: the static patch.* Susskind, one of the originators of the holographic principle [4], addressed the problem from the standpoint of a real observer: holography should be formulated inside the cosmological horizon of a static observer, with all interior information encoded on that horizon at one bit per Planck area,

$$S_{\text{hor}} = \frac{A_{\text{hor}}}{4G_N}. \tag{1.2}$$

In recent work this static patch has been connected to advanced quantum-mechanical models such as double-scaled SYK; the holographic encoding itself, however, remains a postulate rather than a derived correspondence.

1.2.3. *Witten: finite dimensionality.* Also in 2001, Witten [1] analysed the quantum properties of de Sitter, taking up the argument that, because the horizon has finite area, the Hilbert space of quantum gravity in de Sitter is strictly finite-dimensional,

$$\dim \mathcal{H} < \infty. \tag{1.3}$$

This collides directly with traditional conformal field theories, which possess infinitely many states, and imposes severe constraints on any holographic attempt; the observer, in particular, must be included in the description rather than idealised away [2].

1.2.4. *The ensemble and vacuum selection.* Despite these proposals the picture is, as Nico Cooper puts it, always the same [5]. A gravitational theory at weak coupling in a d-dimensional anti-de Sitter interior is captured by a conformal theory at strong coupling on its $(d-1)$-dimensional edge, the cleanest realisations being string backgrounds that flow to supergravity. In low dimensions this breaks down: two-dimensional JT gravity has no single boundary dual, and the bulk is recovered only as an average over an *ensemble* of boundary theories [6, 5], even though quantum gravity should carry its own discrete, individual microstates—much as an ensemble of Hamiltonians reproduces Wigner's nuclear decay rates while any single nucleus is governed by one. This ensemble problem connects directly to the observer problem in de Sitter: the absence of a definite spatial boundary confines the observer to an averaged statistical description within their cosmological horizon, breaking the notion of a single pure quantum state and forcing an effective description that reconciles the finiteness of the horizon with the theoretical existence of fundamental microstates. The most widely held response is to abandon uniqueness altogether: it merges the Everettian many-worlds interpretation of quantum mechanics with anthropic vacuum selection over the string landscape—an estimated $\sim 10^{500}$ metastable de Sitter vacua [7, 8, 9]. Our vacuum is then one branch among many, populated by eternal inflation and fixed by anthropic selection rather than derived, the branches of the wavefunction being identified with the landscape's vacua rather than with a single microstate [10].

1.3. **Hypothesis: self-reference as the common cause.**

1.3.1. *A matter of mathematical representation.* We propose that the problem may be epistemological—a matter of mathematical representation—and not ontological—a matter of vacuum selection in which we do not know which vacuum we inhabit. We propose that, as in the case of the Hilbert–Pólya operator and in several $\mathbb{F}_1$ propositions, the problem is how to describe something without referring to that same something one is describing, to avoid circularity.

Across many scientific and mathematical worlds we encounter geometric structures that are part of a whole identical to themselves: fractals. They occur even in literature, among the rhetorical figures—the *mise en abyme*, for example. Self-reference in mathematics, however, is a profoundly delicate matter, one that wreaks havoc in domains far beyond the de Sitter universe we address here. The way the whole relates to the part appears as a concrete obstruction across many domains: in set theory, as Russell's paradox—the collection of all collections that do not contain themselves; in the foundations of arithmetic, as Gödel's first incompleteness theorem—a true proposition that asserts its own unprovability and so escapes every proof; in the theory of truth, as Tarski's undefinability theorem—the impossibility of defining truth within the language itself; in computation, as Turing's halting problem—the impossibility of deciding whether an arbitrary procedure—itself included—ever halts; and in engineering, as the feedback of a system that must represent itself. We propose that the essence of the problem lies in how to describe the object without referring to the object, akin to the figure–ground problem in the graphic arts and painting.

1.3.2. *The Lawvere–Yanofsky obstruction.* Let us first examine how the problem of self-reference wreaks havoc across diverse domains. In de Sitter holography the observer is forced to classify itself by a two-valued distinction—inside or outside the horizon, bulk or boundary, yes or no—and it is this binary self-reference, not self-reference in general, that becomes obstructed. The obstruction is made precise by the Lawvere fixed-point theorem [11]: in a cartesian closed category a point-surjective map $f\colon T \to Y^T$ forces every endomorphism of Y to have a fixed point, so a single fixed-point-free endomorphism forbids such an f. Yanofsky's diagonal [12] packages the classical paradoxes through

$$g(t) = \alpha\big(f(t,t)\big), \qquad \alpha(x) = 1 - x, \tag{1.4}$$

where the fixed-point-free involution α on the binary register $\{0,1\}$ is precisely what makes the *binary* self-application $f(t,t)$ contradictory. Our hypothesis is that the de Sitter obstruction is an instance of this *type* of circularity, compounded by the dependence of quantum mechanics—as it was originally formulated—on Boolean logic.

1.3.3. *Stratification as prohibition.* Set theory under classical bivalent logic resolves self-reference by *prohibiting* it through stratification. The axiom of extensionality defines objects purely by their membership extension, and the axiom schema of separation restricts comprehension so that no set is formed by a predicate quantifying over the totality of sets—because the predicate $x \notin x$ applied to the class of all sets produces Russell's paradox. The resolution is the cumulative hierarchy: every set has a rank α, with

$$x \in u \implies \operatorname{rank}(x) < \operatorname{rank}(u), \tag{1.5}$$

and no set has a complement that is also a set, so the Boolean algebra of union and intersection is available within each level while global complementation is missing. A morphism translating between two domains A and B would have to occupy some rank α while its arguments and values refer across levels—precisely the cross-level self-application that stratification forbids. The barrier between universes is therefore not an additional restriction imposed on top of the paradox-avoidance mechanism: it *is* that mechanism. Stratification solves self-reference by making cross-level morphisms illegal, and cross-level morphisms are exactly what inter-domain translation requires.

1.3.4. *Boolean logic is not the only logic.* The preceding obstruction is specific to the classical, bivalent setting. Boolean logic, however, is not the only logic that exists, and the resolution we shall pursue lies in replacing prohibition by mediation—a possibility opened by the non-Boolean logics reviewed next.

1.4. **Mathematical antecedents: non-Boolean strategies for self-reference.** The classical, bivalent prohibition of §1.3 is not the only way to confront self-reference; the literature documents several *distinct* strategies, each with its own mechanism, and they should not be conflated into a single route. Historically the constructivist and intuitionist tradition questioned classical logic from outside: Kant's account of number as a construction in the pure intuition of time is a philosophical, not a mathematical, antecedent, followed by Kronecker's finitism, Poincaré's pre-intuitionism, the constructivism of Borel and Lebesgue, and Brouwer's rejection (1908) of the excluded middle for infinite domains. The mechanisms that bear directly on the Lawvere–Yanofsky obstruction, each handling self-reference in a different way, are the following.

(1) **Type-theoretic stratification** (Russell [13], Martin-Löf [14]) proceeds by *prohibition by level*: forbidding $X \in X$ through a hierarchy of universes $U_0 \subset U_1 \subset U_2$ breaks the self-referential cycle at the foundational level. It was the first tower strategy, but, as in §1.3, it does so by making cross-level morphisms illegal.
(2) **Paraconsistent logic** (Priest [15]) proceeds by *tolerance*: contradictions are accepted as non-explosive, so a system may contain a contradiction without becoming trivial, and the diagonal need not be excluded at all.
(3) **Tarski's decidable geometry** [16] proceeds by *removing the capacity for self-reference*: the first-order theory of real closed fields, which contains Euclidean geometry $\mathbb{R}^n$, is complete and decidable precisely because it cannot internally define $\mathbb{N}$. Without Gödel numbering no formula can be assigned a code, and the Yanofsky diagonal $g(t) = \alpha(f(t,t))$ *cannot form*.
(4) **Intuitionism, Heyting algebras and the topos** (Brouwer, Heyting; Curry–Howard; Lawvere) proceed by *mediation*: the internal logic is intuitionistic, valued in a Heyting algebra, and the implication $A \to B$ is realised as an internal object—the exponential B^A, governed by the currying adjunction $\operatorname{Hom}(C \times A,\, B) \cong \operatorname{Hom}(C,\, B^A)$ that also underlies the Lawvere statement (1.4). Self-reference is carried by the exponential rather than forbidden.

What these strategies have in common is structural rather than logical: several are *tower* constructions. Russell's hierarchy of types $U_0 \subset U_1 \subset U_2$ and Martin-Löf's hierarchy of universes both resolve self-reference by ranking objects in an ascending tower of levels indexed by a single rule, and this stratified, level-indexed form recurs throughout the constructivist tradition.

1.5. **Physical antecedents.**

1.5.1. *Throats and vacuum selection.* In holography there already exist constructions that resemble these strategies, but from what we like to call the Tarskian or geometric perspective. In the near-horizon limit of a brane stack the geometry develops an anti-de Sitter *throat* [17] whose radial coordinate plays the role of a renormalisation-group scale: in Poincaré coordinates

$$ds^2 = \frac{L^2}{z^2}\left(-dt^2 + d\vec{x}^2 + dz^2\right), \tag{1.6}$$

the boundary $z \to 0$ is the ultraviolet of the dual theory, with $E_{\text{SYM}} = (L/z)\,E_{\text{proper}}$. The bulk interior is encoded on the boundary through the holographic bound

$$S = \frac{A}{4G_N}, \tag{1.7}$$

one bit per Planck area, matching the N^2 degrees of freedom of the dual gauge theory [18]. In a *warped* throat this count runs with the radial coordinate: in the Klebanov–Strassler/Tseytlin cascade [19, 20] the effective rank grows logarithmically,

$$N_{\text{eff}}(r) \sim g_s M^2 \log\frac{r}{r_s}, \tag{1.8}$$

so the throat *gains degrees of freedom—"gains bits"—as it grows* toward the ultraviolet, terminating in the deep infrared in a confining gauge group with a mass gap through a cascade of Seiberg dualities. This already resembles the Russellian type-theoretic hierarchy; what is missing is a *logical* selection criterion. In the orthodox formulation the rule was a strict duality, one bulk universe to one boundary theory; but in the simplest controllable case, 2d JT gravity, the bulk cannot be matched to a single discrete boundary theory, and the mathematics requires instead a statistical ensemble of random matrices [6], in tension with the discrete microstates expected of quantum gravity. Beyond the anthropic principle—selection by one's own existence—there is no rule isolating a single conformal field theory; this is the vacuum-selection problem, magnified in de Sitter by the landscape of $\sim 10^{500}$ vacua [9].

1.5.2. *M-theory from the superpoint.* A second, independently developed string/M-theoretic construction selects its content logically rather than anthropically. Applying the Lawvere development of Hegel—the *unity of opposites* formalized as an adjunction [21], carried into the cohesive ∞-topos in which the superpoint lives [22]—Huerta and Schreiber [23] prove that the super-Minkowski spacetimes of string/M-theory arise as iterated *maximal invariant central extensions* of the superpoint $\mathbb{R}^{0|1}$—the simplest super-Minkowski spacetime, with neither Lorentz nor spinorial structure. Forming these extensions consecutively discovers the spacetimes that carry superstrings and culminates in the ten- and eleven-dimensional cases,

$$\mathbb{R}^{0|1} \longrightarrow \cdots \longrightarrow \mathbb{R}^{9,1|16} \longrightarrow \mathbb{R}^{9,1|16+\overline{16}} \longrightarrow \mathbb{R}^{10,1|32}, \tag{1.9}$$

each step singled out by super-Lie-algebra cohomology rather than postulated; the full brane content—the D-branes, the M5-brane and their dualities—then follows from the induced higher central extensions of the *brane bouquet.* The dimensional structure is in this sense algebraically necessary, with no anthropic choice. Crucially, the construction is binary at its root. Supergeometry is $\mathbb{Z}_2$-graded: it rests on the cyclic group $\mathbb{Z}_2 = \{0, 1\}$ with addition modulo two $(1 + 1 = 0)$, the most elementary binary structure, which fixes the sign rule of the superalgebra and hence the (anti)commutation of the supercharges that the invariant cocycles encode. This $\mathbb{Z}_2$ is precisely the binary register on which the involution $\alpha(x) = 1 - x$ of §1.3 acts. The sense in which this is binary is worth distinguishing: a classical bit is Boolean, taking the value 0 or 1; a qubit is quantum, a superposition of the two; the superpoint's binary is geometric, its $\mathbb{Z}_2$-grading distinguishing the even, bosonic directions of ordinary geometry (the "0") from the odd, fermionic direction that carries supersymmetry (the "1"), the superpoint $\mathbb{R}^{0|1}$ being the minimal seed consisting of that single odd direction. M-theory begins here because it is the smallest possible seed—a single relation between two distinct kinds of coordinate—from whose iterated extensions the eleven-dimensional fabric is generated. So, although its selection is logical rather than anthropic, the construction remains within the binary register and there is no observer to be found.

1.6. Physico-mathematical antecedents: Manin's $\mathbb{F}_1$ program.

1.6.1. *Quantum tori and numbers as functions.* Between string theory and pure mathematics there is an intermediate, experimental line of research that sought to use the quantum tori of M-theory, or the rings of Borger, to approach the Riemann hypothesis: the $\mathbb{F}_1$-geometry program associated with Yuri Manin. The $\mathbb{F}_1$ ("field with one element") program is the one in which a number of mathematicians—among them Manin, Deninger, Soulé, Kurokawa, Connes and Consani, and Borger—have sought to prove the Riemann hypothesis by transporting Weil's proof of the Riemann hypothesis *for curves over finite fields* to the arithmetic of $\mathbb{Z}$: were $\operatorname{Spec}\mathbb{Z}$ realisable as a "curve" over an absolute base $\mathbb{F}_1$, Weil's geometric argument might carry over.

Manin's quantum-tori strand [24] uses the noncommutative tori that Connes, Douglas and Schwarz [25] showed arise as compactifications of M-theory, placing M-theory and noncommutative/$\mathbb{F}_1$-geometry in contact. Such a torus is the noncommutative space T^2_θ whose algebra A_θ is generated by two unitaries U, V obeying

$$VU = e^{2\pi i\theta}\, UV, \qquad \theta \in \mathbb{R} \setminus \mathbb{Q}, \tag{1.10}$$

the irrational angle θ encoding the real multiplication that Manin attaches to real quadratic fields.

This is particularly significant for the Riemann hypothesis because, as Elvang, Herderschee and Morales [26] showed, the Wilson coefficients that maximal supersymmetry and tree-level factorisation leave free in the gluon effective theory are precisely the carriers of the odd zeta values, $a_{2k-1,0} = g^2\alpha'^{\,2k+1}\zeta(2k+1)$, and their bootstrap converges onto the Veneziano values [27] as the unique stringy completion. The odd zeta values thus tie the two frameworks—arithmetic and string-theoretic—together beyond Manin's program itself.

1.6.2. *A geometric route to the Riemann hypothesis.* In his other program, *Numbers as Functions* [28], Manin develops the conception of numbers as functions. Its arithmetic-differential strand replaces the derivative by the Fermat quotient,

$$\delta_p(a) = \frac{a - a^p}{p}, \qquad \psi_p(a) \equiv a^p \pmod{p}, \qquad \psi_p \circ \psi_q = \psi_{pq}, \tag{1.11}$$

the lift ψ_p being the Frobenius of a Λ-ring [29]. Here Manin approaches the Riemann hypothesis from a geometric perspective intended to transcend the self-reference that proving it entails: one cannot refer to the primes, nor to any zero, without importing circularity. The geometric route is meant to sidestep this by working with the space of the problem rather than its arithmetic statement.

1.6.3. *Riemann's ζ, holography and the primes: beyond Manin's programs.* Both of Manin's frameworks remained unfinished before his death in January 2023; both still lack a *dynamical completion*, and both programs remain open.

Beyond Manin's work, however, further relations have been established between string theory, the spectrum of the primes, the Riemann hypothesis and the problem of holography in de Sitter. Two of the most significant and best known bear directly on the problem: the odd zeta values appear both in the perturbative QED contributions to the electron's anomalous magnetic moment—where $\zeta(3)$ and $\zeta(5)$ enter the three-loop coefficient [30]—and in the Veneziano amplitudes already mentioned. A more recent and striking relation is the one between black-hole interiors and the primes obtained by Hartnoll, Yang and De Clerck [31, 32]: near a spacelike (BKL) singularity the Wheeler–DeWitt wavefunctions become automorphic forms whose L-functions, expanded over their Euler product, take the form of the partition function of a free gas of oscillators labelled by the primes—a "primon gas" [33]—the five-dimensional case requiring the complex (Gaussian and Eisenstein) primes. Finally, and no less significant, is the relation between the GUE (Gaussian

Unitary Ensemble) of random matrices—whose statistics Montgomery found to govern the pair correlation of the zeta zeros, as recognised by Dyson and verified numerically by Odlyzko [34, 35]—and the elusiveness of the Pólya operator: the Hilbert–Pólya conjecture that the zeros form the spectrum of some self-adjoint Hamiltonian.

Read in the light of holography, three points stand out: (i) the Veneziano amplitudes generate the odd zeta values; (ii) AdS/CFT has been most successful precisely for black holes, which have been in turn related to the primes and GUE random matrices; and (iii) the Hermitian operator of the zeros, like the discrete microstate of low-dimensional AdS/CFT, is GUE—itself a statistical description used specifically when no Hamiltonian is known. Taken together, these coincidences strongly suggest that AdS/CFT bears a closer relation to the Riemann hypothesis, at the mathematical level, than it might at first appear; and that, given the relations above, the absence of a single concrete microstate in both cases may share the same mathematical—rather than physical—cause. It is on this that we base our contention that the conjecture of selection among the $\sim 10^{500}$ vacua is mistaken, and that the conflict may be principally epistemological rather than physical.

1.7. **The framework.**

1.7.1. *The proposal: the string as a vanishing point.* The proposal is a reinterpretation of the closed string as an $\mathbb{F}_1$ ring living on an M-theory torus, and the use of that ring as a *vanishing point*—taken literally as the observer. A vanishing point is, literally, how the observer is placed in a drawing: it is the datum that determines how space is disposed on the picture plane, the observational horizon. In this reading the closed string of string theory is reinterpreted as the ring R_{PCF} living on the torus T^2_{PCF}, and that point-like ring is where the observer sits. In the following chapters we show that this reinterpretation—the closed string as an $\mathbb{F}_1$ ring on an M-theory torus, and that ring used as a vanishing point taken literally as the observer—resolves several of the problems holography has faced.

1.7.2. *The vanishing point is the Vitruvian module.* A vanishing point has a classical name: the *Vitruvian module*—an origin, a radius, an angle and a magnitude. In architecture it is the base of a column's shaft, a magnitude tied to a circle, with trigonometry and polygons carrying that one magnitude through an entire building; Brunelleschi and Alberti took it up in the Renaissance to develop linear perspective. It is the ancestor of the isometric lattice and of crystallographic lattices in general—of the Eisenstein and Gaussian lattices in particular—and therefore the ancestor, both geometric and etymological, of the *modulus* of the Argand plane; the ancestor of contemporary moduli spaces, and a pioneer of classification under equivalence. Its four structural data (origin, magnitude as ratio to the module, angular direction, and equivalence under magnitudes) coincide with the data of $e^{i\pi}+1=0$, and it is in this sense that we combine Euler's identity, the modulus of $\mathbb{C}$ and the Vitruvian modulus into a single ring (the unifying isomorphism is established independently in [36]).

As discussed in §6, this approach is related to Lawvere's reinterpretation of Hegel—the unity of opposites formalized as an adjunction [21]—and to the cohesive setting in which Huerta and Schreiber place it in *M-theory from the superpoint* [23]; the mechanisms used here are different. In practice this is not the route of §1.5.2, though it shares its view that logic comes from geometry: Lawvere opened *Quantifiers and sheaves* by holding geometry to be the leading aspect of the unity between logic and geometry, and claimed there that logic is in a sense a special case of it [37]. What is taken here are the components of the vanishing point—the Vitruvian module, and the rhetoric around φ and π that it lets us develop as the geometric degree zero of the contemporary moduli spaces, as we established in [38].

1.7.3. *Exactness: from the drawn point to the ring on the torus.* The difficulty is that a vanishing point, to be formalised, must be exact. The classical vanishing point of drawing is composed with π, trigonometry, polygons, magnitudes and φ, among other ratios—all used approximately in the arts. It must become precise to cross to the AdS/CFT side, and to the crystallographic lattices that give this manuscript its name. Our proposal for achieving this is to treat a string on a torus generating a worldsheet as that vanishing point, but acting as a specific algebraic ring: to put the vanishing point *in the particle*, rather than somewhere in the field, and to let the observer become a geometric superpoint that emerges from pure self-consistency relations intersecting the domains of strings, AdS/CFT and the Riemann ζ. The ring is

$$R_{\mathrm{PCF}} = \mathbb{Z}[\varphi, \varphi^{-1}, \tfrac{1}{2}], \tag{1.12}$$

whose self-similarity is seeded by the fixed-point equation

$$\varphi^2 = \varphi + 1, \qquad \varphi = \tfrac{1+\sqrt{5}}{2} = 2\cos\tfrac{\pi}{5} = \zeta_{10} + \zeta_{10}^{-1}, \tag{1.13}$$

P [phi_sq]

which extends the role of $i^2 = -1$ from $\mathbb{R} \to \mathbb{C}$ to the further embedding $\mathbb{C} \to \mathbb{E}^3$ via $z = \varphi y$. The ring lives on the torus T^2_{PCF}, which maintains a relation between 2D and 3D symmetry without selecting bulk or boundary. The same i that seeds $\mathbb{C}$ also fixes the modulus from the π-side: through $e^{i\pi} = -1$ and the Gaussian integral one has $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, hence the half-factorial $(\frac{1}{2})! = \frac{1}{2}\sqrt{\pi}$, so the common modulus $\mu = \frac{1}{2}$ arises on the i/π side exactly as on the φ side (proved in §2.2, Lemma 2.7 and Theorem 2.8; the $\pi \leftrightarrow \varphi$ bridge is Proposition 2.5). The mechanism of exactness is a descent from arithmetic to pure geometry that we call Tarskian, after the decidability of the first-order theory of real-closed fields: arithmetic statements are recast as geometric ones, and numbers are treated as ratios—in particular as trigonometric ratios through $\varphi = 2\cos(\pi/5)$. The descent is what lets the construction sidestep the direct self-reference that the arithmetic statement of the problem would require, and it yields a single geometric–arithmetic invariant—the unit of scale, one bit—that integrates the two origins of the modulus into one state (made precise in §2.6, where the commutative diagram of Theorem 2.64 shows the eight routes to $\mu = \frac{1}{2}$ coincide). The framework that results sits halfway between set theory and intuitionism: intuitionist in the classical sense—taking the observation of nature as its starting point rather than purely abstract axioms—yet not in conflict with classification under set-theoretic equivalence, so that self-reference is mediated at the intersection of the two systems rather than prohibited. Conceptually it is a self-similar, interdimensional fractal that contains, is part of, and unifies—within a single lattice—the symmetries of $e^{i\pi}$, $\mathbb{N}$, $\mathbb{E}^3$, the primes, the half-integers and $\mathbb{C}$, working as a contemporary Vitruvian module that uses φ, trigonometry and cyclotomic pentagons, mounted on a torus.

The choice of φ rests on a non-paradoxical self-reference: (1.13) is the unique positive solution of $x = 1 + 1/x$, a relation that maps onto itself without contradiction and holds consistently across dimensions—the same relation that, as Pacioli records in his *De Divina Proportione* [39], drawing on Vitruvius, made φ central in art. It is what allows a self-referential observer to be modelled without paradox. We take it up here not to make art, but through the connection already drawn above between the Riemann hypothesis, GUE, holography, string theory and Yuri Manin's $\mathbb{F}_1$ program.

1.7.4. *Powers on one side, the arithmetic spectrum of φ on the other.* To achieve this we interpret the $\mathbb{F}_1$ algebraic ring by joining, from one side, the powers, and from the other, the arithmetic spectrum of φ—from a geometric perspective. On the power side the module is expanded as a tower that is Russellian in form but geometric rather than logical, anchored by the algebraic bridge

$$\varphi^{\lambda_{\log}} = 2, \qquad \lambda_{\log} = \frac{\log 2}{\log \varphi} = \log_{\varphi} 2, \tag{1.14}$$

P [lambda_log, lambda_log_mul_regulator, mersenne_bridge]

so that φ generates the multiplicative spectrum that carries the binary base 2^p. The same tower carries the ternary base, $\varphi^{\log_{\varphi} 3} = 3$, so its growth mediates between the binary register (2) and the ternary register (3), with $\mu = \frac{1}{2}$ as the geometric mean: $\varphi^{\mu\lambda_{\log}} = \sqrt{2}$, $\varphi^{\mu \log_{\varphi} 3} = \sqrt{3}$ (proved in §2.2, Proposition 2.10; the mediated Mersenne form is Proposition 2.36 in §2.7). On the arithmetic side,

φ is the fundamental unit of $\mathbb{Q}(\sqrt{5})$ and the quadratic character χ_5 classifies every rational prime; in the cyclotomic-pentagonal arithmetic of Grisales Herrera [40] this yields a classification of the primes by φ, realised below through the Dedekind zeta of $\mathbb{Q}(\sqrt{5})$. This is the machinery used in the companion Hilbert–Pólya paper [38] to reproduce the multiplicative and additive spectrum of the primes.

1.7.5. *Primes, φ and black holes.* Here we do not approach the relation with the primes from number theory, but the relation between the primes, φ and black holes, and the intersection of those physical objects with AdS/CFT. The empirical trace is broad: φ recurs in hydrogen bonds, superconductivity, quantum phase transitions and photonic systems, and in many lattices—across nearly all domains of science wherever self-organisation or minimum-energy configurations are at play [41], including the inflation factor and the multifractal tight-binding spectrum of the Fibonacci quasicrystal [42] and the gain eigenvalue of four-mode parametric coupling in nonlinear photonic crystals [43]. The gravitational appearance is the striking one. Among the extremal Kerr–Newman black holes whose mass, charge and angular momentum are fixed multiples of the irreducible mass, Sonnino *et al.* identify a distinguished family whose coefficients depend solely on φ [44]: within it lies a unique black hole whose scale parameter and horizon Gaussian curvature at the poles both equal φ—the configuration of highest symmetry admissible in the Kerr–Newman metric, with a proposed bearing on the dark-energy density parameter Ω_Λ of the Kerr–Newman–de Sitter family. Read against the black-hole/prime relation in §1.6.3, where the Wheeler–DeWitt wavefunctions near a five-dimensional spacelike singularity become automorphic L-functions whose Euler product is a primon gas, the two results place black holes, φ and the primes on the same object from complementary sides: the horizon geometry selects φ, and the interior selects the primes. Two arithmetic properties make φ the exact carrier. By Hurwitz's theorem [45] the continued fraction $[1; 1, 1, \dots]$ makes φ the number least well approximated by rationals—maximally obstructed *as a fraction*, yet exact *as a ratio*, $\varphi = 2\cos(\pi/5)$ (1.13). Read geometrically, extremal irrationality is optimal packing—a distribution stepped by φ resonates with no rational sublattice and fills space with maximal uniformity; read informationally, the densest record of distinguishable states, so φ is the ratio at which a region holds the most information it can hold. This we propose as a golden-ratio reading of the holographic principle, and it is what makes φ the natural carrier of the modulus $\mu = \frac{1}{2}$, whose binary entropy $H(\frac{1}{2}) = 1$ is exactly one bit per fundamental cell, saturating the Bekenstein bound (3.9) in §3.1.

1.8. **The physical program.** We begin by talking: this introduction states the problem, the hypothesis, and the framework of the vanishing point. Sections 3 and 4 then demonstrate that the mathematical apparatus assembled in Section 2—the single microstate $|\Omega| = \frac{1}{2}$—makes contact with physics. The aim is not to propose new physics, but to establish that this apparatus meets criteria already fixed in the literature: each physical result is obtained by satisfying a requirement that an accepted framework had left open. The physical starting point is Maldacena's conjecture that "M/string theory on various Anti-deSitter spacetimes is dual to various conformal field theories" [17]: we demonstrate how the mathematical framework allows the holographic and the M-theoretic descriptions to meet in one object, of which the microstate is a projection. The equations are given in the sections indicated below and collected in the demonstrative lines of Sections 3.6 and 4.5.

1.8.1. *Section 2: the $\mathbb{F}_1$ apparatus that makes the vanishing point exact.* Section 2 constructs the $\mathbb{F}_1$ mathematical apparatus that makes the vanishing point exact, building its four structural data as theorems on the torus T^2_{PCF} and the ring R_{PCF}. The magnitude is derived twice over, from the φ side and from the i/π side, and eight independent routes are proved to converge on $\mu = \frac{1}{2}$; the angular datum is fixed by $\varphi = 2\cos(\pi/5)$ and by the eigenvalue triangle, whose component norms are computed rather than posited; the origin and the equivalence under magnitudes are the lattice quotient $T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda$ at $\tau = i$ together with the projection π_{PCF} that carries a ratio into the angular unit. Around these the section develops the arithmetic of $K = \mathbb{Q}(\sqrt{5})$, establishes the uniqueness of the arity that blocks the diagonal, raises the golden tower, assembles the completed zeta as the product over all places and evaluates it at $s = 1$ to obtain the entropy bridge—which turns the magnitude into one bit—and closes with the formal verification.

1.8.2. *Section 3: the framework seen in strings.* Section 3 shows how this framework looks in string theory: all of AdS/CFT and the M-theory type duality web are obtained from that single microstate at each level. There the microstate is confronted with the criteria the string side had left open—its relation to the 't Hooft–Susskind holographic bound [4], to the AdS boundary conditions [17], and to the ER = EPR bridge [46]—with Maldacena's conjecture as the meeting point of the holographic and M-theoretic readings.

1.8.3. *Section 4: the framework seen in de Sitter.* Section 4 shows how the same object looks in de Sitter, and turns to the observer and to gravity: the observer is the projection Π; it does not select a vacuum but accumulates information; and that accumulation yields, through the Landauer–Jacobson–Verlinde chain, Einstein's equations. There the microstate meets Witten's unitarity and entropy conjectures for de Sitter space [1] and the observer requirement of Chandrasekaran, Longo, Penington and Witten [2].

1.8.4. *Section 5: dynamics, operators, degrees of freedom.* Section 5 turns to the dynamics, the operators and the degrees of freedom: the same accumulation, read as degrees of freedom, translates the string amplitude tower to the positive Grassmannian—taking up there the translation already made on the amplitude side, where the Koba–Nielsen integral is the canonical form of a positive geometry [47, 48]. The eigenvalue triangle of that Grassmannian is $\mathfrak{su}(3)$, which makes the dynamics on it a gauge theory and supplies our proposal for the concrete Hamiltonian that low-dimensional holography lacks—from which a Wightman theory is reconstructed, its spectrum discrete and indexed by the same ζ. Section 6 discusses the whole. Section 7 concludes.

TABLE 1. Notation and key definitions.

Symbol / object	Meaning
$\varphi = 2\cos(\pi/5) = \frac{1+\sqrt{5}}{2}$	the golden ratio; generator of the torus
$\lambda_{\log} = \log 2/\log\varphi$	golden logarithmic scale
$\mu = \frac{1}{2}$	the modulus, $\mu = \lvert\Omega\rvert$
σ	tower level; $\sigma = \frac{3}{2}$ at the spectral point
Ω	the microstate; $\lvert\Omega\rvert = \frac{1}{2}$
$T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda,\ \tau = i$	the torus (geometric home)
$R_{\mathrm{PCF}} = \mathbb{Z}[\varphi, \varphi^{-1}, \frac{1}{2}]$	the base ring
$z = \varphi^{\sigma}$	the golden tower / throat coordinate
$N_{\mathrm{modes}} = \lfloor \pi\varphi^{\sigma} \rfloor$	mode count at level σ
ε_0	ultraviolet cutoff (certainty cell)
M_{PCF}	PCF mass scale
$G_N = \frac{1}{2}$	Newton constant
$c = 3$	central charge = colour arity = $\lfloor\pi\rfloor = \chi(\mathbb{CP}^2)$
$S_{\mathrm{BH}} = A/4G_N$	horizon (Bekenstein–Hawking) entropy
$f_{\mathrm{crit}} = \frac{1}{2}$	objectivity threshold
$\Pi : \mathbb{E}^3 \to \mathbb{C}$	the observer projection
Key definitions:	
microstate	the single object Ω, $\lvert\Omega\rvert = \frac{1}{2}$, with no ensemble average
shared signature	$\lvert\Omega\rvert = \frac{1}{2},\ \ell = 1,\ G_N = \frac{1}{2},\ c = 3,\ \mathrm{GKP} = \frac{3}{4}$
golden tower	the levels $z = \varphi^{\sigma}$ of the construction

1.9. **Relation to the companion papers.** Three companion papers—the correspondence paper [36], the $\mathbb{F}_1$-descent paper [49] and the Hilbert–Pólya paper [38]—develop, independently, the identity, arithmetic and operator threads that this construction invokes. None of them is published, so we state the principle we have followed rather than leave it implicit.

Whatever the derivations below use is proved below. Where a result also appears in a companion, it is *duplicated* here with its own proof rather than cited as an authority; the citation then records co-attribution, not dependence. This applies to the pentagonal identity $\varphi = 2\cos(\pi/5)$ (Theorem 2.4, proved here from the Chebyshev polynomial T_5), the mediated Mersenne form (Proposition 2.36), the component norms and the isometric division of the eigenvalue triad (Proposition 2.19, Lemma 2.21), the scale of the lattice (Proposition 2.29), and the completed zeta with its functional equation (Theorem 2.53, Theorem 2.55), whose Archimedean factor the $\mathbb{F}_1$ companion absorbs into a counting correction and which is derived here instead (Remark 2.54). Duplication of this kind is deliberate: each paper of the corpus is meant to be readable alone.

What is deferred is deferred in full, and used nowhere. Three bodies of material belong to the companions and enter no derivation here: the arithmetic operator whose spectrum carries the zeros of ζ, together with the lineage of that problem [38]; the Weil span and the spectral squeeze that the $\mathbb{F}_1$ program aims at the Riemann hypothesis [49]; and the $\mathbb{F}_1$ program proper—Soulé's criterion, the Connes–Consani counting function, and the desiderata by which such programs are judged [49]. Where the text names these, it is delimiting scope, not borrowing a result.

Nothing the derivations use is deferred. The four arithmetic ingredients that an earlier draft of this paper left to the companions—the Euler product as a colimit and the Frobenius lifts of the Λ-structure (§2.8, §2.9), the odd zeta values (Theorem 2.62), the Eisenstein–Gauss transition (Proposition 2.31) and the unifying ring isomorphism (§2.4)—are proved here. The companions are cited for co-attribution and for scope, and for nothing that a derivation below consumes.

2. Methods

This section develops the mathematical core of the framework, our approach to the frameworks of §1.6: Manin's quantum tori $\mathbb{F}_1$ program with Borger's Λ-algebraic ring. Here we develop only what is pertinent to the string; not the whole of the $\mathbb{F}_1$ program, which is developed in a companion paper [49]. What we develop here is presented as a single thread, from i to the two vertices $\mu = \frac{1}{2}$ and $\sigma = \frac{3}{2}$, each step formally verified in Lean 4 (theorem names in brackets); each subsection uses only objects already introduced.

Verification tags: **P**`[...]` formal proof (Lean), **N**`[...]` numerical check, **H**`[...]` hypothesis from the literature. Compound tags (**P**`[...]` **N**`[...]`) mark a claim resting on more than one, and each input is named where it enters.

2.1. The generator, its conjugate, and the binary bridge.

The generator, its conjugate, and the binary bridge. Everything below is read off one quadratic. We fix here what φ is, what its conjugate is, and the one identity of §1 we shall use throughout; nothing in this subsection depends on anything later.

$$\varphi^2 = \varphi + 1, \qquad R_{\mathrm{PCF}} = \mathbb{Z}[\varphi, \varphi^{-1}, \tfrac{1}{2}], \qquad \mathbb{E}^3 = \{(x, y, \varphi y)\}. \tag{2.1}$$

Definition 2.1 (Conjugate, trace, norm, discriminant)**.** Write $\bar\varphi = (1-\sqrt{5})/2$ for the Galois conjugate of φ, the second root of $x^2 - x - 1$. Then

$$\varphi + \bar\varphi = 1, \qquad \varphi\,\bar\varphi = -1, \qquad (\varphi - \bar\varphi)^2 = \Delta_K = 5. \tag{2.2}$$

The norm $N(\varphi) = -1$ exhibits φ as a unit; the discriminant $\Delta_K = 5$ makes 5 the only ramified rational prime, which is what the trichotomy of §2.7 rests on.

P`[phi_trace, phi_norm, galois_pair, phi_discriminant]`

Lemma 2.2 (The Galois conjugate is the involution)**.** *$\bar{\varphi} = 1 - \varphi$; equivalently, $x \mapsto (\varphi + \bar{\varphi}) - x$ and $x \mapsto 1 - x$ are the same map. The Galois conjugation of K and the involution $x \mapsto 1 - x$ are therefore not two analogous symmetries but one map, read on the orbit $\{\varphi, \bar{\varphi}\}$ and on $\mathbb{R}$.*

P `[galois_conj_is_one_sub, galois_involution_is_one_sub]`

Finally we record, without reproving it, the identity established at (1.14): $\varphi^{\lambda_{\log}} = 2$ with $\lambda_{\log} = \log_\varphi 2$. It is the only result of §1 that the present section consumes.

P `[mersenne_bridge]`

Proposition 2.3 (The bridge anchors from outside the lattice)**.** *The exponent $\lambda_{\log}$ is not reachable by descent on the lattice of integer exponents. For no integer n does $\varphi^n = 2$:*

$$\varphi < 2 < \varphi^2 = \varphi + 1, \tag{2.3}$$

and $n \mapsto \varphi^n$ is strictly monotone since $\varphi > 1$, so by (2.3) the 2 falls in the gap between two consecutive powers and is missed by all of them. The identity $\varphi^{\lambda_{\log}} = 2$ therefore holds in $\mathbb{R}$ and at no point of $\mathbb{Z}$. This is what makes (1.14) an anchor*: it fixes the scale from outside rather than being derived inside the tower, which is why the present section consumes it and does not prove it.*

P `[phi_lt_two, two_lt_phi_sq, two_not_in_golden_lattice, anchor_is_exterior]`

2.2. **Three geometric origins of $\mu = \frac{1}{2}$.** The constant $\mu = \frac{1}{2}$ arises from three independent origins on the generator of §2.1: a π-branch (Gamma/Gaussian), a φ-branch (the involution $x = 1 - x$), and an arithmetic branch (the Galois pair of K). These are the seeds.

Theorem 2.4 (Pentagonal identity)**.** *$\varphi = 2\cos(\pi/5)$.*

P `[phi_eq_two_cos_pi_fifth]`

Proof. The Chebyshev polynomial T_5 satisfies $T_5(\cos\theta) = \cos(5\theta)$ with $T_5(x) = 16x^5 - 20x^3 + 5x$. At $\theta = \pi/5$ this gives $T_5(\cos(\pi/5)) = \cos\pi = -1$, so $\cos(\pi/5)$ is a root of $16x^5 - 20x^3 + 5x + 1 = 0$, which factors as $(x+1)(4x^2 - 2x - 1)^2$. Since $0 < \pi/5 < \pi/2$ forces $\cos(\pi/5) > 0$, we have $\cos(\pi/5) \neq -1$; hence $\cos(\pi/5)$ is a root of $4x^2 - 2x - 1 = 0$, whose positive root is $(1 + \sqrt{5})/4 = \varphi/2$. Therefore $\varphi = 2\cos(\pi/5)$; this classical identity is also derived in the correspondence paper [36]. ☐

Proposition 2.5 (Pentagonal bridge $\pi \leftrightarrow \varphi$)**.** *Since $\varphi = 2\cos(\pi/5)$ (Theorem 2.4),*

$$\pi = 5\arccos(\varphi/2). \tag{2.4}$$

P `[pi_eq_five_arccos]`

Proposition 2.6 (Gamma as an infinite integral (Euler))**.** *For $s > 0$,*

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, dt. \tag{2.5}$$

P `[gamma_integral]`

Lemma 2.7 (Gaussian value)**.** *$\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (classical; from the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$).*

P `[gamma_half, gaussian_integral_value, gaussian_weight_phi]`

Theorem 2.8 (Half-factorial (classical))**.**

$$(\tfrac{1}{2})! \;=\; \Gamma(\tfrac{3}{2}) \;=\; \mu\sqrt{\pi}, \qquad \textit{equivalently} \qquad \mu = \frac{(\frac{1}{2})!}{\sqrt{\pi}}. \tag{2.6}$$

P `[half_factorial]`

The Gaussian of Lemma 2.7 enters with its width still free; three separate conditions close it.

Proposition 2.9 (Uniqueness of the self-dual Gaussian)**.** *Let $g_a(x) = e^{-ax^2}$ be the Gaussian family of which Lemma 2.7 uses the case $a = 1$, with the width $a > 0$ now left free, and let $\widehat{g}(\xi) = \int_{\mathbb{R}} g(x)e^{-2\pi i x\xi}\,dx$, so that $\widehat{g_a}(\xi) = \sqrt{\pi/a}\ e^{-\pi^2\xi^2/a}$. Three independent conditions each fix $a = \pi$: normalisation, the amplitude of the Fourier transform, and its width. The self-dual Gaussian is therefore*

$$g(x) = e^{-\pi x^2}, \qquad \widehat{g} = g, \qquad \int_{\mathbb{R}} g = 1. \tag{2.7}$$

***P**[gauss_normalised_iff, fourier_amplitude_iff, fourier_width_iff, selfdual_gaussian_unique]*

Proof. $\int_{\mathbb{R}} g_a = \sqrt{\pi/a}$, so normalisation gives $\sqrt{\pi/a} = 1$, hence $a = \pi$. Self-duality $\widehat{g_a} = g_a$ gives independently $\sqrt{\pi/a} = 1$ (amplitude) and $\pi^2/a = a$ (width); since $a, \pi > 0$, the second reads $(a-\pi)(a+\pi) = 0$ with $a + \pi > 0$, hence again $a = \pi$. The three conditions are distinct and coincide. □

Proposition 2.10 (Mediation half-powers)**.**

$$\varphi^{\mu\lambda_{\log}} = \sqrt{2}, \qquad \varphi^{\mu \log_\varphi 3} = \sqrt{3}. \tag{2.8}$$

***P**[sqrt2_mediation, sqrt3_mediation]*

Proposition 2.11 (φ-branch: the involution and its unique fixed point)**.** $\mu = \frac{1}{2}$ *is the* unique *fixed point of $x \mapsto 1 - x$: for real x, $x = 1 - x$ if and only if $x = \frac{1}{2}$.*

***P**[face_inv_apex]*

Proposition 2.12 (Arithmetic branch: the midpoint of the Galois pair)**.** *By Lemma 2.2 the Galois conjugation of K and the involution $x \mapsto 1 - x$ are the same map; the midpoint of the pair $\{\varphi, \bar{\varphi}\}$ is therefore its fixed point,*

$$\frac{\varphi + \bar{\varphi}}{2} = \frac{1}{2} = \mu, \tag{2.9}$$

by the trace of (2.2). This is the third seed, and it is the one that is arithmetic*: the other two are analytic and elementary, whereas this one is an invariant of the field K itself.*

***P**[face_galois_apex]*

2.3. **Arity uniqueness and the blocking of the diagonal.** The arity 3 and the modulus $\frac{1}{2}$ are forced, and the Lawvere–Yanofsky diagonal is contracted (not prohibited).

Definition 2.13 (Distributed self-reference)**.** A recurrence of depth k is *distributed* when $k \geq 2$ (no component depends directly on itself).

P[distributed_self_reference]

Theorem 2.14 (Fibonacci minimality)**.** *Depth $k = 2$ is minimal, and the unique positive root of $r^2 = r + 1$ is φ. Among natural-coefficient depth-2 recurrences $F_{n+2} = c_1 F_{n+1} + c_2 F_n$, the pair $c_1 = c_2 = 1$ is the unique one whose characteristic root—hence growth ratio—is φ.*

***P**[phi_root_unique, characteristic_root_is_phi]*

***N**[eq:phisq]*

Proposition 2.15 (Spectral constraints force $(\sigma, \mu) = (\frac{3}{2}, \frac{1}{2})$)**.** *$\sigma + \mu = 2$, $\sigma\mu = \frac{3}{4}$, with unique solution $0 < \mu < 1$ giving $\mu = \frac{1}{2}$, $\sigma = \frac{3}{2}$.*

***P**[spectral_uniqueness]*

Remark 2.16 (No-diagonal). $\|\hat{\Omega}\| = \frac{1}{2} < 1$ contracts the diagonal below the contradiction threshold.

2.4. **Construction of PCF and the base ring on the torus.** On the generator and the conjugate of §2.1, the ring of integers and the regulator, then the base ring, the tripartite operator, and the projection that fixes ε_0 out of the regulator.

Proposition 2.17 (The ring of integers and the base ring)**.**

$$\mathcal{O}_K = \mathbb{Z}[\varphi], \qquad R_{\mathrm{PCF}} = \mathbb{Z}\big[\varphi, \varphi^{-1}, \tfrac{1}{2}\big]. \tag{2.10}$$

The two differ by integrality: φ is a root of the monic $x^2 - x - 1$*, so it is an algebraic integer and lies in $\mathcal{O}_K$, whereas $\frac{1}{2}$ is a root of $2x - 1$, which is not monic. R_{PCF} therefore adjoins to $\mathcal{O}_K$ an element that is not an algebraic integer, namely $\frac{1}{2}$; the modulus that the tripartite operator will carry is that same $\frac{1}{2}$, a unit of R_{PCF} and not of $\mathcal{O}_K$.*

P *[phi_is_algebraic_integer, half_not_algebraic_integer]*

Definition 2.18 (Regulator and period)**.**

$$R_K = \log\varphi, \qquad T = 2\pi \log\varphi = 2\pi R_K. \tag{2.11}$$

R_K is the regulator of K in the arithmetic sense—not the ultraviolet regulators of §3.3—and T is the period of the torus constructed below in this subsection. Both are named here because $\varepsilon_0 = \ln\varphi/(6\sqrt{3})$ is derived below out of $\ln\varphi$, and the reader should have the invariant named before it is used.

P [regulator_K_pos, period_eq_two_pi_regulator]

$$\Omega = P \cdot C \cdot F. \tag{2.12}$$

The three component norms are not posited: they are forced by the eigenvalue geometry, and their product is the modulus (Proposition 2.19, (2.13)).

Proposition 2.19 (Component norms from the eigenvalue triangle)**.** *The operator $\hat{\Omega}$ has eigenvalues $\lambda_k = \frac{1}{2}\omega^k$ ($k = 0, 1, 2$, $\omega = e^{2\pi i/3}$), inscribing an equilateral triangle in the circle $|z| = R = \frac{1}{2}$ of $\mathbb{C}$. The component norms are fixed by this geometry:*

$$\|C\| = 1, \qquad \|F\| = \tfrac{\sqrt{3}}{2}, \qquad \|P\| = \tfrac{1}{\sqrt{3}}, \tag{2.13}$$

whence $\|P\|\,\|C\|\,\|F\| = \frac{1}{2}$, which is also the operator norm $\|\hat{\Omega}\|$ and the common modulus $|\lambda_k|$ of each eigenvalue. Three quantities coincide here and are distinguished in Proposition 2.20, where they cease to: the product of the three component *norms, the product of the three eigenvalue* moduli*, and the product of their* real parts*.*

P *[norm_p_eq_tan, norm_f_eq_cos, omega_eigenvalues]*

Proof. *(C)* C is the real eigenvalue $\lambda_0 = \frac{1}{2}$, normalised by the circumradius $R = |\lambda_0| = \frac{1}{2}$: $\|C\| = |\lambda_0|/R = 1$. *(F)* F is the chord between adjacent eigenvalues, $\|F\| = |\lambda_1 - \lambda_0| = \frac{1}{2}|\omega - 1| = \sin(\pi/3) = \frac{\sqrt{3}}{2}$. *(P)* The fundamental right triangle (centre, midpoint of $\lambda_0\lambda_1$, vertex λ_0) has apothem $R\cos(\pi/3) = \frac{1}{4}$ and half-side $\|F\|/2 = \frac{\sqrt{3}}{4}$, so $\|P\| = \dfrac{R\cos(\pi/3)}{\|F\|/2} = \dfrac{1/4}{\sqrt{3}/4} = \frac{1}{\sqrt{3}} = \tan(\pi/6)$. *(Product)* The $\sqrt{3}$ of $\|F\|$ cancels that of $\|P\|$, leaving $\frac{1}{2}$; the cancellation is specific to the equilateral (n=3) triangle. ☐

Proposition 2.20 (Invariants of the eigenvalue triad)**.** *For the triad $\lambda_k = \frac{1}{2}\omega^k$ of Proposition 2.19, with $\varphi^{\lambda_{\log}} = 2$ (1.14):*

$$\operatorname{Re}\lambda_k = \varphi^{-\lambda_{\log}} \cos\Big(\frac{2\pi k}{3}\Big), \qquad k = 0, 1, 2, \tag{2.14}$$

so the real spectrum is $(\frac{1}{2}, -\frac{1}{4}, -\frac{1}{4})$ *and its sum vanishes. The two products do* not *agree, and each exponent records a different face of the same* $\varphi^2 = \varphi + 1$*:*

$$\prod_k |\lambda_k| = 2^{-3}, \qquad \prod_k \operatorname{Re}\lambda_k = 2^{-5} = \varphi^{-5\lambda_{\log}}. \tag{2.15}$$

The base 2 *is the binary bridge* (1.14) *in both. The exponent* 3 *is the arity of the triad—the same* $n{=}3$ *whose cancellation gives* $\frac{1}{2}$ *in Proposition 2.19—and the exponent* 5 *is the pentagon of Theorem 2.4. The two identities that generate the framework meet inside one invariant of the spectrum.*

P [triad_re, triad_trace_zero, triad_two_products, triad_det_is_phi_pow]

Lemma 2.21 (Isometric embedding of the triad)**.** *Let* $v_k = \frac{1}{2}\omega^k$ $(k = 0, 1, 2)$ *be the eigenvalue triad of Proposition 2.19, and set* $\|v\|^2 = \sum_k |v_k|^2$ *and* $V_k = v_k/\|v\|$*. Then*

$$\|v\|^2 = 3 \cdot \tfrac{1}{4} = \tfrac{3}{4}, \qquad \|v\| = \tfrac{\sqrt{3}}{2}, \qquad \sum_{k=0}^{2} \|V_k\|^2 = 1, \tag{2.16}$$

so the column map $V\colon \mathbb{C} \to \mathbb{C}^3$*,* $z \mapsto (V_0, V_1, V_2)\, z$*, is norm preserving:* $V^\dagger V = \mathbf{1}$*.*

P [bulk_boundary_isometry]

Proof. Each $|v_k| = \frac{1}{2}$ (Proposition 2.19), so $\|v\|^2 = 3 \cdot \frac{1}{4} = \frac{3}{4}$ and $\|v\| = \sqrt{3}/2$. Then $\|V_k\|^2 = |v_k|^2/\|v\|^2 = (\frac{1}{4})/(\frac{3}{4}) = \frac{1}{3}$ for each k, and the three sum to 1. For a single column V that sum is $V^\dagger V$. □

Remark 2.22 (Where the two spectral constraints come from)*.* The system solved in Proposition 2.15 is not assumed: both of its constraints are established here. The product is the squared norm of the eigenvalue triad, $\sigma\mu = \|v\|^2 = \frac{3}{4}$ (2.16), and the ratio is the arity, $\sigma/\mu = d = 3$ (§2.3). Eliminating $\sigma = 3\mu$ gives $3\mu^2 = \frac{3}{4}$, hence $\mu = \frac{1}{2}$ and $\sigma = \frac{3}{2}$, and only then $\sigma + \mu = 2$. The same $\frac{3}{4}$ reappears as the colour ratio $1 - \mu_3^2$ and as the GKP entry of the shared signature (§3.5), which is why it is an invariant of the object and not a fitted value.

P [spectral_uniqueness, bulk_boundary_isometry]

Definition 2.23 (PCF projection)**.** For $a, b \in \mathbb{R}$ and $c \neq 0$, the *PCF projection* sends the ratio ab/c onto the P-component of the triad and measures it in the angular unit of the eigenvalue triangle:

$$\pi_{\mathrm{PCF}}(a, b, c) \;=\; \frac{ab}{c} \cdot \underbrace{\|P\|}_{1/\sqrt{3}} \cdot \underbrace{\frac{\pi}{3}}_{\text{angular unit}} \;=\; \frac{ab}{c\sqrt{3}} \cdot \frac{\pi}{3}. \tag{2.17}$$

P [projection_PCF]

Remark 2.24 (The two constants of (2.17) are already fixed)*.* Neither factor is free. The projection norm is $\|P\| = 1/\sqrt{3} = \tan(\pi/6)$, forced by the eigenvalue geometry in Proposition 2.19; and $\pi/3$ is the angular unit of the same equilateral triangle—the half-angle between adjacent eigenvalues $\lambda_k = \frac{1}{2}\omega^k$, already used as the normalising angle in Proposition 2.33, where $\sigma = \zeta(2)/(\pi/3)^2$ and $\cos(\pi/3) = \mu$. So (2.17) is not a fitted form: it is "ratio, projected on P, in units of $\pi/3$", and both constants come from (2.13).

P [oplus_formula]

Proposition 2.25 (Fibonacci sum as projection)**.** $F = P \oplus C := \pi_{\mathrm{PCF}}(P, C, 1) = \dfrac{(PC)\,\pi}{3\sqrt{3}}$.

Theorem 2.26 (Derivation of ε_0)**.** *Apply* (2.17) *to the three data of the module direction—the modulus* $\mu = \sin(\pi/6)$ (2.27)*, the logarithmic generator* $\ln\varphi$ *of the multiplicative tower* φ^σ *of* (2.1)*, and the half-turn* π *accumulated once around the fundamental domain (Proposition 2.28):*

$$\varepsilon_0 = \pi_{\mathrm{PCF}}\big(\sin(\tfrac{\pi}{6}),\, \ln\varphi,\, \pi\big) = \frac{\ln\varphi}{6\sqrt{3}}. \tag{2.18}$$

P *[epsilon0_from_projection]*

Proof. By (2.17) with $a = \sin(\pi/6) = \frac{1}{2}$, $b = \ln\varphi$ and $c = \pi$,

$$\pi_{\mathrm{PCF}}\big(\tfrac{1}{2}, \ln\varphi, \pi\big) = \frac{\frac{1}{2}\ln\varphi}{\pi} \cdot \frac{1}{\sqrt{3}} \cdot \frac{\pi}{3} = \frac{\ln\varphi}{6\sqrt{3}}.$$

The three arguments are not free either: $\sin(\pi/6)$ is the modulus μ of Theorem 2.32 below, $\ln\varphi$ is the logarithmic rate of the only tower the ring (2.1) generates, and π is the monodromy of the fibre, of order two by Proposition 2.28 below. The π of the angular unit and the π of the monodromy cancel, which is why ε_0 is a pure multiple of $\ln\varphi$ and why the scale it fixes below satisfies $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ exactly. □

Remark 2.27 (Torus, scale, and Sierpiński attractor). The microstate is realised on the torus

$$T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda_{\mathrm{PCF}}, \qquad \Lambda_{\mathrm{PCF}} = M_{\mathrm{PCF}}\,\mathbb{Z}[i], \qquad \tau = i, \tag{2.19}$$

whose scale is fixed by ε_0 of (2.18):

$$M_{\mathrm{PCF}} = \frac{6\sqrt{3}\,\pi}{\ln\varphi} = \frac{\pi}{\varepsilon_0} \approx 67.846. \tag{2.20}$$

Equivalently, the *Certainty Principle*

$$\varepsilon_0\, M_{\mathrm{PCF}} = \pi. \tag{2.21}$$

N `[eq:certainty]` The three eigenvalues $\lambda_k = \frac{1}{2}\omega^k$ define an iterated function system of three contractions of ratio $\frac{1}{2}$,

$$T_k(z) = \tfrac{1}{2}(z - v_k) + v_k, \qquad v_k = \tfrac{1}{2}\omega^k, \qquad k \in \{0,1,2\}, \tag{2.22}$$

whose Sierpiński-type attractor has Hausdorff dimension

$$d_H = \frac{\log 3}{\log 2} \approx 1.585, \qquad 2^{d_H} = 3, \tag{2.23}$$

the Moran equation counting the three components P, C, F.

2.5. **The non-orientable fibre is independent of the torus.** The fibre, the isometry type of each map in the golden coupling, and the lattice scale that closes the two.

The base torus of (2.19) carries a module direction and one non-orientable identification of that direction, both fixed by data already in place. The module direction is the phase $\Omega(\tau) = \frac{1}{2}\,e^{i\tau\ln\varphi}$, of constant modulus $|\Omega(\tau)| = \frac{1}{2} = \|\hat{\Omega}\|$ (Proposition 2.19).

Arithmetic type: an Eisenstein–Gauss transition. The eigenvalue triangle $\lambda_k = \frac{1}{2}\omega^k$ ($\omega = e^{2\pi i/3}$) of Proposition 2.19 sits in the Eisenstein lattice $\mathbb{Z}[\omega]$ (hexagonal, 120°); the torus modulus $\tau = i$ sits in the Gauss lattice $\mathbb{Z}[i]$ (square, 90°). The golden coupling $z = \varphi y$ of (2.1) carries the first into the second, and Proposition 2.31 identifies exactly which map is an isometry at each step: the passage $120° \to 90°$ is not a distortion of one lattice into another but a change of the group that acts, and $\tau = i$ is the fixed point of the one that survives. The mediator is φ and no other: from (2.1), $\varphi = 1 + 1/\varphi$, so its continued fraction is $[1; 1, 1, \ldots]$ and φ is the extremal irrational, most poorly approximated by rationals (Hurwitz [45])—the optimality that also places φ in phyllotaxis—and the modulus $\frac{1}{2}$ is carried uniformly on the fibre.

Proposition 2.28 (The fibre is independent of the base algebra)**.** *The non-orientable identification lives in the fibre, not in the algebra of the torus, and is therefore independent of whether that algebra commutes. Carried once around the fundamental domain, the module direction accumulates the half-turn $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ (2.21), acting on the fibre of modulus $\frac{1}{2}$ as $e^{i\pi} = -1$: the reflection $v \mapsto -v$, π and not 2π because $\|\hat{\Omega}\| = \frac{1}{2}$, and orientation-reversing of order two, $(-1)^2 = 1$. This datum is a $\mathbb{Z}/2$ reflection of the fibre; it carries no hypothesis on the base commutator phase $UV = e^{2\pi i\theta}VU$. Hence* *the same fibre couples onto any torus—the commutative torus ($\theta = 0$) and the noncommutative torus T^2_θ that M-theory uses (θ irrational, §1.6)—while preserving $|\Omega| = \frac{1}{2}$ on either.*

P[fibre_independent_of_base, fibre_monodromy, fibre_reflection_order_two]

Proof. The fibre monodromy is $e^{i\pi} = -1$, of order two, $(-1)^2 = 1$, a $\mathbb{Z}/2$ datum that holds with no condition on θ (**P**[fibre_independent_of_base]); the base commutes for $\theta \in \mathbb{Z}$ and does not for θ irrational, the fibre being the same in both. The modulus $\frac{1}{2}$ is Proposition 2.19, constant in τ since $|\Omega(\tau)| = \frac{1}{2}$, and is preserved by the reflection. □

Proposition 2.29 (M_{PCF} is fixed by the ring on the torus, not chosen)**.** *The lattice scale of* (2.19) *is not a free parameter. Two data already in place fix it. The module direction advances at the rate ε_0 of* (2.18) *per unit of lattice length, and carried once around the fundamental domain it must return the fibre to itself through the order-two monodromy of Proposition 2.28—the half-turn π, not 2π, because $\|\hat{\Omega}\| = \frac{1}{2}$. That closure condition,*

$$\varepsilon_0 \, M_{\mathrm{PCF}} = \pi, \tag{2.24}$$

has the unique solution $M_{\mathrm{PCF}} = \pi/\varepsilon_0$, and every factor of it is an invariant of R_{PCF} on the torus:

$$M_{\mathrm{PCF}} \;=\; \frac{\pi}{\varepsilon_0} \;=\; \frac{3\pi}{\mu\,\|P\|\,\ln\varphi} \;=\; \frac{d^2\,\pi}{\|v\|\,\ln\varphi} \;=\; \frac{6\sqrt{3}\,\pi}{\ln\varphi} \;\approx\; 67.846, \tag{2.25}$$

with $d = 3$ the arity of §2.3, $\mu = \frac{1}{2}$ the modulus, $\|P\| = 1/\sqrt{3}$ the projection norm and $\|v\| = \sqrt{3}/2$ the norm by which Lemma 2.21 divides the eigenvalue triad to make $V^\dagger V = \mathbf{1}$, and $\ln\varphi$ the logarithmic rate of the generator of (2.1).

P[M_PCF_eq_pi_div_eps0, epsilon0_from_projection, bulk_boundary_isometry]

Proof. The monodromy is $e^{i\theta}$ with $\theta = \varepsilon_0 M_{\mathrm{PCF}}$ the angle accumulated over one period. Proposition 2.28 requires that this act on the fibre of modulus $\frac{1}{2}$ as $v \mapsto -v$, of order two; on $[0, 2\pi)$ the only such angle is $\theta = \pi$, whence (2.24) and $M_{\mathrm{PCF}} = \pi/\varepsilon_0$, the solution being unique because $\varepsilon_0 > 0$.

For (2.25), (2.17) and (2.18) give $\varepsilon_0 = \mu\|P\|\ln\varphi/3$, so $M_{\mathrm{PCF}} = 3\pi/(\mu\|P\|\ln\varphi)$. The isometric division of Lemma 2.21 has $\|v\| = \sqrt{3}/2$, which is $\|F\|$ of (2.13), and $\|P\|\|F\| = \frac{1}{2} = \mu$ by Proposition 2.19 with $\|C\| = 1$. Hence $\mu\|P\| = \|P\|^2\|v\| = \|v\|/d$, using $\|P\|^2 = 1/3 = 1/d$, and $3\pi/(\|v\|\ln\varphi/d) = d^2\pi/(\|v\|\ln\varphi)$. Substituting $\|v\| = \sqrt{3}/2$ and $d = 3$ gives $18\pi/(\sqrt{3}\ln\varphi) = 6\sqrt{3}\pi/\ln\varphi$. □

Remark 2.30 (What this rules out). Had M_{PCF} been an independent scale, (2.24) would be a normalisation choice and the Certainty Principle a convention. It is neither: the rate ε_0 is fixed by the ring's own data (Theorem 2.26), the required monodromy is fixed by the fibre being of order two (Proposition 2.28), and the lattice scale is what closes the two. In particular M_{PCF} is not tuned to the arity, the modulus or the isometric division—it is *their quotient*, (2.25).

Proposition 2.31 (The golden coupling: which map is an isometry)**.** *Four maps occur in the passage from the eigenvalue triangle to the torus, and the isometry type of each is different and explicit.*

- *(i)* The Farish rotation is an ordinary isometry of $\mathbb{R}^3$. *The cyclic permutation $\rho(x, y, z) = (y, z, x)$ is orthogonal with* $\det\rho = +1$ *and* $\rho^3 = \mathrm{id}$, *fixing the diagonal $a = \frac{1}{\sqrt{3}}(1, 1, 1)$. It preserves the inner product, hence all lengths and distances. The golden coupling is therefore not one plane but the ρ-orbit of* three congruent planes, *congruent in the ordinary metric sense, with normals of equal length $s_\varphi = \sqrt{1 + \varphi^2} = 2\sin\frac{2\pi}{5}$ and equal mutual inclination* $\arccos(-1/\sqrt{5})$.
- *(ii)* The Farish sense enters only in the projection. *Along a the three axes are foreshortened equally, $\|\pi_a(e_k)\| = \sqrt{2/3}$, and appear at $120°$. This is a property of how the figure is* seen, *not of the objects, and it is the sense Farish fixed in 1822 [50].*

(iii) On the torus the order three descends as an ordinary isometry too. $\mathrm{Aut}(\Lambda_{\mathrm{PCF}}) = \mathbb{Z}[i]^\times \cong \mathbb{Z}_4$ *contains no element of order three, so the symmetry changes type rather than disappearing: it acts by* translation *by a three-torsion point of* $\mathbb{T}^2_{\mathrm{PCF}}[3] \cong (\mathbb{Z}/3)^2$*. Translations are isometries of the flat torus, and since* $3t = 0$ *gives* $2t = -t$*, the orbit* $\{p, p+t, p+2t\}$ *is* equilateral in the ordinary metric*, of side* $|t|$*. Both* $\mathbb{Z}_4$ *(rotation, the Gauss face* $\tau = i$*) and* $\mathbb{Z}_3$ *(translation, the ternary face) therefore act by isometries.*

(iv) Only the coupling map itself expands, and by an explicit factor. *The induced metric of* $\iota(x + iy) = (x, y, \varphi y)$ *is*

$$\|\iota(x + iy)\|^2 = x^2 + (1 + \varphi^2)y^2, \tag{2.26}$$

so ι *preserves the real direction and expands the imaginary one by* s_φ*. This does not contradict (i)–(iii): they concern different maps.*

Consequently the passage $120^\circ \leftrightarrow 90^\circ$ *is a change of the group under which the structure is read—in* $\mathbb{R}^3$ *the projection along the Farish axis, on the torus rotation versus translation—and not a morphism carrying one lattice onto another.*

P*[farishRot_cube, farishRot_diag, farishRot_preserves_dot, farishRot_preserves_norm, order_three_not_in_Z4, translation_preserves_difference, three_torsion_equilateral, coupling_metric, coupling_expands_imaginary, coupling_preserves_real]*

Proof. (i) ρ is a permutation matrix: $\rho^\mathsf{T}\rho = \mathbf{1}$, $\det\rho = +1$, $\rho^3 = \mathrm{id}$, $\rho(1,1,1) = (1,1,1)$; and $\langle \rho u, \rho v\rangle = \langle u, v\rangle$ by inspection of the three terms. Congruence of the planes follows because ρ is orthogonal, so $\rho(n^\perp) = (\rho n)^\perp$. (ii) $\langle e_k, a\rangle = 1/\sqrt{3}$ gives $\pi_a(e_k) = e_k - \frac{1}{3}(1,1,1)$, of squared norm $\frac{2}{3}$, and $\langle \pi_a(e_j), \pi_a(e_k)\rangle = -\frac{1}{3}$ for $j \neq k$, so $\cos\angle = -\frac{1}{2}$. (iii) $3 \nmid 4$; $(p + t) - (q + t) = p - q$ in any abelian group, so translation preserves differences and hence the quotient metric; and $3t = 0 \Rightarrow 2t = -t$ makes the three pairwise differences $\pm t$. (iv) is (2.26) evaluated at $x = 1, y = 0$ and at $x = 0, y = 1$. ☐

2.6. **The single microstate, the meta-invariant, and the geometric origin of $\frac{1}{2}$ tied to the Euler product.** The geometric origin of $\frac{1}{2}$ via the angle $\pi/6$, its anchoring to the critical line, and the factorial as connector.

Theorem 2.32 (Trigonometric collapse of the norm product)**.**

$$\|P\|\,\|C\|\,\|F\| = \tan(\tfrac{\pi}{6}) \cdot 1 \cdot \cos(\tfrac{\pi}{6}) = \sin(\tfrac{\pi}{6}) = \tfrac{1}{2}. \tag{2.27}$$

P*[collapse_norm_product, norm_product_half]*

Proposition 2.33 (Angle $\pi/3$ gives both invariants)**.** $\sin(\frac{\pi}{6}) = \cos(\frac{\pi}{3}) = \mu$*, and*

$$\sigma = \frac{\zeta(2)}{(\pi/3)^2} = \frac{\pi^2/6}{\pi^2/9} = \frac{9}{6} = \frac{3}{2}. \tag{2.28}$$

P*[sigma_from_basel, sin_cos_mu]*

Proposition 2.34 (Factorial as connector)**.** $\dfrac{(\frac{1}{2})!}{\sqrt{\pi}} = \mu = \frac{1}{2}$ *(half face), and* $3! = 6 = |S_3|$ *(integer face).*

P*[factorial_face, factorial_six]*

Remark 2.35 ($\frac{1}{2} \leftrightarrow$ Euler product)*.* That the modulus $\frac{1}{2} = \mu$ is anchored to the self-dual line is not an analogy: it is Theorem 2.55 together with Remark 2.56, where the fixed point of $s \mapsto 1 - s$ is shown to be the same $\frac{1}{2}$. The Euler product itself is Proposition 2.47, and it factors through the ternary level $\mu_3 = \frac{1}{2}$.

2.7. **Expansion of the spectrum mediated between i and φ.** Additive spectrum (pentagons) and multiplicative spectrum (Mersenne), joined by the 3/2 structure; and the character χ_5 read off the pentagon itself, with the Fibonacci criterion that makes the classification of the primes effective.

$$\varphi^n = F_n\varphi + F_{n-1} \quad \text{(additive)}, \qquad 3\,\varphi^{\sigma(p)} = 2^p \quad \text{(Mersenne)}. \tag{2.29}$$

Proposition 2.36 (Mediated Mersenne)**.** $3\,\varphi^{\mu p \lambda_{\log}} = 3\,(\sqrt{2})^p$.

P `[mersenne_mediated]`

Definition 2.37 (The quadratic character)**.** χ_5 is the Legendre symbol $\left(\frac{\cdot}{5}\right)$ on $\mathbb{Z}/5\mathbb{Z}$:

$$\chi_5(0) = 0, \qquad \chi_5(1) = \chi_5(4) = +1, \qquad \chi_5(2) = \chi_5(3) = -1, \tag{2.30}$$

multiplicative on $(\mathbb{Z}/5\mathbb{Z})^\times$, with $\sum_{a=0}^4 \chi_5(a) = 0$.

P `[chi5, chi5_even, chi5_mul_on_units, chi5_sum_zero]`

Proposition 2.38 (The pentagon dictates χ_5)**.** *The four pentagonal cosines are exactly $\{\pm\varphi, \pm\varphi^{-1}\}$, so*

$$\bigl|2\cos(\pi a/5)\bigr| = \varphi^{\chi_5(a)}, \qquad \textit{equivalently} \qquad \chi_5(a) = \frac{\log\bigl|2\cos(\pi a/5)\bigr|}{\log\varphi}, \qquad a = 1,2,3,4. \tag{2.31}$$

The character is therefore read off the pentagon itself, through Theorem 2.4 ($\varphi = 2\cos(\pi/5)$), and needs no separate arithmetic input.

P `[norm_cosine_phi, phi_eq_two_cos_pi_fifth]`

Proof. $2\cos(\pi/5) = \varphi$ is Theorem 2.4. The double-angle formula with $\cos(\pi/5) = \varphi/2$ and $\varphi^2 = \varphi + 1$ gives $2\cos(2\pi/5) = \varphi - 1 = \varphi^{-1}$, and $\cos(\pi - x) = -\cos x$ carries the remaining two cases: $\cos(3\pi/5) = -\cos(2\pi/5)$ and $\cos(4\pi/5) = -\cos(\pi/5)$. Taking absolute values gives $\varphi, \varphi^{-1}, \varphi^{-1}, \varphi$ against $\chi_5 = +1, -1, -1, +1$. □

Proposition 2.39 (The logarithmic signature of φ)**.**

$$\sum_{a=1}^{4} \chi_5(a)\, \log\bigl(2\sin(\pi a/5)\bigr) = -2\log\varphi. \tag{2.32}$$

The whole content is $\sin 2x = 2\sin x\cos x$: with $\sin(3\pi/5) = \sin(2\pi/5)$ and $\sin(4\pi/5) = \sin(\pi/5)$ the sum collapses to $2\log\bigl(\sin(\pi/5)/\sin(2\pi/5)\bigr)$, and $\sin(2\pi/5)/\sin(\pi/5) = 2\cos(\pi/5) = \varphi$. So the sine reads the same character the cosine read in (2.31), and φ is the value in both.

P `[sin_ratio_eq_two_cos, pentagon_log_signature]`

Proposition 2.40 (Fibonacci criterion and the trichotomy)**.** *For every rational prime $q \neq 2, 5$,*

$$F_q \;\equiv\; \left(\frac{q}{5}\right) \;=\; \chi_5(q) \quad (\mathrm{mod}\ q), \tag{2.33}$$

and the value is determined by the residue class modulo 20: $\chi_5(q) = +1$ (split) for $q \equiv 1, 9, 11, 19$, $\chi_5(q) = -1$ (inert) for $q \equiv 3, 7, 13, 17$, and $\chi_5(5) = 0$ (ramified), 5 being the only ramified prime by $\Delta_K = 5$ (2.2). This is what makes χ_5 an effective *classification of the primes of $\mathbb{Q}(\sqrt{5})$: the class of q is computed from the Fibonacci sequence of (2.29), without factoring anything in $\mathcal{O}_K$. for the thirteen primes decided in Lean and the twenty-three verified numerically;*

P `[mersenne_bridge, phi_pow_fib]`

the general congruence for every $q \neq 2, 5$ is Lucas' [51]

2.8. **The rings as a tower (our reinterpretation of the Λ-structure of Borger).** The multiplicative tower generated by φ carries a Frobenius lift for each prime, and the lifts compose as the primes multiply. Both facts are established here.

Definition 2.41 (The golden monoid and its Fibonacci coordinates)**.** Let $\langle\varphi\rangle = \{\varphi^n : n \in \mathbb{Z}\}$ be the multiplicative monoid generated by the unit φ of (2.1), and let F_n be the Fibonacci sequence, $F_0 = 0$, $F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$—the depth-two recurrence singled out in Theorem 2.14. The coordinates of the tower in the basis $\{\varphi, 1\}$ hold in the following generality: in *any* commutative ring containing an element α with $\alpha^2 = \alpha + 1$, induction on n gives

$$\alpha^{n+1} = F_{n+1}\,\alpha + F_n \qquad (n \geq 0), \tag{2.34}$$

of which the additive spectrum (2.29) is the case $\alpha = \varphi$. Nothing of $\mathbb{R}$ or of $\sqrt{5}$ enters: only the quadratic relation and the recurrence. This is why the same coordinates govern the tower in R_{PCF}, in $\mathcal{O}_K$, and modulo q—which is what Proposition 2.40 exploits.

P [binet_general, phi_pow_fib]

Definition 2.42 (Frobenius lift on the golden monoid)**.** For each prime p define

$$\psi_p : \langle\varphi\rangle \to \langle\varphi\rangle, \qquad \psi_p(\varphi^n) = \varphi^{pn}, \qquad \text{so} \qquad \psi_p(\varphi) = \varphi^p = F_p\,\varphi + F_{p-1} \tag{2.35}$$

by (2.34).

P [psi_golden, psi_on_powers, psi_golden_fib]

Remark 2.43 (Two maps, one symbol). The ψ_p of (2.35) is *not* the map $a \mapsto a^p \bmod p$ recalled in §1.6, which is a ring endomorphism in the sense of Manin and Buium [28, 29]. Ours is exact rather than reduced modulo p, and it is an endomorphism of the multiplicative *monoid* $\langle\varphi\rangle$, not of the ring R_{PCF}: additivity is not preserved, since $\psi_p(\varphi + 1) = \psi_p(\varphi^2) = \varphi^{2p} \neq \varphi^p + 1$ for $p \geq 2$. The restriction to the multiplicative level is the point of the descent, not a limitation of it.

Proposition 2.44 (The lifts compose as the primes multiply)**.** *For all $p, q \geq 1$ and all $x \in \langle\varphi\rangle$,*

$$\psi_p \circ \psi_q = \psi_{pq}, \qquad \psi_1 = \mathrm{id}, \tag{2.36}$$

so $p \mapsto \psi_p$ is a monoid homomorphism from $(\mathbb{N}_{\geq 1}, \times)$ into $\mathrm{End}_{\mathrm{Mon}}\langle\varphi\rangle$. This is the Λ-structure of Borger [52] in its multiplicative form, and it is what makes the tower carry the multiplicative structure of the Euler product (2.39)*.*

P *[psi_functorial, psi_one]*

Proof. On a generator, $\psi_p(\psi_q(\varphi^n)) = \psi_p(\varphi^{qn}) = \varphi^{pqn} = \psi_{pq}(\varphi^n)$, and every element of $\langle\varphi\rangle$ is such a generator; $\psi_1(\varphi^n) = \varphi^n$. ☐

The remaining ingredient—that the Dedekind zeta of $\mathbb{Q}(\sqrt{5})$ is the *Euler colimit* of the partial Euler products over finite sets of primes, $\zeta_{\mathbb{Q}(\sqrt{5})}(s) = \varinjlim_S E(S)(s)$—is Proposition 2.50 below, and it is what Theorem 2.53 uses on the finite side.

2.9. **The zeta values, the Euler product, and the completed zeta.** The zeta values, even and odd; the Dedekind zeta of K with its local factors and its Euler colimit; the Archimedean place; and the completed zeta as the product over all places, with its functional equation. This is the first of two movements—building ξ; §2.10 evaluates it at $s = 1$. The bridge (2.4) ties π to φ throughout.

$$\zeta(s) = \frac{\zeta_{\mathbb{Q}(\sqrt{5})}(s)}{L(s, \chi_5)}, \qquad L(1, \chi_5) = \frac{2\log\varphi}{\sqrt{5}}. \tag{2.37}$$

P [eq:local-dedekind]

Theorem 2.45 (Basel value)**.** $\zeta(2) = \pi^2/6$ *(Euler).*

P *[basel_value]*

Theorem 2.46 (General even zeta value (Euler))**.** *For every integer $k \geq 1$,*

$$\zeta(2k) = (-1)^{k+1} \frac{B_{2k}\,(2\pi)^{2k}}{2\,(2k)!}, \tag{2.38}$$

with B_{2k} the Bernoulli numbers; $k = 1$ recovers $\zeta(2) = \pi^2/6$ (Theorem 2.45).

P[even_zeta_bernoulli]

N[thm:basel]

Proof. Euler's cotangent expansion $\pi z \cot(\pi z) = 1 - 2\sum_{k\geq 1} \zeta(2k)\, z^{2k}$ $(0 < |z| < 1)$ and the Bernoulli generating function $\frac{w}{e^w - 1} = \sum_{n\geq 0} \frac{B_n}{n!} w^n$ are linked by $\pi z \cot(\pi z) = \pi i z + \dfrac{2\pi i z}{e^{2\pi i z} - 1}$. Substituting the second series with $w = 2\pi i z$ and using $(2i)^{2k} = (-1)^k 4^k$, the coefficient of z^{2k} on the right is $(-1)^k B_{2k}(2\pi)^{2k}/(2k)!$; matching it to $-2\zeta(2k)$ gives (2.38). The pentagonal, χ_5-twisted refinement, with generalised Bernoulli numbers B_{2k,χ_5}, extends this computation; it is taken up in §A.8 below and carried out independently in the $\mathbb{F}_1$-descent companion [49]. ☐

The remaining ingredients of ξ are the product over primes and the factor that completes it.

Proposition 2.47 (Euler product)**.** *For* $\operatorname{Re} s > 1$*,*

$$\sum_{n\geq 1} n^{-s} = \zeta(s) = \prod_p \left(1 - p^{-s}\right)^{-1}, \tag{2.39}$$

by unique factorisation in $\mathbb{Z}$.

P[regge_dirichlet_eq_zeta, regge_tower_is_euler_product]

Definition 2.48 (The Dedekind zeta of K)**.**

$$\zeta_K(s) \;=\; \sum_{\mathfrak{a}} N(\mathfrak{a})^{-s} \;=\; \prod_{\mathfrak{p}} \left(1 - N(\mathfrak{p})^{-s}\right)^{-1}, \qquad \operatorname{Re} s > 1, \tag{2.40}$$

the sum over integral ideals of $\mathcal{O}_K = \mathbb{Z}[\varphi]$ (2.10) and the product over prime ideals, N the absolute norm. This is the object (2.37) factorises; it is defined here rather than assumed.

P[ideal_norm_k, ideal_euler_k, euler_from_ideals_K]

Proposition 2.49 (Splitting data and local factors)**.** *Every rational prime p falls in one of the three classes of Proposition 2.40, with $\sum_i e_i f_i = [K : \mathbb{Q}] = 2$ in each:*

$$\begin{array}{lllll} \chi_5(p) = +1 & \textit{split} & g = 2 & e = 1,\ f = 1 & N(\mathfrak{p}) = p \\ \chi_5(p) = -1 & \textit{inert} & g = 1 & e = 1,\ f = 2 & N(\mathfrak{p}) = p^2 \\ \chi_5(p) = 0 & \textit{ramified} & g = 1 & e = 2,\ f = 1 & N(\mathfrak{p}) = p \end{array} \tag{2.41}$$

Writing $u = p^{-s}$, the product over the prime ideals above p is

$$f_p(s) = \prod_{\mathfrak{p}|p} \left(1 - N(\mathfrak{p})^{-s}\right)^{-1} = \begin{cases} (1-u)^{-1} & \chi_5(p) = 0, \\ (1-u)^{-2} & \chi_5(p) = +1, \\ (1-u^2)^{-1} & \chi_5(p) = -1, \end{cases} \tag{2.42}$$

The von Mangoldt function of K weights a prime power by the log of its norm,

$$\Lambda_K(\mathfrak{p}^k) = \log N(\mathfrak{p}), \tag{2.43}$$

so the weight is $\log p$ at a split prime and $2\log p$ at an inert one. In all three cases the local factor factors as

$$f_p(s) = (1-u)^{-1} \cdot \left(1 - \chi_5(p)u\right)^{-1}, \tag{2.44}$$

which is (2.37) place by place. The inert case is the identity $(1-u)(1+u) = 1 - u^2$.

P[degree_formula_K, ideal_norm_values_K, euler_from_ideals_K, von_mangoldt_k_values, local_dedekind_ramified_K, local_dedekind_split_K, local_dedekind_inert_K]

Proposition 2.50 (The Euler product as a colimit). *Let $\mathcal{P}$ be the poset of finite sets of primes under inclusion and $E(S)(s) = \prod_{p\in S} f_p(s)$. Then*

$$\zeta_K(s) = \varinjlim_{S\in\mathcal{P}} E(S)(s), \qquad \operatorname{Re} s > 1. \tag{2.45}$$

Existence *is coefficientwise stabilisation: the n-th coefficient is already determined once S contains the prime factors of n, and enlarging S never alters it.* Uniqueness *is extensionality: a Dirichlet series is determined by its coefficients. This is the* Euler *colimit, not the framework colimit of §3.5. (absolute convergence for* $\operatorname{Re} s > 1$ *[53]*

Proposition 2.51 (The Archimedean local factor). *The completing factor is the Mellin transform of the self-dual Gaussian (2.7), i.e. the local zeta integral at the real place,*

$$\Gamma_{\mathbb{R}}(s) \;:=\; \pi^{-s/2}\Gamma\big(\tfrac{s}{2}\big) \;=\; 2\int_0^\infty g(x)\, x^{s-1}\, dx, \tag{2.46}$$

so it is produced by the object of §2.2 and not imported: the π is that of Lemma 2.7 and the Γ that of the Euler integral (2.5). Summed over the integer direction of the Gauss lattice $\Lambda_{\mathrm{PCF}} = M_{\mathrm{PCF}}\,\mathbb{Z}[i]$ of (2.19), the same weight gives

$$\Theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad \Theta(1/t) = \sqrt{t}\,\Theta(t) \quad (t>0), \tag{2.47}$$

which is the involution $\tau \to -1/\tau$ of the torus (2.19) written in the Mellin variable t, with unique fixed point $t = 1$, i.e. $\tau = i$; (2.47) is Jacobi's theta transformation [54], and reading $\Gamma_{\mathbb{R}}$ as the local zeta integral at the real place is Tate's [55]. The Boltzmann weight e^{-nt} of the primon gas fails (2.47) and is therefore excluded by that involution.

P*[gamma_r_from_tate, theta_from_jacobi, theta_fixed_point_unique]*

Proof. Substituting $u = \pi x^2$ in (2.46) gives $2\int_0^\infty e^{-\pi x^2} x^{s-1} dx = \pi^{-s/2}\int_0^\infty e^{-u} u^{s/2-1} du = \pi^{-s/2}\Gamma(s/2)$, which is (2.5) read with $A = \pi$ and exponent $s/2$. For (2.47), Poisson summation applied to $f(x) = g(x\sqrt{t})$, whose transform is $\widehat{f}(\xi) = t^{-1/2} g(\xi/\sqrt{t})$ by Proposition 2.9, gives $\Theta(1/t) = \sqrt{t}\,\Theta(t)$ directly. The fixed point of $t \mapsto 1/t$ on $t > 0$ is $t = 1$, since $(t-1)(t+1) = 0$ with $t + 1 > 0$. The exclusion is a computation: $\sum_{n\geq 1} e^{-nt} = (e^t - 1)^{-1}$ does not satisfy (2.47). □

Remark 2.52 (The value at the self-dual point). The classical evaluation $\Theta(1) = \pi^{1/4}/\Gamma(\frac{3}{4})$ (Gauss; see [54]) together with the reflection formula $\Gamma(\frac14)\Gamma(\frac34) = \pi\sqrt{2}$ gives

$$\Theta(1) = \sqrt{2}\,\eta(i) = \varphi^{\mu\lambda_{\log}}\,\eta(i), \qquad \eta(i) = \frac{\Gamma(\frac14)}{2\,\pi^{3/4}}, \qquad \Gamma_{\mathbb{R}}(\tfrac12) = 2\sqrt{\pi}\,\eta(i), \tag{2.48}$$

with $\varphi^{\mu\lambda_{\log}} = \sqrt{2}$ of Proposition 2.10. (the two classical evaluations enter as classical [54])

Theorem 2.53 (The partition is the product over all places). *Write the framework's spectral partition as the product of the Archimedean factor (2.46) with the finite places, the latter being the Dirichlet series (2.39):*

$$\Xi_{\mathrm{PCF}}(s) \;=\; \underbrace{\Gamma_{\mathbb{R}}(s)}_{\textit{Archimedean}} \cdot \underbrace{\sum_{n\geq 1} n^{-s}}_{\textit{finite places}} \;=\; \Gamma_{\mathbb{R}}(s)\,\zeta(s) \;=\; \xi(s) \qquad (\operatorname{Re} s > 1). \tag{2.49}$$

Equivalently $\Xi_{\mathrm{PCF}}(s) = \int_0^\infty t^{s/2-1}\,\omega(t)\,dt$ with $\omega(t) = \frac12(\Theta(t) - 1)$, so the partition is the Mellin transform of the framework's own lattice sum. Reading ξ as the product over all *places, the Archimedean one included, is the formulation of López Peña and Lorscheid [56]. The completed zeta is therefore not posited: the finite places are the Euler product (2.39) and the Archimedean place is the self-dual Gaussian (Proposition 2.51); §3.3 reads the first as the Regge tower and §3.1 the second as the worldline weight.*

P[framework_partition, framework_partition_eq_completed_zeta, places_assembly]

Proof. The finite-place leg is Proposition 2.47; the Archimedean leg is Proposition 2.51. Their product is $\pi^{-s/2}\Gamma(s/2)\zeta(s)$, which is $\xi(s)$ by definition of the completed zeta. For the integral form, apply (2.46) at each level n with $A = \pi n^2$ and sum over $n \geq 1$, exchanging sum and integral by absolute convergence for $\operatorname{Re} s > 1$. □

Remark 2.54 (Consistency with the $\mathbb{F}_1$ companion). $\mathbb{Q}(\sqrt{5})$ is real quadratic, hence has two real places, and χ_5 is even ($\chi_5(-1) = \chi_5(4) = +1$), so each real place contributes $\Gamma_\mathbb{R}$ and not the odd factor. The Dedekind factorisation $\zeta_K = \zeta \cdot L(\cdot, \chi_5)$ of Proposition 2.49 therefore distributes one $\Gamma_\mathbb{R}$ to each factor, and the one attached to ζ is exactly (2.46). The companion's Euler colimit is indexed by finite sets of *primes*; the Archimedean place is the one it leaves out, absorbed there into the counting correction κ_K of the Connes–Consani type [57]. Proposition 2.51 supplies it.

P[chi5_even]

Theorem 2.55 (Functional equation and the self-dual line)**.** *The completed zeta,*

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s),$$

satisfies

$$\xi(1-s) = \xi(s), \tag{2.50}$$

and the reflection $s \mapsto 1 - s$ fixes the single point $s = \frac{1}{2}$ of $\mathbb{C}$, while the line

$$\{s \in \mathbb{C} : \operatorname{Re} s = \operatorname{Re}(1-s)\} = \{s \in \mathbb{C} : \operatorname{Re} s = \tfrac{1}{2}\} \tag{2.51}$$

is the locus on which it preserves the real part. The point and the line are distinct statements about the same involution, and it is the line that the cocone of §2.11 takes as a leg.

P[functional_equation_fixed_line, fixed_line_is_face_phi, places_assembly]

Proof. By Theorem 2.53, $\xi(s) = \int_0^\infty t^{s/2-1}\omega(t)\,dt$ for $\operatorname{Re} s > 1$, with $\omega(t) = \frac{1}{2}(\Theta(t) - 1)$. Splitting at $t = 1$ and applying (2.47) on $(0,1)$—Riemann's argument [53]—,

$$\xi(s) = \frac{1}{s(s-1)} + \int_1^\infty \left(t^{s/2-1} + t^{(1-s)/2-1}\right)\omega(t)\,dt, \tag{2.52}$$

whose right-hand side is manifestly invariant under $s \leftrightarrow 1 - s$: the involution exchanges the two exponents and fixes $1/(s(s-1))$. This gives (2.50), and by analytic continuation for all s. The completing factor $\pi^{-s/2}\Gamma(s/2)$ is not imported: it is the Archimedean local factor (2.46), the π of Lemma 2.7 with the Γ of (2.5), and it carries at every s the same π that Theorem 2.45 gives at $s = 2$. The map $s \mapsto 1 - s$ is an involution of $\mathbb{C}$; its fixed set is $\{s : 2s = 1\}$, i.e. the single value $s = \frac{1}{2}$. For the line, $\operatorname{Re}(1-s) = 1 - \operatorname{Re} s$, so $\operatorname{Re} s = \operatorname{Re}(1-s)$ holds precisely when $2\operatorname{Re} s = 1$: this is (2.51), and by (1.14) its abscissa is $\varphi^{-\lambda_{\log}}$. The line is fixed *pointwise* not by $s \mapsto 1 - s$ but by the antiholomorphic $s \mapsto 1 - \bar{s}$; that reading is §2.13 below. □

Remark 2.56 (The self-dual line is the modulus). The $\frac{1}{2}$ of (2.50) is the same $\frac{1}{2}$ that §2.2 fixed as the unique fixed point of $x \mapsto 1 - x$ (Proposition 2.11) and as the midpoint of the Galois pair (Proposition 2.12): the self-dual line of the completed zeta *is* the modulus $\mu = \frac{1}{2}$, and by (2.51) its abscissa is $\varphi^{-\lambda_{\log}}$. Three involutions—$x \mapsto 1 - x$ on $\mathbb{R}$, the Galois conjugation of K, and $s \mapsto 1 - s$ on $\mathbb{C}$—have here one fixed value; by Lemma 2.2 the first two are the same map.

2.10. **The fundamental formula and the entropy bridge.** With ξ built, the second movement evaluates it at $s = 1$: the class number, the residue, the value $L(1, \chi_5)$, and the bridge that turns it into one bit.

Proposition 2.57 (Class number one)**.** *The Minkowski bound of K is*

$$M_K = \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^{r_2}\sqrt{|\Delta_K|} = \frac{\sqrt{5}}{2} \approx 1.11803, \qquad (n, r_1, r_2, \Delta_K) = (2, 2, 0, 5), \tag{2.53}$$

and $M_K < 2$. Granting Minkowski's theorem [58]—every ideal class contains an integral ideal of norm at most M_K—that ideal has norm a positive integer below 2, hence norm 1, hence is the unit ideal. Therefore

$$h_K = 1. \tag{2.54}$$

P *[class_number_one_K, minkowski_bound_lt_two]*

Theorem 2.58 (The fundamental formula)**.**

$$L(1,\chi_5) = \frac{2\log\varphi}{\sqrt{5}} = \frac{T}{\pi\sqrt{5}} \approx 0.4304089410. \tag{2.55}$$

Granting the Dirichlet class-number formula [59],

$$\lim_{s\to1}(s-1)\zeta_K(s) = \frac{2^{r_1}(2\pi)^{r_2}h_K R_K}{w\sqrt{\Delta_K}} = \frac{4\cdot 1\cdot\log\varphi}{2\sqrt{5}} = \frac{2\log\varphi}{\sqrt{5}}, \tag{2.56}$$

with $h_K = 1$ (2.54), $R_K = \log\varphi$ (2.11), $w = 2$, $\Delta_K = 5$ (2.2), $r_1 = 2$, $r_2 = 0$; then (2.37) and $\lim_{s\to1}(s-1)\zeta(s) = 1$ *give (2.55). The arithmetic downstream of the formula is proved; the formula itself is the input.*

P *[L1_from_class_number_formula, L1_eq_period_over_pi_sqrt5]*

Theorem 2.59 (The entropy–L-value bridge)**.** *With $\lambda_{\log} = \log 2/\log\varphi$, so that $\varphi^{\lambda_{\log}} = 2$ and $\varphi^{-\lambda_{\log}} = \frac{1}{2}$,*

$$\frac{S_{\mathrm{BH}}}{k_B} = \lambda_{\log} R_K = \lambda_{\log}\cdot\frac{\sqrt{5}}{2}L(1,\chi_5) = \log 2. \tag{2.57}$$

The regulator cancels exactly: $\lambda_{\log}\log\varphi = \log 2$, and $(\sqrt{5}/2)L(1,\chi_5) = \log\varphi = R_K$ by (2.55). The one bit per Planck area that §3.2 and §4.4 carry is therefore not a coincidence with $\mu_3 = \frac{1}{2}$: it is the regulator of $\mathbb{Q}(\sqrt{5})$, measured in the unit $\lambda_{\log}$.

P *[lambda_log_mul_regulator, sqrt5_half_L1_eq_regulator, entropy_bridge]*

Remark 2.60 (Three roles of one invariant). $R_K = \log\varphi$ occurs three times over: as the class-number residue (2.56), as the exponent of the torus period $T = 2\pi\log\varphi$ (2.11), and through (2.57) as the entropy scale. It is also the rate out of which $\varepsilon_0 = \ln\varphi/(6\sqrt{3})$ is built (2.18). One arithmetic invariant of K, four appearances.

Proposition 2.61 ($\frac{1}{2}$ is where the packing is maximal)**.** *For every $p \in (0,1)$ the binary entropy satisfies*

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p) \;\le\; 1, \tag{2.58}$$

with equality exactly at $p = \frac{1}{2}$. The modulus is therefore not merely a point where the entropy happens to equal one bit: it is the point where it is largest.

P *[entropy_slack_nonneg, binary_entropy_le_one, binary_entropy_half]*

Proof. $\log x \le x-1$ for $x > 0$ applied to $x = 1/(2p)$ and $x = 1/(2(1-p))$ gives $p\big(-\log 2p\big) \le (1-2p)/2$ and $(1-p)\big(-\log 2(1-p)\big) \le (2p-1)/2$; the two right-hand sides cancel, so $p\log 2p + (1-p)\log 2(1-p) \ge 0$, which is (2.58) after dividing by $\log 2$. Equality in $\log x \le x-1$ holds only at $x = 1$, i.e. only at $p = \frac{1}{2}$. ∎

Theorem 2.62 (Odd zeta values over the tower)**.** *For $k \ge 1$, $L(2k+1,\chi_5) \ne 0$ and*

$$\zeta(2k+1) = \frac{\zeta_K(2k+1)}{L(2k+1,\chi_5)}. \tag{2.59}$$

Together with Theorem 2.46 this closes the two families: π produces the even values, and φ through χ_5 produces the odd ones, both over the same pentagonal points $\{a/5\}$. It is what the loop hierarchy of §3.3 rests on, and it is derived here rather than deferred.

P *[odd_zeta_ratio]*

2.11. **The commutative diagram: convergence on μ and σ.** The constant $\frac{1}{2}$ is the apex of a cocone: eight routes coincide on it. The argument is convergence, not a chain.

$$
\begin{tikzcd}
\frac{(\frac{1}{2})!}{\sqrt{\pi}} \arrow[rddddd] & \\
\frac{\Gamma(\frac{3}{2})}{\Gamma(\frac{1}{2})} \arrow[rdddd] & \\
\|P\|\|C\|\|F\| \arrow[rddd] & \\
\cos(\frac{\pi}{3}) \arrow[rdd] & \\
\varphi^{-\lambda_{\log}} \arrow[rd] & \\
\frac{\varphi+\bar{\varphi}}{2} \arrow[r] & \mu = \frac{1}{2} \\
\mathrm{fix}(x{=}1{-}x) \arrow[ru] & \\
\mathrm{fix}(\mathrm{Re}\, s{=}\,\mathrm{Re}(1{-}s)) \arrow[ruu] &
\end{tikzcd}
$$

Remark 2.63 (One principle, five fixed points). None of the last three legs is a separate mechanism. Each of the framework's constants is the fixed point of an involution: $a = \pi$ of the Fourier transform (Proposition 2.9), $\tau = i$ of $\tau \mapsto -1/\tau$ on the torus (2.19), $\frac{1}{2}$ of the Galois conjugation of $\mathbb{Q}(\sqrt{5})$—whose fixed field is $\mathbb{Q}$ and whose midpoint is (2.9)—and $\frac{1}{2}$ of both $x \mapsto 1 - x$ (Proposition 2.11) and $s \mapsto 1 - s$ (Theorem 2.55). The cocone above collects the *five* that land on the modulus; the π of (2.7) is the same principle applied to the weight rather than to the modulus. By Lemma 2.2 the Galois conjugation and $x \mapsto 1 - x$ are one map, so the five involutions are four distinct maps landing on one value.

A further symmetry is *not* of this kind, and the contrast is informative: the order three of the eigenvalue triad is not the fixed point of an involution but a symmetry that *changes type* on descending—a rotation of $\mathbb{R}^3$ above, a translation by a three-torsion point of the torus below (Proposition 2.31). It is the same three that Proposition 2.20 reads as the exponent of $\prod_k |\lambda_k| = 2^{-3}$. Five fixed points and one change of type.

Theorem 2.64 (Modulus diagram commutes)**.** *Six values and two identifications coincide on the apex:*

$$\frac{(\frac{1}{2})!}{\sqrt{\pi}} = \frac{\Gamma(\frac{3}{2})}{\Gamma(\frac{1}{2})} = \|P\|\|C\|\|F\| = \cos(\tfrac{\pi}{3}) = \varphi^{-\lambda_{\log}} = \frac{\varphi + \bar{\varphi}}{2} = \mu = \tfrac{1}{2},$$

and $x = 1 - x \iff x = \mu$ *on* $\mathbb{R}$, $\mathrm{Re}\, s = \mathrm{Re}(1 - s) \iff \mathrm{Re}\, s = \mu$ *on* $\mathbb{C}$*. The non-elementary legs are established at* (2.6) *(the Gamma value),* (2.27) *(the norm product),* (1.14) *(the binary bridge* $\varphi^{\lambda_{\log}} = 2$*, whence* $\varphi^{-\lambda_{\log}} = \frac{1}{2}$*) and* (2.9) *(the Galois midpoint, by Lemma 2.2); the angle* $\cos(\frac{\pi}{3})$ *and the involution fixed point are elementary; the self-dual line of* ξ *is* (2.51)*, derived above rather than cited.*

Two of the eight come from the same quadratic read on its two faces: the binary $\varphi^{-\lambda_{\log}}$ *through* (1.14)*, and the arithmetic* $\frac{\varphi+\bar{\varphi}}{2}$ *through* (2.2)*. That they agree is not a further coincidence but the statement that the two readings of* $\varphi^2 = \varphi + 1$ *close on the same value.*

P *[mu_diagram_commutes, mu_faces_pairwise_eq]*

Remark 2.65 (The microstate is unique: no landscape to select). The eight routes of Theorem 2.64 leave no freedom in the modulus: the value $\mu = \frac{1}{2}$ is forced at once by the Gamma value, the norm product, the angle $\cos(\frac{\pi}{3})$, the binary bridge, the involution fixed point and the self-dual line of the completed zeta. The microstate is therefore the *unique* self-consistent solution, not one option among many. This is what dissolves the vacuum-selection problem of §1.2.4: there is no $\sim 10^{500}$ landscape to choose from and no anthropic selection to invoke—the single microstate is fixed by the convergence itself, and the abstract's "unique self-consistent solution" is established here. The uniqueness proved here is that of the modulus $\mu = \frac{1}{2}$ as the algebraic fixed point of the eight coincident routes (Theorem 2.64); its reading as the absence of a physical vacuum landscape is the interpretive step developed in §4, where the observer builds—rather than selects—the geometry.

$$\begin{tikzcd}
\dfrac{\zeta(2)}{(\pi/3)^2} \arrow[dr] & \\
\dfrac{|\mathrm{rot}\, S_3|^2}{|S_3|} \arrow[r] & \sigma = \tfrac{3}{2}
\end{tikzcd}$$

Lemma 2.66 (Orders in S_3 and the geometric leg)**.** *The equilateral triangle of eigenvalues $\lambda_k = \frac{1}{2}\omega^k$ (Proposition 2.19) has symmetry group the symmetric group $S_3 \cong D_3$. Its order is $|S_3| = 3! = 6$, and its rotation subgroup is the cyclic group $C_3 = \{1, r, r^2\}$ of order $|\mathrm{rot}\, S_3| = 3$. Hence*

$$\frac{|\mathrm{rot}\, S_3|^2}{|S_3|} = \frac{3^2}{6} = \frac{9}{6} = \tfrac{3}{2}, \tag{2.60}$$

so the geometric leg equals σ rather than defining it.

P*[factorial_six, s3_orders, sigma_geom_from_S3]*

Theorem 2.67 (Sum diagram commutes)**.** *Both routes coincide on $\sigma = \frac{3}{2}$: the analytic leg gives $\dfrac{\zeta(2)}{(\pi/3)^2} = \dfrac{\pi^2/6}{\pi^2/9} = \dfrac{9}{6}$ (2.28), and the geometric leg gives $\dfrac{|\mathrm{rot}\, S_3|^2}{|S_3|} = \dfrac{9}{6}$ (Lemma 2.66); both equal $\sigma = \frac{3}{2}$.*

P*[sigma_diagram_commutes]*

Theorem 2.68 (Methods master)**.** *Both diagrams commute and the spectral invariants follow: $\sigma + \mu = 2$, $\sigma/\mu = 3$; and the arithmetic they carry closes on the same modulus, the completed zeta $\xi = \Gamma_{\mathbb{R}}\zeta$ of Theorem 2.53 being self-dual exactly on $\mathrm{Re}\, s = \mu$ (Theorem 2.55).*

P*[section2_master, places_assembly, fixed_line_is_face_phi, functional_equation_fixed_line, s_duality_exact, s_duality_fixed_point_unique, theta_from_jacobi]*

Remark 2.69 (The sum and the real places). Recorded as an exact coincidence, with nothing asserted: the invariant $\sigma + \mu = 2$ of Theorem 2.68 equals $r_1 = 2$, the number of real places of $K = \mathbb{Q}(\sqrt{5})$—which is what fixes the factor $\Gamma_{\mathbb{R}}^2$ in the completed zeta and the doubling $\kappa_K = 2\kappa_{\mathbb{Q}}$ of §A.8. No causal claim is attached and none is used below.

Remark 2.70 (One bit, three readings of the same $\frac{1}{2}$). Equation (2.57) is not an analogy between an arithmetic invariant and a physical one; it is one quantity read three ways, and the three are the three things the framework has been saying about $\frac{1}{2}$ all along.

It is *self-referential.* The regulator measures K by its own fundamental unit: $R_K = \log \varphi$ and φ generates the units of $\mathcal{O}_K$ (2.2), so the field is measured in the scale it itself produces. On the geometric side the same $\frac{1}{2}$ is the fixed point of the involution $s \mapsto 1 - s$ (Theorem 2.55), and of the involution $x \mapsto 1 - x$ that Theorem 2.64 assembles with seven further routes to the same value—a fixed point, which is the non-paradoxical self-reference of §1.6 and Definition 2.13, where the recurrence is distributed at depth two and so never depends directly on itself.

It is *maximal*. By Proposition 2.61, $\frac{1}{2}$ is where the binary entropy attains its largest value: one bit is not one value among many but the most a single site can carry. That is what makes the holographic bound a bound, and it is why $f_{\text{crit}} = \frac{1}{2}$ of §4.4 is a threshold rather than a coincidence.

And it is *arithmetic*. Through $\lambda_{\log} R_K = \log 2$ that maximal bit *is* the regulator of $\mathbb{Q}(\sqrt{5})$, measured in the logarithmic index. A quantity that refers to itself, that is the most information a site can pack, and that is an invariant of a number field, are here the same quantity; the framework does not relate them, it finds them to be one.

2.12. **The conductor, the scale, and the unit of spacing.** Section 2.7 fixed the residue class modulo 20 that decides χ_5 (2.33). What that 20 *is* has not been said, and it is needed before any statement about the spacings of zeros: a spacing is a number only once a unit is fixed, and if the unit were free the statement would be about the unit rather than about the zeros.

Proposition 2.71 (The conductor is derived)**.** *The modulus* 20 *is not chosen. It is the least common multiple of the two periods already in the framework—the* 4 *of the turn,* $i^4 = 1$*, and the* 5 *of the pentagon (Theorem 2.4)—and these are coprime, so the lcm is the product:*

$$\operatorname{lcm}(4,5) = 20, \qquad \gcd(4,5) = 1. \tag{2.61}$$

P *[conductor_is_derived]*

Proposition 2.72 (The two factors are one four and one five)**.** *The* 4 *of the turn and the* 4 *that the binary register carries are not two numbers that agree. They are one, and the identification is a reduction rather than an analogy. In* $\mathbb{F}_5$ *the element* 2 *satisfies the equation of the turn,*

$$2^2 = -1 \ \ \textit{in } \mathbb{F}_5, \qquad i^2 = -1 \ \ \textit{in } \mathbb{C}, \tag{2.62}$$

and the two cycles agree term by term: $2^k \bmod 5$ *runs* $2,4,3,1$ *where* i^k *runs* $i,-1,-i,1$*. The reduction that carries one to the other exists because* 5 *splits in the Gaussian integers,* $(2+i)(2-i) = 5$*; under* $\mathbb{Z}[i] \to \mathbb{Z}[i]/(2-i) \cong \mathbb{F}_5$ *the image of* i *is* 2*. What makes this possible is* $5 \equiv 1 \pmod 4$*, so that* $|(\mathbb{Z}/5)^\times| = 4$*, and* 5 *is the only prime with* $p - 1 = 4$*. The* 4 *of* (2.61) *is therefore* $\operatorname{ord}(i)$*, the conductor of* $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$*, and the period of the binary register is that same* 4 *seen through the reduction—not a property of the factor* 4 *itself, where* $2^m \equiv 0$ *for* $m \geq 2$ *and there is no period at all.*

P *[two_sq_eq_neg_one_mod_five, five_splits_gaussian, conductor_four_is_order_of_turn, binary_period_lives_in_the_five_component, four_cocone]*

Proposition 2.73 (Each factor governs one character, and only one)**.** *The factorisation* $20 = 4 \cdot 5$ *separates the two characters exactly. Writing* r_4, r_5 *for the two reductions, the pair is injective—the Chinese remainder theorem in the form needed here—and each factor determines its own character while leaving the other free:*

$$r_4(a) = r_4(b) \ \wedge \ r_5(a) = r_5(b) \implies a = b, \tag{2.63}$$

with r_4 *determining* χ_4 *and* r_5 *determining* χ_5*, and neither determining the other. The two witnesses are what keep this from being empty:* 1 *and* 13 *agree mod* 4 *and differ in* χ_5*;* 1 *and* 11 *agree mod* 5 *and differ in* χ_4*. Consequently the classes* $\{1,9,11,19\}$ *and* $\{3,7,13,17\}$ *of §2.7 are not tabulated lists but the fibres of* r_5*, and the* χ_5 *of* (2.31) *agrees with the quadratic fibre:* $\chi_5(n) = +1$ *exactly when* $n \equiv \pm 1 \pmod 5$*.*

P *[conductor_splits, fiber_mod4, fiber_mod5, chi5_is_the_mod5_fiber, mod4_does_not_determine_chi5, mod5_does_not_determine_chi4, conductor_attribution]*

Proposition 2.74 (The scale is injective in the conductor)**.** *For* q, T *with* $2\pi < qT$ *put* $\operatorname{sc}(q,T) = 2\pi/\log(qT/2\pi)$*. Distinct conductors give distinct scales at the same height:*

$$q_1 \neq q_2 \implies \operatorname{sc}(q_1, T) \neq \operatorname{sc}(q_2, T). \tag{2.64}$$

Hence "the unfolded spacings have mean one" selects one conductor: it is a statement about the zeros, not about a choice of units. Without it the bound of Definition 2.81 would say nothing.

P[spacing_scale, scale_injective_in_conductor]

Proposition 2.75 (The envelope carries both faces). *With* $\operatorname{env}(q,T)=\frac{T}{2\pi}\log\frac{qT}{2\pi e}$*, whose derivative is the density that unfolds the ordinates,*

$$\log\frac{qT}{2\pi e}=\underbrace{\log q}_{\text{field}}+\log T-\underbrace{\log 2\pi e}_{\text{universal}},\tag{2.65}$$

and only the field part distinguishes: $q_1\neq q_2\Rightarrow\log q_1\neq\log q_2$*. With* $q=5$ *the first term is the* φ*-face of §2.11 and the third the* π*-face. The counting law* $N(T)=\operatorname{env}(T)+S(T)$ *and the unit* u_n *it derives are asymptotic and enter as hypothesis arguments; what is proved is that the conductor entering them is forced.*

P[envelope, envelope_splits, field_part_distinguishes, counting_law, derived_unit, the_unit_is_derived]

Proposition 2.76 (The radial reading). *For* $\rho\neq 0$,

$$\left|1-\tfrac{1}{\rho}\right|=1\iff\operatorname{Re}\rho=\tfrac{1}{2}:\tag{2.66}$$

the critical line is exactly the preimage of the unit circle under $\rho\mapsto 1-1/\rho$.

P[li_modulus_on_line]

2.13. **The extreme case, and the individual the ensemble replaces.** The abstract began with an ensemble: the pair correlation of the zeros of ζ follows GUE statistics, and so does the boundary side of low-dimensional holography. An ensemble is what one writes for want of an individual. This section has produced the individual—the modulus $\mu=\frac{1}{2}$ of Theorem 2.64. What remains is to say, with statements of our own, what the ensemble was standing in for; and the answer is not a statistic but a pair.

Definition 2.77 (The companion of the functional equation). For $\rho\in\mathbb{C}$ set

$$\sigma\tau(\rho)=1-\bar{\rho},\tag{2.67}$$

the reflection $s\mapsto 1-s$ of Theorem 2.55 composed with complex conjugation. The two commute—1 is real—so $\sigma\tau$ is an involution, and it is *antiholomorphic*.

P[sdual_mate, sdual_conj_commute, sdual_mate_involutive]

Proposition 2.78 (The self-dual line, as a fixed set). $\sigma\tau$ *preserves the ordinate and reflects the abscissa,* $\Im\sigma\tau(\rho)=\Im\rho$ *and* $\operatorname{Re}\sigma\tau(\rho)=1-\operatorname{Re}\rho$*; hence*

$$\sigma\tau(\rho)=\rho\iff\operatorname{Re}\rho=\varphi^{-\lambda_{\log}}.\tag{2.68}$$

This is the sense in which the line (2.51) is self-dual*: not merely the locus where* $s\mapsto 1-s$ *preserves the real part, but the set fixed pointwise by an involution—the antiholomorphic one. The holomorphic reflection fixes a single point of* $\mathbb{C}$*; the line is what its composition with conjugation fixes. Its abscissa is the apex of Theorem 2.64.*

P[sdual_mate_im, sdual_mate_re, sdual_mate_fixed_iff]

Proposition 2.79 (The extreme case has arity two). *If* $\operatorname{Re}\rho\neq\varphi^{-\lambda_{\log}}$ *then* $\sigma\tau(\rho)\neq\rho$ *and* $\Im\sigma\tau(\rho)-\Im\rho=0$*: off the line there are* two *distinct points sharing one ordinate, at separation exactly zero. The configuration that GUE statistics make improbable is therefore not a statistic here but a concrete pair. The phenomenon the ensemble describes—*level repulsion*, the suppression of small spacings—is on this side a statement of arity two, decided pointwise on one pair.*

P[extreme_case_is_binary]

Proposition 2.80 (Angular and radial bound the same event). *For* $\rho\neq 0$*, by (2.66),*

$$\sigma\tau(\rho)\neq\rho\iff\left|1-\tfrac{1}{\rho}\right|\neq 1.\tag{2.69}$$

The angular reading (arity two: a pair with zero separation) and the radial one (arity zero: a modulus off unity) are two descriptions of one event. No hypothesis is needed: the radial reading is Proposition 2.76.

P `[binary_iff_radial]`

Definition 2.81 (Measured repulsion)**.** Say Z *repels* at $m > 0$ if any two distinct elements differ in ordinate by at least m, the ordinates unfolded by the unit of §2.12:

$$\forall\, \rho \neq \tau \in Z: \quad |\Im\tau - \Im\rho| \;\geq\; m. \tag{2.70}$$

This is *level repulsion*: the suppression of small spacings that GUE statistics describe and Poisson statistics do not. It is measured, not proved: over the 237 spacings of $\gamma_1 = 14.13$ to $\gamma_{238} = 453.99$, unfolded by the unit of §2.12, the minimum is 0.2911 and none falls below 0.25, let alone 0.10, where independence would put 22.6 and the Gaussian ensemble 0.25. The unfolded mean is 0.9979 with ζ's own conductor $q = 1$ against 1.8828 with the foreign $q = 20$, and every other conductor breaks it as well (q=2: 1.2026, q=5: 1.4733)—which is Proposition 2.74 showing in the data: the scale is injective in q, so the ordinates identify which conductor belongs to which L-function. Only the *sign* of m enters anything below.

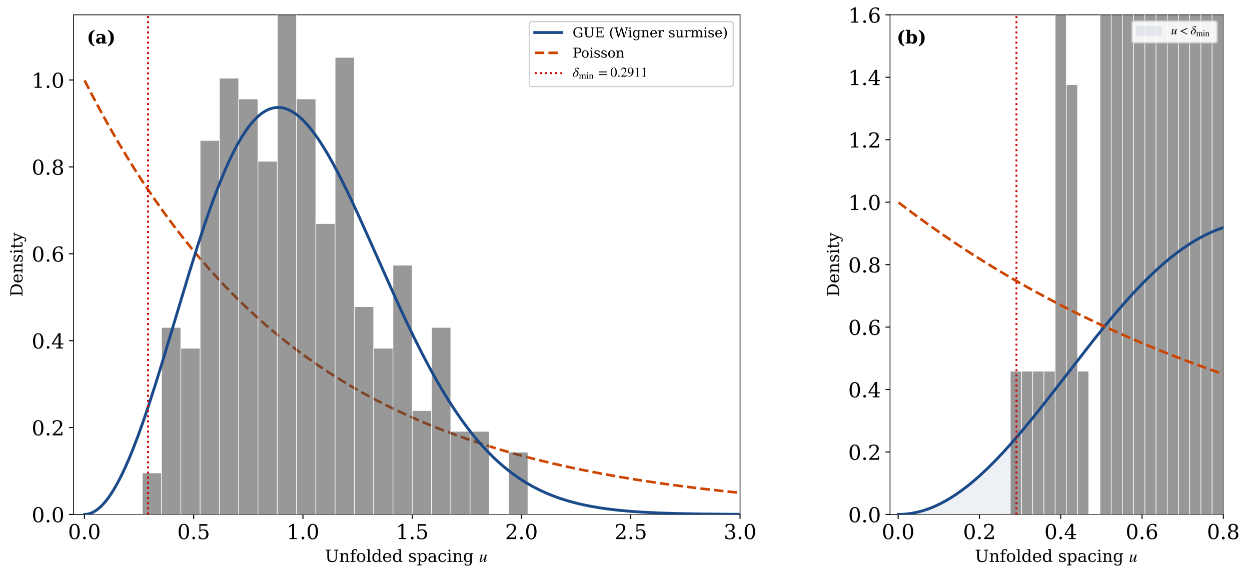

FIGURE 1. Level repulsion in ζ zeros (Def. 2.81): (a) histogram of 237 unfolded spacings (γ_1=14.13 to γ_{238}=453.99) against the GUE Wigner surmise and the Poisson exponential; (b) zoom on the repulsion region, with $\delta_{\min}$=0.2911 marked. The shaded zone $u<\delta_{\min}$ is where GUE suppresses spacings that Poisson permits.

Proposition 2.82 (Repulsion excludes the extreme case)**.** *If Z repels at some $m > 0$ then no two distinct elements share an ordinate; and since off the line the pair $\{\rho, \sigma\tau(\rho)\}$ has separation exactly zero (Proposition 2.79), every $\rho \in Z$ closed under $\sigma\tau$ satisfies $\operatorname{Re}\rho = \varphi^{-\lambda_{\log}}$. The configuration the ensemble makes improbable and the one repulsion excludes are the same configuration.*

P `[repulsion_excludes_zero_spacing, repulsion_excludes_the_extreme_case]`

Theorem 2.83 (The weak form, and what would suffice)**.** *Closure under $\sigma\tau$ alone gives*

$$\left(\forall \rho \in Z: \;\operatorname{Re}\rho = \varphi^{-\lambda_{\log}}\right) \iff \textit{no two distinct elements of } Z \textit{ share an ordinate.} \tag{2.71}$$

The right-hand side is strictly weaker than Definition 2.81: repulsion implies it, not conversely. It is what any future argument would discharge in place of the metric bound.

P[E_iff_no_shared_ordinate]

Proposition 2.84 (Repulsion gives the modulus)**.** *Under the same hypotheses,* $|1-1/\rho| = 1$ *exactly. The measured repulsion does not merely constrain the modulus: it* produces *it.*

P[repulsion_gives_the_modulus]
N[prop:repulsion-modulus]

Theorem 2.85 (Zeros on the apex, with its inputs in view)**.** *Let* $Z \subset \mathbb{C}$ *be closed under* $\sigma\tau$ *and repel at some* $m > 0$ *(Definition 2.81). Then every* $\rho \in Z$ *satisfies* $\operatorname{Re}\rho = \varphi^{-\lambda_{\log}}$*, the apex of Theorem 2.64.*

Two inputs are open *and are named here rather than hidden. Closure needs the conjugation half of* (2.67)*,* $\xi(\bar{s}) = \overline{\xi(s)}$*, not established in this paper; and the repulsion is measured. Everything between them is proved, and of the measurement only the* sign *enters.*

Remark 2.86 (What the argument does not use). Theorem 2.85 uses the geometry of $\mathbb{C}$ and the sign of one bound. It uses no joint density, no sine kernel, no form factor, no asymptotic statistics of the spacings. That is the point of stating the extreme case at arity two: the exclusion the criterion needs is about *one pair*, and a pair is an individual. Where the ensemble was standing in for an object, the object here is the modulus, and what the statistics described is a consequence of its geometry rather than a substitute for it. The chain that discharges the two open inputs, and the measured bound itself, belong to the companion papers [38, 49]; note that they are stated there for $L(s, \chi_5)$, whose functional equation is a different theorem from (2.50), proved separately in [49].

Definition 2.87 (The ensemble's own object)**.** The statistics invoked in §1 are those of the GUE pair-correlation kernel

$$K(u) = 1 - \left(\frac{\sin \pi u}{\pi u}\right)^2. \tag{2.72}$$

It is written here—the attribution is at §1—so that the object the paper opened with is a node of this development rather than a gap in it. It vanishes as $u \to 0$, which is the suppression of small spacings that Definition 2.81 measures, and tends to 1 for large u, which is decorrelation. Nothing below uses it.

Remark 2.88 (Envelope yes, fluctuations no). The ensemble stands on a distinction between the smooth spectral density and the fluctuations around it. A deterministic operator built without reference to ζ determines the envelope—the global distribution of zeros—but the pair-correlation statistics reside in the fluctuations, and reproducing them would require either an exact spectrum or a dynamical mechanism generating the level repulsion of Definition 2.81. *This is intrinsic to the construction, not a gap in the work*: it is why the present subsection proves a statement about *one pair* rather than about a distribution. The joint density and the form factor are declared in the development and used in no signature; that is the check that nothing here rests on them. ∅

Remark 2.89 (A geometric origin for the constraint). What the framework does supply is a candidate *origin* for the constraint that makes such statistics possible, and it is already proved above: the ratio σ/μ is the arity $d = 3$ (Remark 2.22), the modulus contracts strictly, $\|\hat{\Omega}\| = \frac{1}{2} < 1$ (Remark 2.16), and the same 3 is the exponent of $\prod_k |\lambda_k| = 2^{-3}$ in Proposition 2.20. The reading that eigenvalue correlations in this regime arise from *geometric constraint*—the no-diagonal condition—rather than from dynamical instability is advanced in the companion paper [38]; it is a proposal there and not a theorem here, and is recorded as such. What is established in this section is the regime, not the reading.

Remark 2.90 (Not the tower spacing). One caution, now that spacings are in play: the tower has its own spacing, the regulator $R_K = \log\varphi$ of (2.11), numerically 0.481212, while the unit that unfolds ordinates at $T = 100$ is 1.435589 (Proposition 2.74). They share the symbol φ and are different objects.

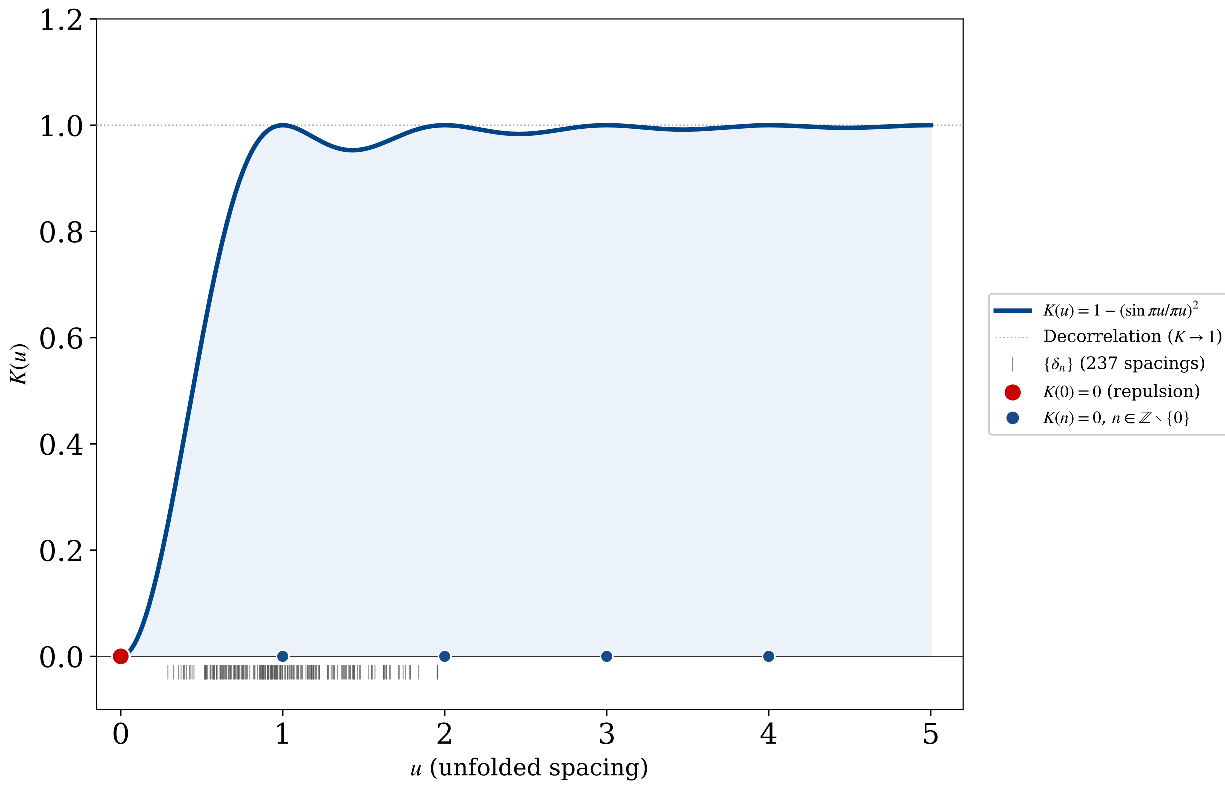

FIGURE 2. The GUE pair-correlation kernel $K(u) = 1 - (\sin \pi u/\pi u)^2$ of Def. 2.87. Red point: $K(0)$=0 (repulsion). Blue points: $K(n)$=0, $n \in \mathbb{Z} \setminus \{0\}$ (no long-range correlation). The kernel tends to 1 for large u (decorrelation). The rug plot shows the 237 measured unfolded spacings, clustering away from zero in the repulsion region.

P [`tower_spacing_is_the_regulator`]

2.14. **Formal verification.** Each result of this section carries, in brackets, the name of its Lean 4 theorem. The development that accompanies the paper is the self-contained file `CW6_complete_v2.lean`. It compiles under `leanprover/lean4` v4.29.0 with Mathlib v4.29.0: `lake build` completes with no errors and no warnings, and the file declares *no* `axiom` and contains no `sorry`. The file closes with the framework colimit of §3.5 below (namespace `PCFColimit`): the frameworks the paper treats—Polyakov, Maldacena, ER=EPR, Huerta–Schreiber, Kiely—as partial spectral objects together with the PCF core, whose fields are populated by theorems of this same file (`modulus_Omega`, `holographic_area`, `kk_at_PCF`, `two_pow_hausdorff`, the P/C/F norms), the cocone into the vertex T_{PCF} (`cocone_property`), and the universal property that makes the vertex unique (`colimit_universal`, `framework_colimit`). The external results the file uses enter as explicit hypothesis arguments in the types of the theorems that consume them—where the reader can see them—rather than as axioms; the classical analytic inputs (`gamma_r_from_tate`, `theta_from_jacobi`) and the combinatorial ones (`ladder_from_scott`, `e8_dim_from_kissing`, `positroid_from_kp`, `no_separable_parts`) are carried that way. Two further inputs enter the same way in §2.13 and are recorded here because they are the only open ones in that subsection: `XiConjClosed`, the closure $\xi(\bar{s}) = \overline{\xi(s)}$ of the zeros under conjugation, which is *not* established in this paper; and `MinSpacing`, a positive minimal ordinate separation, which is measured rather than proved. Of the second only its sign is used. Neither is an axiom: both are hypothesis arguments in the type of Theorem 2.85. One decision procedure uses `native_decide` rather than `decide`—the sweep of $16 \times 17 \times 18$ triples behind Proposition 4.20—which extends the trusted base to Lean's compiler as well as its type kernel; every other proof rests on the kernel alone. The golden ratio is defined, $\varphi := (1+\sqrt{5})/2$, and the two facts that anchor the construction are theorems proved from that definition, not assumptions. The relation $\varphi^2 = \varphi + 1$ (`phi_sq`) is derived from the definition and is (1.13). The modulus $\|\hat{\Omega}\| = \frac{1}{2}$ (`modulus_Omega`) is likewise a theorem. Its value is reached by the routes assembled in Theorem 2.64, its component structure is Proposition 2.19, and the trigonometric collapse of the norm product is Theorem 2.32; the three quantities that coincide there and cease to coincide afterwards are separated in Proposition 2.20.

Compilation note (single point of record). The development declares no axiom: every classical input is a hypothesis argument in the type of the theorem that consumes it. `lake build` completes against Mathlib, confirming the spelling of every library lemma the proofs call. The lemma names, grouped by what they serve, are recorded here so that a future Mathlib rename breaks the build visibly rather than passing silently:

- *Completed zeta and the places* (§2.9): `Gamma`$_\mathbb{R}$, `Gamma`$_\mathbb{R}$`_ne_zero_of_re_pos`, `riemannZeta_def_of_ne_zero`, `zeta_eq_tsum_one_div_nat_cpow`, `riemannZeta_eulerProduct_tprod`, `completedRiemannZeta_one_sub`, `integral_gaussian`, `Real.sq_sqrt`, `Real.Gamma_eq_integral`.
- *The generator and its conjugate* (§2.1), *the arithmetic seed* (§2.2) *and the cocone* (§2.11): `Real.sq_sqrt`, `Real.Gamma_add_one`, `Real.Gamma_pos_of_pos`.
- *The eigenvalue triad* (Proposition 2.20): `Complex.exp_ofReal_mul_I_re`, `Complex.exp_nat_mul`, `Complex.exp_two_pi_mul_I`, `Real.cos_pi_sub`, `Real.rpow_mul`, `Real.rpow_natCast`, `Real.rpow_neg`.
- *The companion and the extreme case* (§2.13): `map_sub`, `map_one` on `starRingEnd`, `Complex.conj_re`, `Complex.conj_im`, `Complex.ext_iff`.
- *Theta and Poisson* ((2.47)): the file defines `Theta` directly without Mathlib's `jacobiTheta`.
- *Pentagon and Fibonacci* (§2.7, §2.8): `Real.cos_pi_sub`, `Real.cos_two_mul`, `Nat.fib`. The general form of (2.33) would need Frobenius in `ZMod q`; it is stated here by cases.
- *Even zeta values* (Theorem 2.46): the name of the even-value lemma, and **whether it is stated with `bernoulli` or with `bernoulli'`**. The two differ only at B_1 and agree at every even index ≥ 2, so the statement is the same either way; the identifier is not.
- *Group orders* (Lemma 2.66): the index-two fact for A_n, `two_mul_card_alternatingGroup` or its equivalent.
- *Matrices* (Theorem 5.5): `Matrix.mul_inv_rev`, `Matrix.transpose_nonsing_inv`.
- *Earlier blocks*: `Gamma_one_half_eq`, `riemannZeta_two`, `Real.sqrt_eq_rpow`, `rpow_natCast`, `Gamma_add_one`, `arccos_cos`, and for the M6 uniqueness `Nat.Prime.irrational_sqrt`, `Irrational.intCast_mul`, `Int.not_irrational`.

The numerical companion is the pytest suite in `tests/` (`pnpm run verify:py`), keyed by the TEX labels: of the 162 labelled equations, 148 carry a Lean proof and 119 a numerical check, 102 both—evaluating identities at up to 25-digit precision and, for correspondences with measurement, comparing against the measured value. The figures are generated by the scripts deposited with the source.

3. Derivations

Building on the single microstate established in §2, this section presents one demonstrative line: the object generates a tower, and the tower generates the duality web. The line runs in the order 3.1 → 3.2 → 3.3 → 3.4 → 3.5, summarised as a 20-step spine at the end (§3.6). The node that stitches §2 to here is the function Γ; every quantitative statement below is verified numerically at 25-digit precision where the identity requires it (§2.14).

3.1. **The fundamental object in physics.** The single object is the half-integer Γ value, the carrier of the string microstate. The seed inherited from §2 is

$$\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}. \tag{3.1}$$

The worldline phase and its modulus are

$$\Omega(\tau) = \tfrac{1}{2}\, e^{i\tau \ln \varphi}, \qquad |\Omega| = \tfrac{1}{2}. \tag{3.2}$$

The wavefunction is Gaussian, with the half-factorial normalisation $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and a weight fixed by the scale quantum,

$$\int_{\mathbb{R}} e^{-x^2}\, dx = \sqrt{\pi}, \qquad \mu = \varphi^{-\lambda_{\log}} = \tfrac{1}{2} \quad (\varphi^{\lambda_{\log}} = 2). \tag{3.3}$$

Remark 3.1 (The wavefunction is the self-dual Gaussian). The Gaussian weight of (3.3) is the self-dual one fixed in §2.2, Proposition 2.9: normalised, and equal to its own Fourier transform, $g(x) = e^{-\pi x^2}$ (2.7). The physical reading added here is that this g is the worldline wavefunction of the microstate; the width is not a modelling choice but the fixed point of the Fourier transform, the same self-duality that §2.4 used to fix $\tau = i$.

P [selfdual_gaussian_unique]

The one-dimensional worldline is promoted to the two-dimensional worldsheet, governed by the Polyakov action [60]

$$S_P = -\frac{1}{4\pi\alpha'} \int d^2\sigma\, \sqrt{-h}\, h^{ab}\, \partial_a X^\mu\, \partial_b X^\nu\, \eta_{\mu\nu}. \tag{3.4}$$

Proposition 3.2 (The worldsheet inherits the non-orientable fibre). *The worldsheet carries the non-orientable fibre of the base torus (Proposition 2.28, §2.5): the module direction of $\Omega(\tau) = \frac{1}{2}\, e^{i\tau \ln\varphi}$ (3.2), carried once around the fundamental domain, closes with the orientation-reversing half-turn $e^{i\pi} = -1$. The worldsheet is therefore non-orientable and one-sided—the half-turn carries the module's interior edge to its exterior—so the bulk–boundary relation the torus carries admits no privileged side, independently of which torus carries the fibre.*

P *[fibre_monodromy, fibre_reflection_order_two, fibre_independent_of_base]*

The Schwinger parametrisation, used to assemble propagators and amplitudes, follows from the Γ integral by the change of variable $t = xA$,

$$A^{-n} = \frac{1}{\Gamma(n)} \int_0^\infty x^{n-1}\, e^{-xA}\, dx \qquad (A > 0,\ n > 0). \tag{3.5}$$

Remark 3.3 (The role of φ in the object). Since $\pi = 5 \arccos(\varphi/2)$ (§2), both $\sqrt{\pi}$ and the half-factorial are functions of φ; and $\mu = \frac{1}{2} = \varphi^{-\lambda_{\log}}$. The object is φ-generated, not posited.

3.2. **The meta-invariant: the holographic quantum.** The meta-invariant emerges from the structure already in place: the worldsheet phase $\Omega(\tau) = \frac{1}{2}\, e^{i\tau\ln\varphi}$ of §3.1 (3.2), carried on the torus T^2_{PCF} of §2.4 (2.19), whose single modulus $|\Omega| = \frac{1}{2}$ is the norm product fixed in §2.4 (2.12). The bridge from this object to the rest of physics is that $|\Omega| = \frac{1}{2}$ *is* the holographic bit (one bit per Planck area); that this bit is the regulator of $\mathbb{Q}(\sqrt{5})$ measured in $\lambda_{\log}$, and not a coincidence with μ_3, is Theorem 2.59. The spectral invariants are

$$d = 3\ (= |S_3| = 3! = 6), \qquad \mu = \tfrac{1}{2}, \qquad \sigma = \tfrac{3}{2}, \qquad \sigma + \mu = 2. \tag{3.6}$$

The two $\frac{3}{2}$'s of the corpus—the Γ argument of (2.6) and the spectral exponent here—are one value for a discriminating reason: the first is $1 + \mu$, the second 3μ, and $1 + \mu = 3\mu$ has the unique solution $\mu = \frac{1}{2}$; at arity n the Basel route (2.28) gives $n^2/6$, which meets $\frac{3}{2}$ only at $n = 3$.

P`[three_halves_unique, arity_control_three]`

The spectral angle and the uniqueness of the invariants. At scale σ the worldline subtends the spectral angle $\alpha(\sigma) = \arctan(\varepsilon_0\varphi^\sigma)$, normalised by $\varepsilon_0 M_{\mathrm{PCF}} = \pi$. The invariants (3.6) are not chosen but forced: the pair (σ, μ) with $\sigma + \mu = 2$ and $\sigma\mu = \frac{3}{4}$ has the unique solution $(\sigma, \mu) = (\frac{3}{2}, \frac{1}{2})$, i.e. $\sigma = \frac{3}{2}$ the same arity 3 and modulus $\frac{1}{2}$ forced by the No-Diagonal theorem of §2.3. Which *integer* levels the angle is read at is not settled here: it is fixed by the level interval of §4, Proposition 4.20.

Proposition 3.4 (The tangent of the spectral angle is the tower)**.** *For every σ,*

$$\tan\alpha(\sigma) = \varepsilon_0\varphi^\sigma, \qquad \textit{hence} \qquad \frac{\tan\alpha(\sigma+1)}{\tan\alpha(\sigma)} = \varphi, \tag{3.7}$$

so the angle is not an auxiliary parameter: its tangent is the multiplicative spectrum that φ generates, climbing at the one rate φ of (2.1) *— the same spectrum §3.3 will read as the tower. The surface* $\sin\alpha(\sigma_1)\cos\alpha(\sigma_2)$ *has the closed form*

$$\sin\alpha(\sigma_1)\cos\alpha(\sigma_2) = \frac{\varepsilon_0\varphi^{\sigma_1}}{\sqrt{\left(1+\varepsilon_0^2\varphi^{2\sigma_1}\right)\left(1+\varepsilon_0^2\varphi^{2\sigma_2}\right)}}. \tag{3.8}$$

What the certainty principle $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ (2.21) *contributes is the* scale *of the argument, $\varepsilon_0 = \pi/M_{\mathrm{PCF}}$, and nothing else: the bound $\alpha(\sigma) < \pi/2$ is a property of* arctan *and holds for every real argument, so it is not the certainty relation that keeps the angle off the right angle. The bound is used for one thing—it gives $\cos\alpha(\sigma) > 0$, the hypothesis under which Corollary 3.17 divides by* $\cos\alpha$.

P*[spectral_angle_tan, spectral_angle_tower_ratio]*

Proof. $\tan\arctan x = x$ with $x = \varepsilon_0\varphi^\sigma > 0$ gives the first identity, and the quotient of two consecutive values is $\varphi^{(\sigma+1)-\sigma} = \varphi$, the ε_0 cancelling. For (3.8), $\sin\arctan x = x/\sqrt{1+x^2}$ and $\cos\arctan y = 1/\sqrt{1+y^2}$; multiplying the two gives the display. The bound is $\arctan x < \pi/2$ for all finite x. □

With the binary entropy $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$, the modulus saturates the Bekenstein bound [61, 62],

$$\frac{1}{4G_N} = \tfrac{1}{2} = |\Omega|, \qquad H\left(\tfrac{1}{2}\right) = 1 \text{ bit}, \qquad S_{\mathrm{BH}} = \frac{A}{4G_N}. \tag{3.9}$$

Remark 3.5 (The role of φ in the meta-invariant). $\mu = \frac{1}{2} = \varphi^{-\lambda_{\log}}$ is the scale quantum ($\varphi^{\lambda_{\log}} = 2 =$ one bit). The invariants $d = 3$, $\mu = \frac{1}{2}$, $\sigma = \frac{3}{2}$ come from the φ/S_3 geometry of §2; they are not postulated.

Thread (the invariants force the holographic principle). The holographic principle is read off the invariants, not assumed:

1. the modulus $\mu = \frac{1}{2} = |\Omega|$ has binary entropy $H(\frac{1}{2}) = 1$, so each fundamental cell (each eigenvalue) carries exactly one bit ((3.6)→(3.9));
2. the $d = 3$ eigenvalues give 3 bits $= c = 3$ (Brown–Henneaux [63]: three eigenvalues × one bit);
3. and $1/(4G_N) = \mu = \frac{1}{2}$ turns $S_{\mathrm{BH}} = A/(4G_N)$ into the area law.

Hence information = one bit per Planck area (the holographic bound), read from the invariants. (Numerically verified: (2.21), (4.1), (3.21).)

This is the holographic bound that Susskind, following 't Hooft, drew from unproven assumptions about the nonperturbative behaviour of string theory: one discrete degree of freedom per Planck area [4, 18]. Here it is not assumed but derived—the bit $H(\frac{1}{2}) = 1$, the central charge $c = 3$, and the area law $S_{\mathrm{BH}} = A/4G_N$ are read off the single modulus $|\Omega| = \frac{1}{2}$.

3.3. **The object as generator of the tower and the amplitudes.** Built on the golden tower of §2.8 (the Borger Λ-structure on φ^σ) and the mediated spectrum of §2.7, the tower mediates *between* a binary register (the half-factorial $/\,\Gamma\,/\,\pi$; i, d=1→2) and a ternary one (S_3/φ; d=2→3), with $\mu = \frac{1}{2}$ as the geometric mean. The Huerta–Schreiber superpoint is purely binary ($\mathbb{Z}_2$) [23]; PCF spans and mediates both. The relation is sharper than a contrast of registers. The binary $\mathbb{Z}_2$ of the superpoint is not isolated in the framework: the binary register and the turn are one object, since 2 satisfies in $\mathbb{F}_5$ the equation i satisfies in $\mathbb{C}$ (2.62) and is its image under the reduction at a prime over 5 (Proposition 2.72). Its fourth order is therefore one of the two factors of the conductor 20 derived in §2.12, the other being the 5 of the pentagon (Theorem 2.4), the two being coprime so that $\operatorname{lcm}(4,5) = 20$ (2.61). The $\mathbb{Z}_{20}$ that decides χ_5 (2.33) therefore factors as $\mathbb{Z}_4 \times \mathbb{Z}_5$, and each factor governs one character and only one (2.63), neither determining the other (Proposition 2.73): the binary side of the superpoint and the pentagonal side of φ are its two components. That is the precise sense in which PCF mediates rather than merely spans—the two registers are not juxtaposed but are the coprime factors of a single conductor, and the mediation $\mu = \frac{1}{2}$ of (3.11) below is the geometric mean between them. The M-theory object enters here: the HS tower projects from the φ^σ tower. The dimensional chain is

$$\mathbb{R}_{d=1} \xrightarrow{i^2=-1} \mathbb{C}_{d=2} \xrightarrow{\varphi^2=\varphi+1} \mathbb{E}^3_{d=3}. \tag{3.10}$$

Mediation by $\mu = \frac{1}{2}$ produces the two registers,

$$\varphi^{\mu\lambda_{\log}} = \sqrt{2}\ \ \text{(binary)}, \qquad \varphi^{\mu \log_\varphi 3} = \sqrt{3}\ \ \text{(ternary)}. \tag{3.11}$$

P [sqrt2_mediation, sqrt3_mediation]

The tower and its mode count are

$$z(\sigma) = \varphi^\sigma, \qquad N_{\text{modes}}(\sigma) = \lfloor \pi\,\varphi^\sigma \rfloor \tag{3.12}$$

N [eq:tower-modes]

This is where the geometric encoding replaces the Fourier expansion of the string as the device that fixes the mode content: (3.12) counts modes by the cell capacity π of (2.21) at each level of the tower rather than by harmonic order. The two are not in conflict—the count is finer-grained, and it is tied through §3.4 to the information the horizon carries.

Remark 3.6 (Fibonacci-adjacent, and where it parts company). The count (3.12) is *adjacent* to the Fibonacci sequence, not equal to it: $|N_{\text{modes}}(\sigma) - F| \le 1$ for the nearest Fibonacci F, with equality $N_{\text{modes}}(\sigma) = F$ for $\sigma \le 5$ — the values $3, 5, 8, 13, 21, 34$ — and a genuine departure at the tower ceiling, where $N_{\text{modes}}(6) = \lfloor \pi\varphi^6 \rfloor = 56$ while the Fibonacci value is 55. The distinction matters: the factor π of (3.12) is the cell capacity of (2.21) and not a Fibonacci index, so the staircase is π-scaled and only *shadows* the recurrence of Theorem 2.14. Reading the tower as Fibonacci would put 55 at the tower ceiling $\sigma = 2n$, where the saturation entropy is largest.

P [fib_adjacent_le_one, nmodes_six_ne_fib]

N [thm:fib-min]

Remark 3.7 (The scale operator of the tower, and which spectrum is which). The tower $z = \varphi^\sigma$ of (3.12) defines the diagonal scale operator

$$H|\sigma\rangle = m_0\,\varphi^\sigma, \qquad \operatorname{spec} H = \{0\} \cup \{m_0\varphi^\sigma\}_{\sigma \ge 0}, \tag{3.13}$$

self-adjoint (real diagonal) and bounded below, with the vacuum at 0; its gap and the reconstruction it supports are established in §5.

P [tower_ham_is_hermitian, tower_ham_bounded_below]

Two spectra must be kept apart here, and the framework supplies both from the same pair π, φ. The golden tower (3.12) is the *radial and scale* tower: its successive ratio tends to φ and its physical mass reading is the Kaluza–Klein one, the interior KK mass carrying the golden identity $\varphi^2 + \varphi^{-2} - 2 = 1$ (§A.7, (A.10))—the gravity side. The *confining* mass spectrum, on the gauge side, is the Regge tower $\alpha' M_n^2 = n$ of (3.15), whose integer pole positions index the Dirichlet series (2.39) and hence ζ; its mass ratios are therefore $M_n/M_1 = \sqrt{n}$, not φ^σ. The two are distinct spectra but not unrelated: the constant of the confining tower is the Regge slope α', the string tension $T = 1/2\pi\alpha'$, and §5 shows this tension welded to the golden tower level by level—so the radial and confining towers are two readings of one scale, related through the tension rather than conflated.

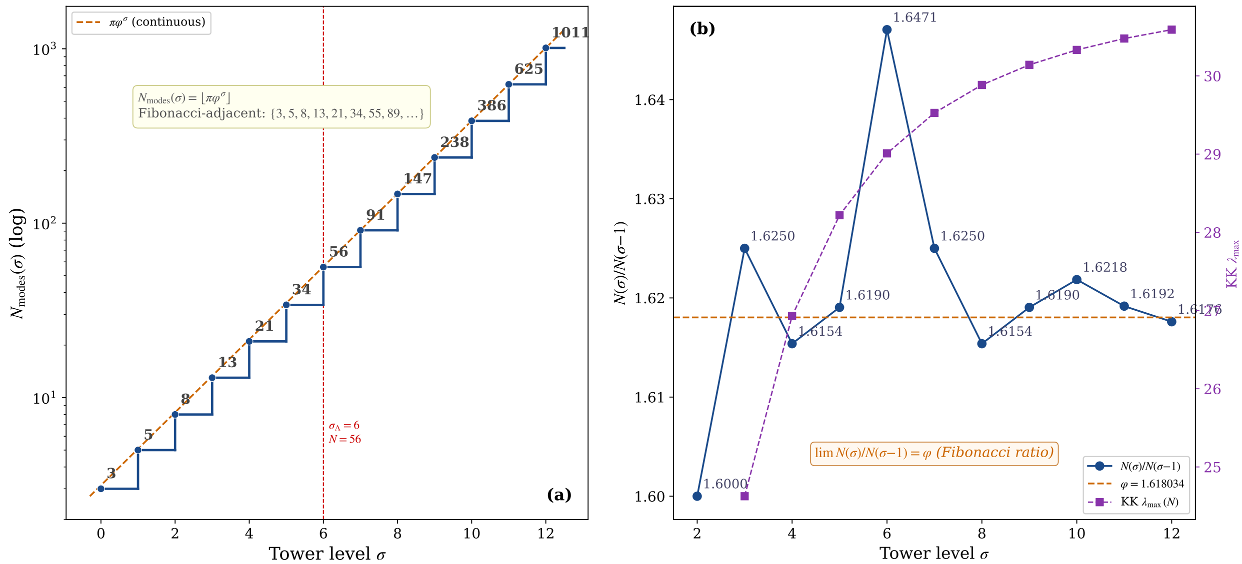

FIGURE 3. The mode tower $N_{\text{modes}}(\sigma) = \lfloor \pi\varphi^\sigma \rfloor$ (3.12): (a) the Fibonacci-adjacent staircase $\{3, 5, 8, 13, 21, 34, 56, \dots\}$, equal to the Fibonacci values for $\sigma \leq 5$ and departing at $\sigma = 6$ where $56 \neq 55$ (Remark 3.6); (b) the successive ratio $N_{\text{modes}}(\sigma)/N_{\text{modes}}(\sigma - 1) \to \varphi$ alongside the Kaluza–Klein spectrum.

The Huerta–Schreiber brane-bouquet tower [23] maps to the PCF tower under a functor F,

$$\mathbb{R}^{0|1} \to \mathbb{R}^{1,1|1} \to \cdots \to \mathbb{R}^{9,1|16} \to \mathbb{R}^{10,1|32} \xrightarrow{F} \{\varphi^\sigma\}_\sigma, \tag{3.14}$$

the dynamics being presupposed on the HS side ($H = L_0 + \bar{L}_0$) and derived on the PCF side. Both sides are chains, so F is the order-preserving level map—superpoint $\to$ NS5 at $\sigma = 0, \dots, 5$ on the hypercube H_5 (§3.4, App. A.5)—and what it carries is the modulus, $|\Omega| = \frac{1}{2}$, $|\Omega|^2 = \frac{1}{4}$, proved at the corner (App. A.4). Tree amplitudes follow from π (binary, via Γ) and φ (ternary, via Regge): the Veneziano amplitude [27]

$$A_4 = \frac{\Gamma(-\alpha' s)\,\Gamma(-\alpha' u)}{\Gamma\big(1 - \alpha'(s+u)\big)} \tag{3.15}$$

has simple poles at $\alpha' s = n$, $n \geq 1$ (the Regge tower). The pole *positions* are the integer spectrum, so they index the tower's Dirichlet series; the residues are the Regge polynomials of degree $n - 1$ (carrying spin $\leq n - 1$), nonzero at every level, which certifies that every integer level is populated. That Dirichlet series is $\zeta(s)$, which reorganises into the Euler product by Proposition 2.47 of §2.9, The Dirichlet series whose index set the poles reproduce is (2.39), proved in §2.9; what is established here is the identification of that index set with the Regge tower.

Remark 3.8 (The tower's Dirichlet series is the Euler product of §2.9). The Euler product (2.39) of §2.9 is here read on the string side. The arithmetic supplies the identity; the amplitude supplies that the pole positions $\alpha' s = n$ of (3.15) are exactly its index set, with nonzero residue at every level. The physical content is the identification of the index set, not the arithmetic identity.

P `[regge_dirichlet_eq_zeta, regge_tower_is_euler_product]`

The pentagonal origin $\varphi = 2\cos(\pi/5)$ generates this structure: the Regge tower (string side) and the Euler product (arithmetic side) are the same object seen from two angles. The uniqueness of the ultraviolet completion selected here—that under supersymmetry and the standard consistency conditions the Veneziano form is essentially forced—is a recent result of Elvang and collaborators [26]. Loops are likewise φ-organised; the one-loop partition function is finite,

$$Z_{\mathrm{PCF}}(i) = \frac{e^{-3\pi/2}}{|\eta(i)|^6}, \tag{3.16}$$

with $\eta(i)$ the value established in (2.48).

Remark 3.9 (The one-loop constant is the lattice sum at the self-dual point). The $\eta(i)$ of (3.16) is not an independent constant: by Remark 2.52 of §2.9 it is the Gaussian lattice sum evaluated at the fixed point of $t \mapsto 1/t$, $\Theta(1) = \varphi^{\mu\lambda_{\log}}\eta(i)$ (2.48), with $\varphi^{\mu\lambda_{\log}} = \sqrt{2}$ the binary register of Proposition 2.10. The finiteness of (3.16) and the convergence of the completed zeta are therefore the same fact, read at $\tau = i$ and at general t.

P `[theta_from_jacobi, theta_fixed_point_unique]`

The tower carries a loop hierarchy—one loop = colour, two loops = generations—resting on two proved anchors: the colour ratio $1-\mu_3^2 = \frac{3}{4}$, and the odd zeta values over the tower (Theorem 2.62), with Apéry's $\zeta(3)$ [64] as the two-loop constant. What is *not* derived is the assignment itself—that one loop reads as colour and two as generations—which remains structural.

P `[colour_ratio]`

The ultraviolet is regulated in two complementary ways: the modular partition (3.16) (global, on the torus) and dimensional regularisation at the vertex (local),

$$\Gamma\left(\tfrac{\varepsilon}{2}\right) = \frac{2}{\varepsilon} - \gamma_E + O(\varepsilon). \tag{3.17}$$

Remark 3.10 (The role of φ in the tower). $\varphi^{\mu\lambda_{\log}} = \sqrt{2}$ ties the binary register and $\varphi^{\mu\log_\varphi 3} = \sqrt{3}$ the ternary one, with $\mu = \frac{1}{2}$ as mediator — distinguishing PCF from the purely binary $\mathbb{Z}_2$ chain of HS.

Thread (amplitudes and loops from π and φ via the tower).

1. *tree (Veneziano):* in (3.15) the Γ factors carry the π/binary side ($\Gamma(\frac{1}{2}) = \sqrt{\pi}$) and the poles at $\alpha' s = n$ are the Regge tower (the φ/ternary side);
2. the Regge tower *is* the Euler product (2.39), linking amplitude and primes;
3. *loops:* the one-loop partition (3.16) is finite; the colour/generation hierarchy rests on two proved anchors, $1-\mu_3^2 = \frac{3}{4}$ and the odd zeta values (Theorem 2.62), with $\zeta(3)$ as the two-loop constant, while the assignment of loop order to colour and generations remains structural;
4. the modes at each level are $N_{\mathrm{modes}}(\sigma) = \lfloor \pi\varphi^\sigma \rfloor$ (3.12).

(Numerically verified: (2.7), (5.8), (2.45), (3.16), (3.12).)

Thread (the two ultraviolet regulators coexist).

1. *modular/η:* the partition (3.16) is finite because $\eta(i)$ is well defined — modular invariance regulates the ultraviolet on the torus, with no divergence;
2. *dimensional/$\Gamma(\varepsilon/2)$:* (3.17) extracts the $2/\varepsilon$ pole at the local vertex;
3. they are complementary: η handles the worldsheet (global, finite partition), $\Gamma(\varepsilon/2)$ the vertex (local pole); both give the same finite physics.

3.4. **All AdS/CFT from a single microstate, at each level.** The whole AdS/CFT correspondence follows from one microstate (the object of §3.1), not from an ensemble — and it holds at each level of the tower. This per-level statement is the dependency the web (§3.5) rests on. The single microstate is the object Ω with $|\Omega| = \frac{1}{2}$ (its realisation as a self-adjoint operator is §5); being one state and not an ensemble, it resolves the firewall ambiguity (AMPS) without averaging. The throat and its warped metric are

$$ds^2 = e^{2A(z)}\,\eta_{\mu\nu}\,dx^\mu dx^\nu + dz^2, \qquad z(\sigma) = \varphi^\sigma \text{ of (3.12)}. \tag{3.18}$$

The five-dimensional throat: AdS_5, S^5 and the ladder. The throat (3.18) is the AdS_5 geometry whose curvature invariants are fixed by the modulus: $R = -20$, $\Lambda_5 = -6$, $G_{\mu\nu} = 6$, saturating the Breitenlohner–Freedman bound (Fig. 4); computed in §A.1.

P [R_scalar_pcf, lambda5_pcf, G_AB_pcf, BF_value]

Its companion five-sphere carries the Hopf chain $S^1 \hookrightarrow S^5 \to \mathbb{CP}^2$ with $\chi(\mathbb{CP}^2) = 3 = n$

P [hopf_latitude, hopf_from_clifford]

The same arity-3 of §2.3; the five-dimensional case is precisely the one requiring the Gaussian and Eisenstein primes (§1). The discrete levels $\sigma = 0, \ldots, 5$ (superpoint $\to$ NS5, §1.5.2) sit on the Hamming hypercube $H_5 = \{0,1\}^5$, $|H_5| = 2^5$, with $\Lambda_\sigma = \varphi^\sigma \Lambda_{\mathrm{PCF}}$ and the saturation entropy $S(\sigma) = \pi\varphi^\sigma$ (3.12).

P [hypercube_card, conductor_attribution, four_cocone, S_tower]

The bulk–boundary map is an isometry—proved here as Lemma 2.21, and independently in the companion [38]—used in §5, and it obeys the GKPW dictionary [17, 65, 66],

$$V : \mathcal{H}_{\mathrm{bulk}} \to \mathcal{H}_\partial, \qquad V^\dagger V = \mathbf{1}, \qquad Z_{\mathrm{grav}}\big[\phi|_\partial\big] = \big\langle e^{\int_\partial \phi_0 \mathcal{O}} \big\rangle_{\mathrm{CFT}}. \tag{3.19}$$

The isometry as an algebra–isometry (bulk–boundary) duality. The embedding V (3.19) is realised on the two faces of one modulus. The eigenvalue triangle $\lambda_k = \frac{1}{2}\omega^k$ ($\omega = e^{2\pi i/3}$) of Prop. 2.19 (2.13) is the Eisenstein lattice $\mathbb{Z}[\omega]$ (*algebra* face: 120°, S_3 symmetry, A_2 root system $= \mathfrak{su}(3)$); the same modulus on the i/π side, where $(\frac{1}{2})! = \frac{1}{2}\sqrt{\pi}$ fixes $\mu = \frac{1}{2}$ (§2.4), is the Gauss lattice $\mathbb{Z}[i]$ (*isometry* face: $\tau = i$, $\ell = 1$).

P [eisenstein_omega, omega_eigenvalues, a2_minimal_vectors_exactly_six]

The φ-mediation $S_3 \to \mathbb{Z}_4$ between them preserves $|\Omega| = \frac{1}{2}$ and fixes $\tau = i$: this is $V^\dagger V = \mathbf{1}$ read on the lattice — bulk (geometry, $\mathbb{Z}[i]$) and boundary (algebra, $\mathbb{Z}[\omega]$) are two readings of one triangle, neither privileged, exactly as (3.19) requires. The GKPW dictionary of (3.19) enters cited [17, 65, 66].

P [holographic_area, hopf_from_clifford, hopf_latitude, G_AB_pcf, lambda5_pcf, R_scalar_pcf, BF_value, omega_eigenvalues, a2_minimal_vectors_exactly_six, eisenstein_omega, hypercube_card]

Standard AdS/CFT integrates out the radial coordinate z (it privileges a boundary); here the isometry singles out neither side, so PCF keeps both bulk and boundary ((3.19); "forget z" only on the Maldacena side). The modulus is scale invariant, so the holographic relation holds at every level:

$$|\Omega|_\sigma = \tfrac{1}{2} \qquad \forall\, \sigma. \tag{3.20}$$

The self-similarity is the physical face of the Frobenius lift (2.35): both act on the golden monoid through the one composition law $\psi_p(\varphi^n) = (\varphi^n)^p$, exponents multiplying while the modulus stands. The gravitational data follow from the same $\mu = \frac{1}{2}$, the $\frac{3}{4}$ of the signature being the $\|v\|^2$ of (2.16) read as the colour ratio (Remark 2.22):

P [bulk_boundary_isometry, spectral_uniqueness, bulk_boundary_isometry]

$$S_{\mathrm{BH}} = \frac{1}{4G_N} = \mu = \tfrac{1}{2}, \qquad c = \frac{3\ell}{2G_N} = 3 \quad (\ell = 1,\ G_N = \tfrac{1}{2}). \tag{3.21}$$

This closes the circle to §3.1: the Newton constant is the worldline modulus itself, $G_N = |\Omega(\tau)|$ for every τ (3.2), and the central charge singles that value out—$3\ell/(2G) = 3$ iff $G = \frac{1}{2}$.

Remark 3.11 (The role of φ in AdS/CFT). Everything follows from the microstate Ω (§3.1) and the tower (§3.3); the modulus is $\mu = \frac{1}{2}$ at every level, and the AdS/CFT signature ($\ell{=}1, G_N{=}\frac{1}{2}, c{=}3, \mathrm{GKP}{=}\frac{3}{4}$) is exactly the microstate's invariants.

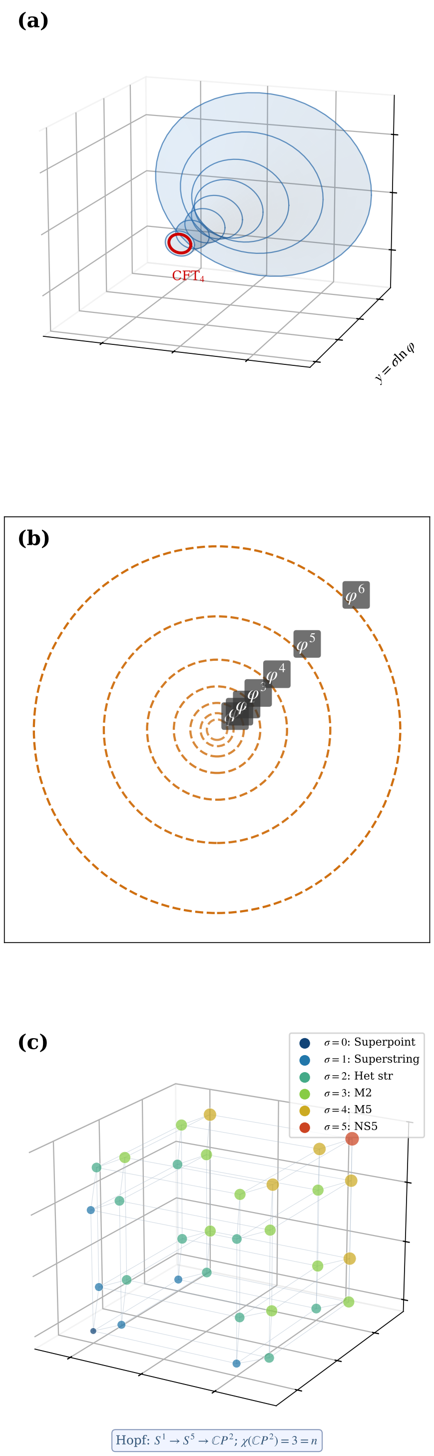

FIGURE 4. AdS$_5$ funnel $z = \varphi^\sigma$ (3.18) $\leftrightarrow$ the lattice tower Λ_σ $\leftrightarrow$ the Hamming hypercube H_5 of the M-theory ladder (superpoint $\to$ NS5); φ^σ connects all three: (a) the funnel with $\ell = 1$, $G_N = \frac{1}{2}$; levels bottom-to-top are superpoint (σ=0), superstring (σ=1), het string (σ=2), M2-brane (σ=3), M5-brane (σ=4), NS5-brane (σ=5), 10-brane (σ=6); (b) the Eisenstein lattice $\Lambda_\sigma = \varphi^\sigma \Lambda_{\mathrm{PCF}}$ in $\mathbb{C}$ with concentric φ-rings; (c) the Hamming cube $H_5 = \{0, 1\}^5$ coloured by Hamming weight = H-S level.

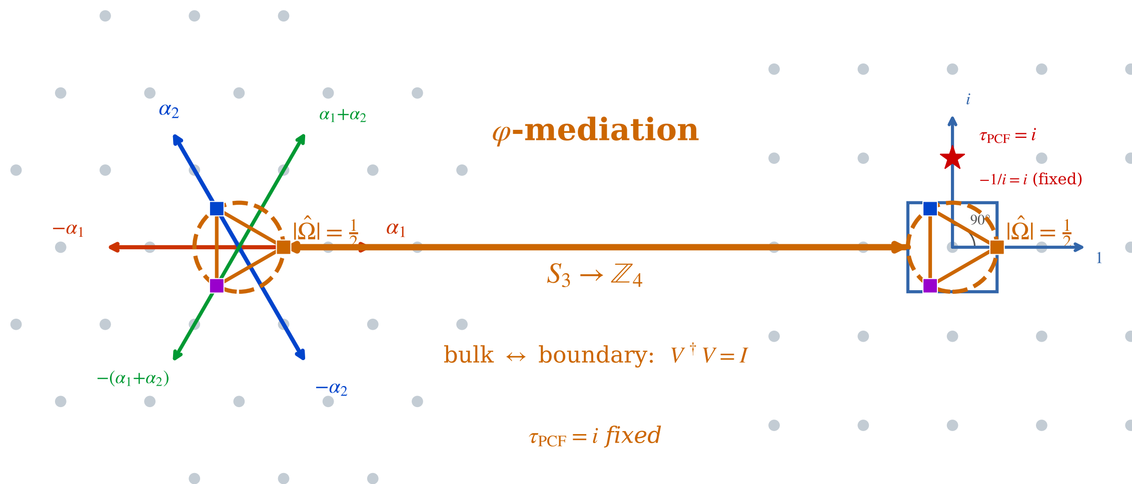

FIGURE 5. Isometry $\leftrightarrow$ algebra (bulk–boundary): left, the Eisenstein lattice $\mathbb{Z}[\omega]$ (120°, S_3 symmetry, $A_2 = \mathfrak{su}(3)$, algebra/boundary); right, the Gauss lattice $\mathbb{Z}[i]$ (90°, $\mathbb{Z}_4$ symmetry, $\tau = i$, $\ell = 1$, isometry/bulk). They are φ-mediated ($S_3 \to \mathbb{Z}_4$) with $|\Omega| = \frac{1}{2}$ preserved — the lattice realisation of $V : \mathcal{H}_{\mathrm{bulk}} \to \mathcal{H}_\partial$, $V^\dagger V = \mathbf{1}$ (3.19).

Thread (all AdS/CFT from one microstate).

(1) the isometry V with $V^\dagger V = \mathbf{1}$ embeds bulk into boundary preserving the norm, so the duality holds from *one* state, without averaging ((3.2)$\to$(3.19));
(2) the modulus is scale invariant, $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$, so the holographic relation holds at *each* level of the throat, not only at the boundary (3.20);
(3) from the same $\mu = \frac{1}{2}$: $S_{\mathrm{BH}} = 1/(4G_N) = \frac{1}{2}$ (area law) and $c = 3$ (Brown–Henneaux) — the gravity-side entropy and central charge come from the microstate invariant (3.21);
(4) hence GKPW [65, 66] and Ryu–Takayanagi [67] are consequences of the isometric embedding of the microstate.

Thus the entire AdS/CFT correspondence comes from one microstate, not an ensemble. (Numerically verified: $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$ via (2.21), GKP $= \frac{3}{4}$, $S_{\mathrm{BH}} = \mu$, $c = 3$.)

Maldacena left the AdS boundary conditions unspecified—“we leave this interesting problem for the future” [17]. Here they are determined: the modular parameter is fixed at $\tau = i$, the fixed point of S-duality and the scale by the certainty principle $\varepsilon_0 M_{\mathrm{PCF}} = \pi$.

P `[certainty_principle]`

3.5. **The M-theory type duality web from the microstate.** The M-theory type duality web projects from the microstate (§3.1) through the φ^σ tower (§3.3); Witten's eight/eleven-dimensional object is not required here. Witten *posits* that object and *conjectures* that the web follows from it [68]; PCF starts from a *demonstrated* microstate and *generates* the tower, of which the web is a projection. The web follows from the tower with HS (§3.3) and from per-level AdS/CFT (§3.4), both already stated. The shared signature, derived in App. A.4 from the web's two corners, is that the M-theory object and AdS/CFT are projections of the microstate, carrying

$$\underbrace{|\Omega| = \tfrac{1}{2},\ |\Omega|^2 = \tfrac{1}{4}}_{\text{Huerta–Schreiber (M-theory)}} \qquad \underbrace{\ell = 1,\ G_N = \tfrac{1}{2},\ c = 3,\ \mathrm{GKP} = \tfrac{3}{4}}_{\text{Maldacena (AdS/CFT)}}, \tag{3.22}$$

which equal the microstate invariants $\mu = \frac{1}{2}$. That the two corners are one and the same object is Maldacena's conjecture that "M/string theory on various Anti-deSitter spacetimes is dual to various conformal field theories" [17]—the definition of M-theory he anticipated extending to five non-compact dimensions: here it is realised, both corners being projections of the single microstate $|\Omega| = \frac{1}{2}$. The torus on which they meet is the noncommutative torus that Connes, Douglas and Schwarz identify with M-theory compactification [25] and that Manin attaches to the $\mathbb{F}_1$ program [24]; the non-orientable fibre it carries is independent of that algebra (Prop. 2.28). The dualities that engage the tower are

$$\begin{aligned}
&\text{(S-duality)} && \tau \to -\tfrac{1}{\tau} \iff \varphi \to -\tfrac{1}{\varphi} = \bar{\varphi}, && (3.23)\\
&\text{(T-duality)} && R \to \alpha'/R, && (3.24)\\
&\text{(IIA strong coupling)} && \sigma = 10 \to \sigma = 11 \quad \text{(M-theory)}, && (3.25)\\
&\text{(ER = EPR)} && \text{entanglement} \leftrightarrow \text{geometry}, && (3.26)\\
&\text{(de Sitter swampland)} && \frac{|\partial_\sigma V|}{V} = \ln\varphi, && (3.27)\\
&\text{(asymptotic safety)} && \text{UV fixed point.}\,\mathbf{P}\,\texttt{[c_tendsto_two]} && (3.28)
\end{aligned}$$

Remark 3.12 (S-duality is the involution of §2.9). The S-duality (3.23) is, in the Schwinger variable, the involution $t \mapsto 1/t$ under which the Gaussian lattice sum transforms with weight $\frac{1}{2}$, $\Theta(1/t) = \sqrt{t}\,\Theta(t)$ (2.47), and whose unique fixed point $t = 1$ is $\tau = i$ (Proposition 2.51). What §2.9 proves is the transformation; what (3.23) adds is that the string's electric–magnetic duality and the Galois conjugation $\varphi \to -1/\varphi$ are that same involution read on the coupling and on the field.

P `[theta_from_jacobi, theta_fixed_point_unique]`

Remark 3.13 (The IIA→11D bridge as Witten's Kaluza–Klein tower). The bridge $\sigma = 10 \to \sigma = 11$ in (3.25) corresponds to Witten's Kaluza–Klein mechanism: at Type IIA strong coupling the states in the tower are the Kaluza–Klein modes of eleven-dimensional supergravity on $\mathbb{R}^{10} \times S^1$, of mass $\sim 1/r$ [68]. The φ^σ tower plays that role here. The golden Kaluza–Klein structure ($\varphi^2+\varphi^{-2}-2 = 1$; $G_5 \to G_4 = \frac{1}{4}$; the Breitenlohner–Freedman comparison; discrete-spectrum stability) is proved in Appendix A.7.

Remark 3.14 (The role of φ in the duality web). The dualities are projections of the microstate's φ^σ tower, not of a posited object; the modulus is $\mu = \frac{1}{2}$ at every level, and S-duality is the Galois conjugation $\varphi \to -1/\varphi$.

Remark 3.15 (The web is reconstructed from its two corners). The web this paper uses has two corners only—Huerta–Schreiber M-theory and Maldacena AdS/CFT—and both are shown in Appendix A.4 to carry the microstate modulus $|\Omega| = \frac{1}{2}$; the dualities are the symmetries of the φ^σ tower that fix it. The wider string-theory content (U-duality, E_8, the five string theories) is neither used nor needed here.

Thread (microstate → tower → web).

(1) the dualities are written in σ/φ, as projections of the tower: S as $\varphi \to -1/\varphi$ (Galois conjugate), T as $R \to \alpha'/R$, IIA→11D as $\sigma = 10 \to 11$, ER=EPR, and the de Sitter swampland $|\partial_\sigma V|/V = \ln\varphi$ ((3.23)–(3.27));

(2) the shared signature (3.22): the *framework colimit* proves the M-theory object (modulus $\frac{1}{2}$) and AdS/CFT ($\ell{=}1, G_N{=}\frac{1}{2}, c{=}3, \text{GKP}{=}\frac{3}{4}$) are projections with the microstate invariants ($\mu = \frac{1}{2}$)—this is the colimit of *frameworks*, not the Euler colimit of §2.8;

(3) hence the web is reconstructed from its two corners (App. A.4): both carry $\mu = \frac{1}{2}$, and the dualities fix the modulus; Witten's eight-dimensional object is not required.

Thus microstate → tower → web (HS and per-level AdS/CFT). The quantitative pieces ($\varphi \to -1/\varphi$, the swampland constant $\ln\varphi$ (the regulator R_K, Definition 2.18), the T-self-dual radius, and the shared HS/Maldacena signature) are checked numerically ((4.11) **N**`[eq:obs-swampland]`, (4.17) **N**`[eq:obs-jacobson]`) and formally in `CW6_complete_v2.lean` (§2.14); the framework colimit itself—the five frameworks and the PCF core as partial spectral objects, the cocone into the vertex T_{PCF}, and its universal property—is proved in the same file (`cocone_property`, `colimit_universal`, `framework_colimit`).

The remaining duality, ER = EPR, is where the microstate supplies what Cool Horizons left open. Maldacena and Susskind observe that the ER bridge "may be a highly quantum object that is yet to be independently defined", and that they "do not have an independent argument for its smoothness" [46]. In PCF the bridge is both defined and analytic: it is the cocycle $T(\sigma_1,\sigma_2)$ of the proposition below, smooth in the analytic sense because its denominator $1+\varepsilon_0\varphi^{\sigma}$ is strictly positive at every level, carried, with its geometry supplied by the non-orientable fibre, on the M-theory noncommutative torus rather than on a general quantum object.

Proposition 3.16 (ER = EPR from the modulus)**.** *The entanglement of the two hemispheres and the geometric bridge between them are one invariant, $\mu_3^2 = \frac{1}{4}$.* (EPR.) *Since $|\Omega(\tau)| = \frac{1}{2}$ for every τ* (3.2)*, the two-point entanglement modulus is*

$$|\Omega(\tau_A)\,\overline{\Omega(\tau_B)}| = |\Omega(\tau_A)|\,|\Omega(\tau_B)| = \mu_3^2 = \tfrac{1}{4} \qquad (\forall\,\tau_A,\tau_B),$$

which is the holographic area factor $\mu_3^2 = \frac{1}{4}$. (ER.) *The bridge amplitude between two levels of the throat, $T(\sigma_1,\sigma_2) = \dfrac{1+\varepsilon_0\varphi^{\sigma_1}}{1+\varepsilon_0\varphi^{\sigma_2}}$, satisfies $T(\sigma_1,\sigma_2)\,T(\sigma_2,\sigma_1) = 1$ and $T(\sigma_1,\sigma_2)\,T(\sigma_2,\sigma_3) = T(\sigma_1,\sigma_3)$, so the bridges form a group: a round trip is the identity and composition is additive in σ.* (ER = EPR.) *Both are fixed by the one scale-independent modulus $\mu_3 = \frac{1}{2}$ ($|\Omega|_\sigma = \frac{1}{2}$ $\forall\sigma$,* (3.20)*): the entanglement between P (south) and F (north), mediated by C,* is *the bridge that joins them through the equatorial axis. Entanglement is geometry, derived from the modulus, not posited.*

P*[bridge_cocycle, bridge_inverse, bridge_compose]*

The bridge and its $\frac{1}{4}$ identity are shown in Fig. 6.

Proof. (EPR) $\Omega(\tau) = \frac{1}{2}e^{i\tau\ln\varphi}$ has $|\Omega(\tau)| = \frac{1}{2} = \mu_3$ for all τ (3.2); the product rule for moduli gives $|\Omega(\tau_A)\overline{\Omega(\tau_B)}| = \mu_3^2 = \frac{1}{4}$, independent of τ_A,τ_B because μ_3 comes from the S_3 arity (Proposition 2.19), not from the phase. *(ER)* $T(\sigma_2,\sigma_1)$ is the reciprocal of $T(\sigma_1,\sigma_2)$, whence $T(\sigma_1,\sigma_2)T(\sigma_2,\sigma_1) = 1$; the telescoping of numerator and denominator gives $T(\sigma_1,\sigma_2)T(\sigma_2,\sigma_3) = T(\sigma_1,\sigma_3)$. *(*ER = EPR*)* The levels are equivalent because $\varepsilon_0\varphi^{\sigma}\cdot M_{\mathrm{PCF}}\varphi^{-\sigma} = \pi$ at every σ (2.21), so $|\Omega|_\sigma = \frac{1}{2}$ is scale-invariant (3.20); the entanglement invariant μ_3^2 and the bridge group thus share one origin. □

Corollary 3.17 (The bridge is the spectral angle)**.** *With $\alpha(\sigma)$ of Proposition 3.4,*

$$T(\sigma_1,\sigma_2) = \frac{1+\tan\alpha(\sigma_1)}{1+\tan\alpha(\sigma_2)} = \frac{\sin\bigl(\alpha(\sigma_1)+\frac{\pi}{4}\bigr)\,\cos\alpha(\sigma_2)}{\sin\bigl(\alpha(\sigma_2)+\frac{\pi}{4}\bigr)\,\cos\alpha(\sigma_1)}. \tag{3.29}$$

The first equality is (3.7)*; the second is $\sqrt{2}\,\sin(\alpha+\frac{\pi}{4})/\cos\alpha = 1+\tan\alpha$, and the $\frac{\pi}{4}$ enters for one reason only:* $\tan\frac{\pi}{4} = 1$ *is the* 1 *that the cocycle adds to $\varepsilon_0\varphi^{\sigma}$. So the* ER = EPR *bridge and the spectral angle are one object, and the trigonometric form that Fig. 6 plots against the algebraic one is an identity, not a numerical coincidence.*

P*[bridge_eq_tan_form, bridge_pi_quarter, spectral_angle_tan]*

Proof. $\sin(\alpha+\frac{\pi}{4}) = (\sin\alpha+\cos\alpha)/\sqrt{2}$, so $\sqrt{2}\sin(\alpha+\frac{\pi}{4})/\cos\alpha = 1+\tan\alpha$ whenever $\cos\alpha \neq 0$, which holds since $\alpha(\sigma) < \pi/2$. Forming the quotient at σ_1 and σ_2 cancels the $\sqrt{2}$, and $\tan\alpha(\sigma_i) = \varepsilon_0\varphi^{\sigma_i}$ returns the algebraic form of Proposition 3.16. □

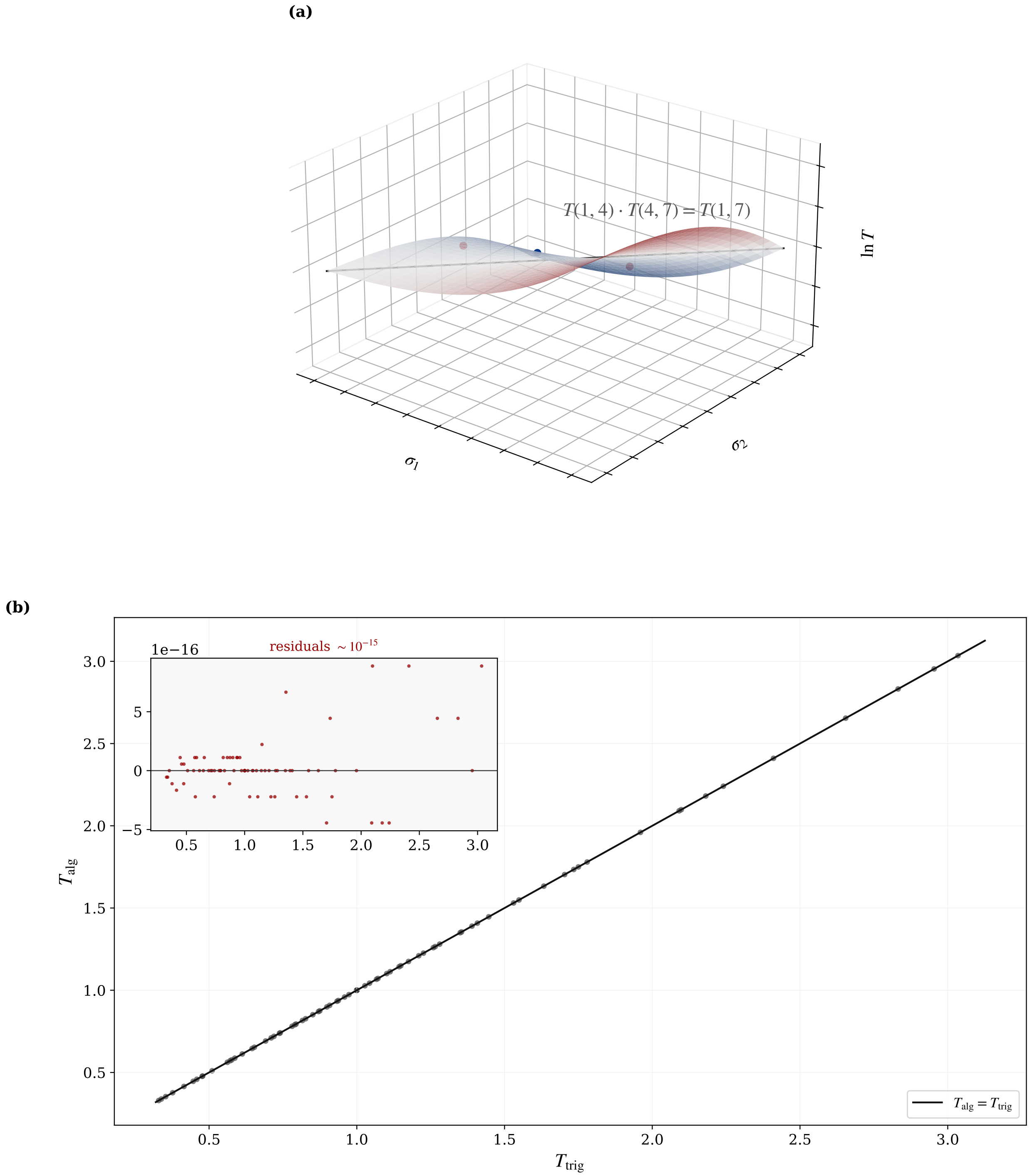

FIGURE 6. ER = EPR from the modulus (Prop. 3.16): (a) the bridge amplitude $T(\sigma_1, \sigma_2)$ as a cocycle, $T(1,4)\,T(4,7) = T(1,7)$; (b) the algebraic and trigonometric forms agree, $T_{\text{alg}} = T_{\text{trig}}$, to machine precision — which is Corollary 3.17, (3.29), and not a numerical accident.

Proposition 3.18 (Two towers, one microstate)**.** *Carrying boundary degrees of freedom into the bulk requires, in the standard treatments, a tower: the Virasoro algebra with its central charge on the holographic side, and the chain of maximal invariant central extensions of the superpoint, $\mathbb{R}^{0|1} \to \cdots \to \mathbb{R}^{10,1|32}$, on the M-theory side (§A.5 below). Each is built independently and each is costly; covering what they must cover takes a great deal of structure. Neither is contradicted here. What the colimit* (3.22) *adds is that they are two routes to one vertex, and this is visible in four places at once.*

First, strings and holography commute in c. *The central charge arrives twice—from the worldsheet* (3.4) *and from Brown–Henneaux $c = 3\ell/(2G_N)$ with the derived $\ell = 1$ and $G_N = \frac{1}{2} = |\Omega|$* (3.21)*—and the two arrivals are the same component of the vertex, not two values that agree. The one free parameter of the Virasoro tower is therefore the microstate modulus: $3\ell/(2G_N) = 3$ holds only at $G_N = \frac{1}{2}$.*

Second, the ladder is counted by the turn. *Its 32 supercharges are $|H_5| = 2^5$, and the 2 that counts them satisfies in $\mathbb{F}_5$ the equation the turn satisfies in $\mathbb{C}$* (2.62)*: the binary side of the ladder and the $\mathbb{C}$ level of the dimensional chain* (3.10) *are one generator, reduced. Field theory and group theory meet at the same place: the two quadratic fields $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{5})$ have coprime conductors 4 and 5, and their product 20 carries a character group whose two factors govern one character each and no more* (2.63).

Third, one recurrence covers the ladder. *The tower advances by $S(\sigma+1) = \varphi S(\sigma)$ and the bulk–boundary ratio is π/m_0 at every level ((5.9) below); in any other base the ratio would drift and no level-wise isometry could intertwine them. With no level privileged, the many discrete steps are an index on one flow rather than a stack of separate structures—which is why $\sigma = 10 \to 11$* (3.25) *is one more level and not one more theory.*

Fourth, dimension three is reached by an isometry, not by an algebra. *Frobenius's theorem leaves only 1, 2, 4 for a finite-dimensional associative real division algebra, and 3 is not among them [69]; yet the chain* (3.10) *reaches $\mathbb{E}^3$. There is no contradiction because no product in dimension three is ever asked for: what reaches it is the norm-preserving—hence injective, hence faithful—embedding $V\colon \mathbb{C} \to \mathbb{C}^3$ of Lemma 2.21. The algebraic bound and the geometric reach are different questions, and the isometry is what carries one to the other, as the Frobenius lift* (2.35) *carries the same modulus on the arithmetic side.*

Taken together: the complexity each tower needs on its own is the cost of not having the microstate. for the four items; the Virasoro and brane-bouquet constructions themselves enter cited [63, 68].

3.6. **The demonstrative line (20-step spine, seeded by §2).** Each step uses only previous steps.

§2→§3.: P1 $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (3.1) (§2.2; seed) → P2 the self-dual Gaussian (2.7), its Mellin the Archimedean factor (2.46) and its lattice sum completing ζ into ξ (2.49) (§2.9).

3.1 object.: P3 $\Omega, |\Omega| = \frac{1}{2}$ (3.2) (§2.4) → P4 Gaussian, $\mu = \frac{1}{2}$ (3.3) (§2.2; width fixed by Prop. 2.9) → P5 worldsheet/Polyakov (3.4).

3.2 meta-invariant.: P6 invariants $d{=}3, \mu{=}\frac{1}{2}, \sigma{=}\frac{3}{2}$ (3.6) (§2.3, §2.11) → P7 $1/(4G_N) = \frac{1}{2} = |\Omega|$, $H(\frac{1}{2}) = 1$ (3.9) (§2.4).

3.3 tower + HS.: P8 chain $i \to \varphi$ (3.10) (§2.7) → P9 $\sqrt{2}/\sqrt{3}$ by μ (3.11) (§2.2) → P10 tower φ^σ, N_{modes} (3.12) (§2.8) → P11 HS↔tower (3.14) → P12 Veneziano (3.15) → P13 loops $Z_{\text{PCF}}(i)$ (3.16) → P14 both ultraviolet regulators (3.17).

3.4 AdS/CFT per level.: P15 throat $z(\sigma) = \varphi^\sigma$ (3.18) (§2.8) → P16 isometry $V^\dagger V = \mathbf{1}$, GKPW (3.19) → P17 $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$ (3.20) (§2.4) → P18 $S_{\text{BH}} = \mu$, $c = 3$ (3.21) (§2.11).

3.5 web (final).: P19 shared signature (3.22) → P20 web (3.23)–(3.28), whose S-duality is the involution that P2 already carries (Remark 3.12).

The commutative diagram supports this line with precise structure. The thirty-one nodes form a directed acyclic graph with four roots—the seed $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (3.1), the Schwinger parametrisation

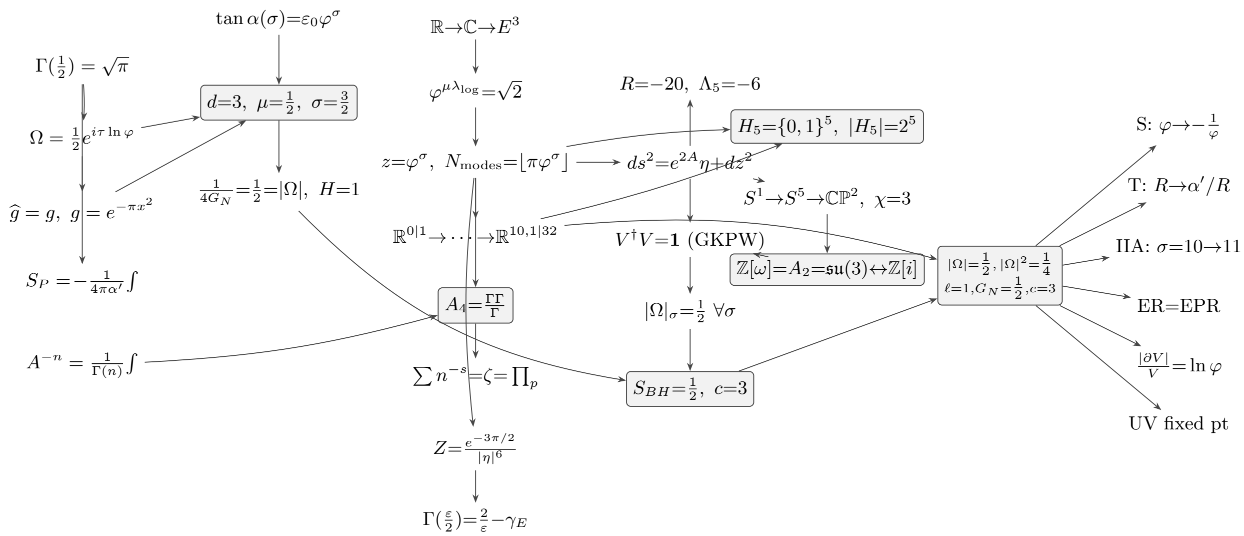

FIGURE 7. Commutative spine of §3 (P1→P20): the thirty-one nodes carrying the twenty steps, arranged by subsection (the object, §3.1; the meta-invariant, §3.2; the tower and amplitudes, §3.3; AdS/CFT per level, §3.4; the duality web, §3.5), with the thirty-four demonstrative arrows. Shaded nodes are the six apices, where two or more independent chains converge on the same value; the forced invariants $d{=}3, \mu{=}\frac{1}{2}, \sigma{=}\frac{3}{2}$, with three incoming routes, are the maximal one. Seeded by the arithmetic of $K = \mathbb{Q}(\sqrt{5})$ in §2; mathematical, physical and conceptual layers coincide on one diagram.

(3.5), the spectral angle (3.7) and the dimensional chain (3.10)—from which all else derives, with 34 demonstrative edges. A topological order exists—the steps can be stated so that none depends on a later one—which guarantees the line reads without forward reference. Six nodes are apices, where two or more independent chains converge; the forced invariants $d{=}3, \mu{=}\frac{1}{2}, \sigma{=}\frac{3}{2}$, with three routes, are the maximal one. Those apices are the sense in which the construction is *overdetermined*: the invariants are not posited and then shown consistent, but forced by more independent conditions than there are degrees of freedom to satisfy them. This is what anchors the proposal in what is already observed—it brings holography to de Sitter by overdetermination rather than by postulate. The observer steps that close the introduction's problem (the objectivity threshold $f_{\text{crit}} = \frac{1}{2} = \mu$ and the Cramér–Rao saturation $F_{\max}^{-1} = \frac{1}{4} = \mu^2$) open §4 Implications. Here P19 is what makes P20 (the web) defensible without re-deriving M-theory or positing Witten's object: §3 closes with its demonstrative line complete (Fig. 7).

4. Implications for holography in de Sitter space

Sections 2 and 3 derived AdS/CFT from a single microstate, with no ensemble average. The present section shows that this apparatus meets the criteria for holography in de Sitter space already fixed in the literature. The aim is not to propose new physics: the mechanisms it uses—the tensors of general relativity, the Polyakov worldsheet, Kaluza–Klein compactification, the warped throat, Jacobson's horizon thermodynamics—are established results. What is shown is that these principles already coincide with the microstate; much of the section is therefore a review of the known principle followed by how the object realises it.

4.1. **What the section does, and against what it is measured.** Holography in de Sitter, as Witten poses it [1], requires that general relativity be embedded in a more precise theory that determines the cosmological constant Λ, and in which the observer is described by the theory rather than introduced from outside—because in a cosmology with $\Lambda > 0$ there is no asymptotic region to which an external observer may retreat.

The yardstick is the classification of von Neumann factors into three types—used in this context by Chandrasekaran, Longo, Penington and Witten—three modes of existence of the same field. *Type I* is that of discrete objects—particle, path integral, string counted one by one, operator of definite spectrum—with minimal projections. *Type II* is that of observation—the trace, the crossed product of the algebra by its modular flow—where observing turns an algebra without trace into one with a trace. *Type III* is that of the local field—the wedge, the horizon, the worldsheet, the spacetime—without normal trace or projections: non-locality.

Of these types, what the section *proves* is the signature of each, not the technical classification: type I as discrete spectrum (**P** `[eigenvalues_modulus_half, witten_finite_rank]`); type II as an invariant the flow preserves—the trace in concrete form (**P** `[omega_flow_invariant, tower_scale_group]`); type III as the non-factorisation of the horizon into interior$\otimes$exterior (Proposition 4.1). The technical names (II_1, III_1) and the crossed product are taken from CLPW by citation; the signatures are proved here. Signature first, name second.

The criteria come from three distinct sources, which should not be conflated. From **Witten 2001** (*Quantum Gravity in De Sitter Space*, W01) come the determination of Λ, the finiteness of the Hilbert space with $S = \ln N$, the isometry-group obstruction, the pairing $\langle f|i\rangle$, the CPT map, the two conjectures—of unitarity and of entropy—and the antipodal singularity. From **Chandrasekaran, Longo, Penington and Witten 2022** (CLPW) come the observer described by the theory, the maximally mixed vacuum state, and the construction of the type II algebra by crossed product from the type III. The thermal nature of the static patch predates both: it is the result of **Gibbons–Hawking**. Throughout, each criterion is tagged with which of the three sources states it.

The thesis is that the framework realises the three types on a single object, the microstate Ω of modulus $|\Omega| = \frac{1}{2}$, and thereby satisfies the requirements Witten left open. The three types are not a label per subsection but three threads crossing the whole section: type I runs through the particle (§4.2), its path integral, and the discrete string of the tower (§4.4)—to it belongs the arithmetic operator of the zeros, referred to the Hilbert–Pólya companion and not proved here (the tower's own Hamiltonian, self-adjoint and with a gap, is Proposition 5.16 and Theorem 5.18); type II runs through the ER=EPR bridge (§4.2), observation (§4.3) and the tracial flow (§4.4); type III runs through the worldsheet (§4.2) and the spacetime with gravity (§4.4). The modulus $\frac{1}{2}$, the projection Π, the entropy S, the centre C and the conjugation Θ recur across movements; each recurrence declares its register and its anchor. The repetition is the weave of a multi-thread demonstration.

4.2. **The non-local field and the particle that emerges from it.**

The worldsheet, the type III field, and its link to gravity. General relativity requires that the laws not depend on coordinates; the tensor guarantees this, because a tensor equation true in one frame is true in all. In string theory this invariance lives on the worldsheet: the string sweeps a surface whose dynamics is the Polyakov action, written on the worldsheet metric h_{ab}—a tensor—and invariant under diffeomorphisms and Weyl rescalings; its constant is the tension $T = 1/2\pi\alpha'$. The worldsheet is the type III field of the section: a local wedge without normal trace, and it is the same field that will reappear as the curved spacetime of §4.4—the type III face that opens and closes §4. Its non-locality is geometric, not postulated. The worldline $\Omega(\tau) = \frac{1}{2}e^{i\tau \ln\varphi}$, promoted by Polyakov, closes onto the torus $T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda$ with modular parameter $\tau = i$ fixed—the self-dual point of S-duality, not chosen—and φ generates the lattice. A torus is compact and boundaryless, and a boundaryless domain admits no question of what happens at the edge of one region against another, because there is no edge at which to separate (**P** `[fibre_independent_of_base]`). By the modulus $\frac{1}{2}$, the loop around the torus accumulates a half rotation, $e^{i\pi} = -1$: the sheet is a Möbius band, non-orientable, with no privileged interior side. The particle is the bulk–boundary interface this field carries:

$$\Pi\colon \mathbb{E}^3 \longrightarrow \mathbb{C}, \tag{4.1}$$

whose Hausdorff dimension is (2.23), established in §2.4. This Π is the projection of the first movement: in closed form $\pi_{\mathrm{PCF}}(a,b,c) = (ab/c)\,\pi/(3\sqrt{3})$ (2.17), the one constant that fixed ε_0 (2.18) and the Fibonacci sum (Proposition 2.25).

P`[projection_closed_form, epsilon0_from_projection]`

The horizon of Π does not factorise into interior $\otimes$ exterior, which is the type III signature and the reason there is no firewall:

Proposition 4.1 (Holographic–relativistic horizon (no firewall)). *The projection $\Pi\colon \mathbb{E}^3 \twoheadrightarrow \mathbb{C}$ creates a horizon that is at once* holographic *(information fully encoded on the boundary) and* relativistic *(the Galois conjugation $\varphi \leftrightarrow \bar{\varphi}$ exchanges the two sides like the two wedges of a Rindler spacetime): Π is surjective, with $\ker \Pi = \{(0,0,0)\}$ and $\Pi^{-1}(p) \neq \varnothing$ for every $p \in \mathbb{C}$. No averaging over microstates is needed: the microstate is one state and not an ensemble, as established in §3.4 (its realisation as a self-adjoint operator being §5)—so the AMPS firewall ambiguity [70] does not arise in this setting: with a single microstate there is no ensemble of typical states for the argument to average over (§1.2.4).*

P`[omega_flow_invariant, bridge_cocycle, bridge_inverse, bridge_compose]`

Three independent facts confirm the non-separability, and all three fall on $\frac{1}{2}$. At every scale, $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$ (**P**`[omega_flow_invariant]`). At every separation, the entanglement is $|\Omega(\tau_A)\bar{\Omega}(\tau_B)| = \mu_3^2 = \frac{1}{4}$, and this ER=EPR bridge is derived, not postulated: the operators $T(\sigma_1,\sigma_2)$ form a group (Proposition 3.16).

P`[bridge_cocycle, bridge_inverse, bridge_compose]`

The conjugation $\Theta = \mathrm{CPT}$, the antilinear Galois map $\varphi \leftrightarrow \bar{\varphi}$, swaps the wedges $P \leftrightarrow F$ and fixes C.

W6 — the antilinear Θ/CPT map (W01). Witten defines the Hilbert space non-perturbatively through an antilinear map Θ (a CPT operation) that renders the bilinear pairing Hermitian (W01, eq. 18). It is the Galois conjugation $\varphi \leftrightarrow \bar{\varphi}$: Θ swaps the two Rindler wedges and fixes the centre, with $\Theta^2 = 1$; it preserves the modulus, $\|\Theta z\| = \|z\|$, hence is antiunitary.

P`[cpt_swaps_P_to_F, cpt_fixes_C, cpt_involution, cpt_preserves_modulus]`

W7 — the $\mathrm{SO}(1,n{-}1)$ *obstruction (W01).* With N finite the de Sitter isometry group cannot act on the Hilbert space; being non-compact it has no finite-dimensional unitary representations (W01). Its place is taken by the discrete $\mathbb{Z}_3$ symmetry of the cube roots of unity with the scale invariance $|\Omega|_\sigma = \frac{1}{2}$ [1].

P`[eisenstein_cube, omega_conj]`

The whole geometry as a tensor: the warp generates curvature. This spacetime is tensorial in its entirety. The throat that compactifies the scale has a warped metric with radial coordinate $z(\sigma) = \varphi^\sigma$ and non-vanishing warp gradient: with $A = -\log z$, so $A' = -1$, the standard machinery—from g to Christoffel, to Riemann, to Ricci—turns that gradient into curvature computed, not assumed: $R_{\mu\nu} = -(A'' + d\,A'^2)\,g_{\mu\nu} = -4\,g_{\mu\nu}$, so the throat is an Einstein space (§A.1; **P**`[R_scalar_pcf, R_AB_einstein]`). The densification of the warp *is* the Ricci tensor. The extrinsic side closes it: the constant-σ leaves, their embedding and second fundamental form are tied to the intrinsic curvature by Gauss's equation—de Sitter is the hyperboloid whose intrinsic curvature is the wedge of the extrinsic curvature of the embedding, $R_{\mu\nu} = 3H^2 g_{\mu\nu}$, developed in full in §A.2 (**P**`[ds_ricci_from_gauss]`).

From that same geometry the particle emerges, discrete (type I).. The warp is sustained by a stress–energy tensor: Einstein's equation with source balances the throat geometry against a matter tensor whose spatial part is pressure. The tensor that in gravity describes the metric is the same that describes the tension—both words share the root *tendere*—and it is that tension that sustains the throat. Geometrically the throat is a densifying centre: on the Riemann sphere, with zero at one pole and infinity at the other, the golden tower produces concentric shells growing outward and a centre compressed without bottom, since $e^{2A} = z^{-2} = \varphi^{-2\sigma}$ diverges as $z \to 0$. The centre is superpoint $\mathbb{R}^{0|1}$-like, the object without metric or spin structure, with a single odd direction (§1.5.2). Its $\mathbb{Z}_2 = \{0,1\}$ grading distinguishes the even, bosonic directions (the "0") from the single odd, fermionic one (the "1"); the centre of the tension is that odd direction: fermionic. The involution $\alpha(x) = 1 - x$ swaps 0 and 1 and has no fixed point between them—which makes the binary register self-contradictory—but a unique fixed point at $x = \frac{1}{2}$: the extremes $\{0,1\}$ are the binary register—the bit, the boson, the fermion—and the self-conjugate midpoint $\frac{1}{2}$ is at once the modulus $|\Omega|$ and the fermion's *spin* $\frac{1}{2}$. The particle is thus coded by three readings of the same fact: $\frac{1}{2}$ (modulus, spin, fixed point of $1-x$); the $0|1$ of the fermionic superpoint; and the tension T_{AB} that produced it.

From this object follows its one-dimensional description: the worldline and its path integral, the Gaussian normalised by the half-factorial, $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, $(\frac{1}{2})! = \Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ (**P** `[gamma_half]`), computed where it is stated. The particle is then a type I operator: real discrete spectrum, minimal projections, integer dimension. Its internal structure is the spin-star,

$$\underbrace{S + E_1 + \cdots + E_N}_{\text{central+environment}} \xrightarrow{N=2} \underbrace{C + P + F}_{\text{PCF}}, \qquad F_{\max} = N^2 = 4. \tag{4.2}$$

with arithmetic $1+2=3$: the binary fermion, mediated by $\frac{1}{2}$, has become the tripartite particle. And the geometry that generated the particle fixes its charges: the eigenvalue triangle is the Eisenstein lattice $\mathbb{Z}[\omega] = A_2 = \mathfrak{su}(3)$; the Clifford latitude gives $SU(2)$; the Hopf fibre, $U(1)$; the weak mixing angle at unification is a ratio of mode counts, $\sin^2\theta_W|_{\text{GUT}} = N(0)/N(2) = \frac{3}{8}$, which runs to the measured 0.231 at M_Z through the MSSM β-functions (§A.3).

P `[gauge_dim_su3, hopf_from_clifford, weinberg_angle_gut]`

Geometry and particle are the same object. The order of these facts is fixed by the diagram. The projection $\Pi\colon \mathbb{E}^3 \to \mathbb{C}$ is the *root* of the whole downstream chain, and it is at once the geometry of spacetime—the sheet and torus projected—and the particle. This reflects the *self-similarity* of the object: the same modulus $\frac{1}{2}$ at every scale level, seeded by $\varphi^2 = \varphi + 1$ extending $i^2 = -1$ from the plane to space. The particle carries the same geometry in its information and, as it accumulates entropy, keeps it and moves within the spacetime geometry it itself is. There is no stage with geometry and a later one with the particle: they are the same object at two scale readings. It is this accumulation of information, carried by the self-similar geometry, that the next movement reads as an observer (§4.3): container and content, bulk and boundary, geometry and particle are not separate—the same non-separation the Möbius band already fixed.

W1 — the observer described by the theory (CLPW). CLPW require (§2.5) that the observer be described by the theory, not introduced from outside. Π is internal—no external region from which to look—and self-similar, so the observer is the geometry itself read at its centre. *Proof:* (4.1) with the bulk–boundary isometry of §3; Proposition 4.1; [2].

P `[cw3_observer_items_core]`

W2 — finite-dimensional Hilbert space (W01). The finite horizon area requires finite dimension; here the spectrum is the triangle of three eigenvalues and the mode count is integer. (4.2).

P `[witten_finite_rank]`

4.3. **The particle as observer.** What turns the particle into an observer is the accumulation of information, described here by an internal Wick rotation, not imposed. When the centre is taken as the time axis, its square changes sign, $(i\,t)^2 = -t^2$ (**P** `[wick_squares_flip_sign, lorentzian_signature_sum]`), and past and future become the two light cones, with the apex—the present—as observer (**P** `[C_fixed_PF_rotate, causal_trichotomy]`). The full metric is formalised in §4.4; here the algebra of the rotation, which is self-contained, suffices. The

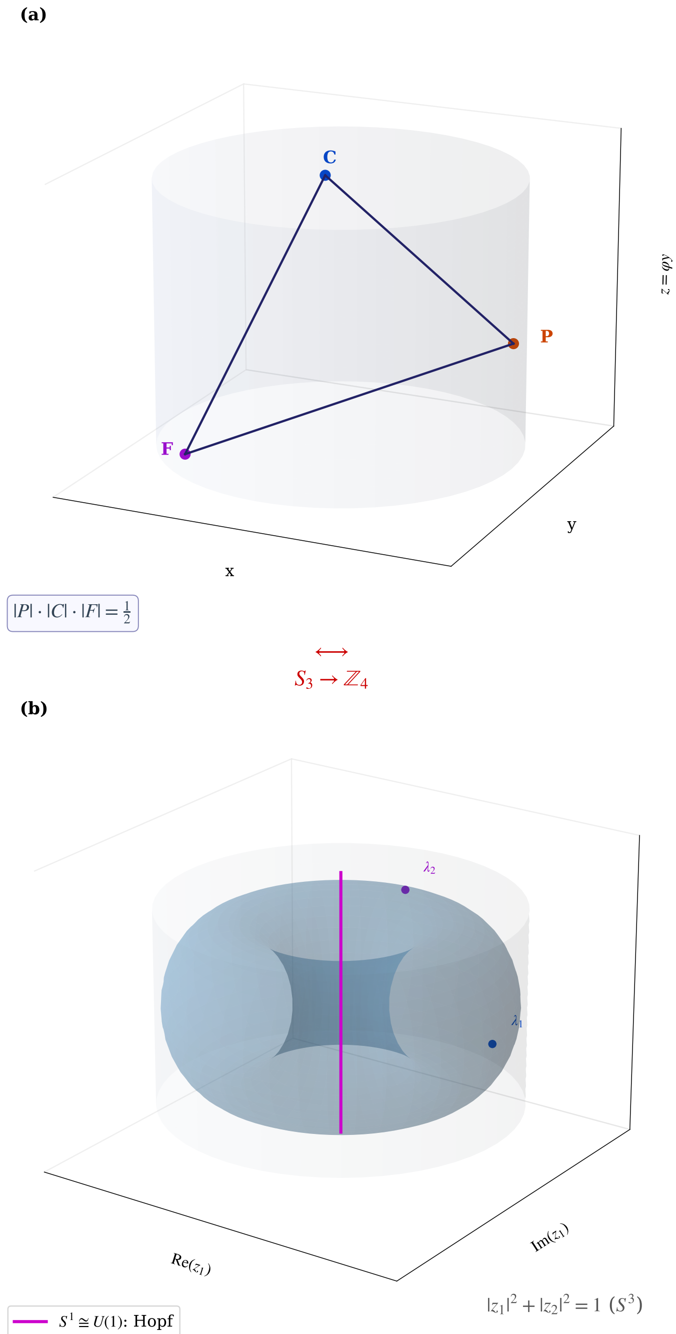

FIGURE 8. From $\varphi^2 = \varphi + 1$: (a) the cylinder C_0 with $\mathrm{P}, \mathrm{C}, \mathrm{F}$ at $120°$ ($\|P\|\,\|C\|\,\|F\| = \frac{1}{2}$) closes to (b) the torus $T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda$, $\tau = i$ ($|z_1|^2 + |z_2|^2 = \frac{1}{4}+\frac{3}{4} = 1$, the S^3 constraint), whose Clifford embedding $T^2 \hookrightarrow S^3$ and Hopf fibre give the gauge factors $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ of App. A.3.

observer accumulates: its Fisher information grows to the ceiling the Cramér–Rao bound permits,

$$\tau_F = \frac{\tau_D}{\sqrt{2f}}, \qquad \tau_F = \tau_D \iff f = \tfrac{1}{2}. \tag{4.3}$$

$$\mathrm{Var}(\theta) \geq F_\theta^{-1}, \qquad \underbrace{F_{\max}^{-1} = \tfrac{1}{4}}_{\text{Kiely: Cramér–Rao}} \quad = \underbrace{|\Omega|^2 = \mu_3^2 = \tfrac{1}{4}}_{\text{PCF area factor}}. \tag{4.4}$$

$$R_\delta \sim N \ \ \text{(all fragments agree)} \longleftrightarrow \text{objective record}, \tag{4.5}$$

$$F_\theta(t) \xrightarrow{N\to\infty} F_{\max}\big(1 - e^{-(t/\tau_F)^2}\big), \quad F_{\max} = 4, \qquad F_\Omega = 4\mu_3^2 = 1 \text{ bit}, \tag{4.6}$$

and sees exactly the half,

$$|P| \cdot |C| \cdot |F| = \tfrac{1}{2} = |\Omega|. \tag{4.7}$$

The "half" is geometric: the eigenvalue dipole places P at the south pole and F at the north, with C the equatorial axis ($|C| = 1$), and $|P|\,|F| = \frac{1}{4}$ while $|C| = 1$ gives the product $\frac{1}{2}$; the $\sqrt{3}$ of $|F|$ cancels that of $|P|$—a cancellation proper to the equilateral triangle, arity three (Proposition 2.19, Proposition 2.15: $\sigma + \mu = 2$, $\sigma\mu = \frac{3}{4}$ with unique solution $\mu = \frac{1}{2}$, $\sigma = \frac{3}{2}$.

W5 — the initial–final pairing observable from within (W01). Witten computes the pairing $\langle f|i\rangle$ as calculable but not observable from outside (W01). It becomes the internal pairing $\langle F|P\rangle = \bar{F}P/C$, with $|\langle F|P\rangle| = \frac{1}{2}$, observable because Π is inside and no external region remains.

W3 — the unitarity conjecture (W01). Witten conjectures that the Hermitian pairing is positive definite (W01). The spectrum is positive and $|\Omega| = \frac{1}{2} > \frac{1}{4}$, so the object does not collapse. (4.7).

The coincidence with Kiely's quantum Darwinism is forced, not numerical. The objectivity threshold is $f_{\text{crit}} = \frac{1}{2}$; the redundancy grows as the mode number; the certainty condition is $\varepsilon_0 M = \pi$:

$$\underbrace{f_{\text{crit}} = \tfrac{1}{2}}_{\text{Kiely: objectivity}} \quad = \underbrace{|\Omega| = \mu_3 = \tfrac{1}{2}}_{\text{PCF}}, \tag{4.8}$$

$$\varepsilon_0 \cdot M_{\text{PCF}} = \pi \qquad \text{(cell capacity } = \pi \text{ bits)}, \tag{4.9}$$

Proposition 4.2 (No separable parts before observation)**.** *For a pure two-party state let $p_1 \geq p_2 \geq \cdots \geq 0$ be the eigenvalues of the reduced state, $\sum_i p_i = 1$ (these are the squared Schmidt coefficients, $p_i = \lambda_i^2$; the criterion below holds in this convention and not for the λ_i themselves). The entanglement modulus is the top eigenvalue, $|\Omega| = p_1$, and it satisfies $|\Omega| = |\Omega|^2$ exactly when $p_1 = 1$—that is, when every other p_i vanishes, the reduced state is pure, the Schmidt rank is one and the state is a product—and $|\Omega| > |\Omega|^2$ otherwise. The No-Diagonal fixes $|\widehat{\Omega}| = \frac{1}{2}$ in the balanced (maximally entangled) case (Proposition 2.15), so*

$$|\Omega| = \tfrac{1}{2} \ > \ \tfrac{1}{4} = |\Omega|^2, \qquad H(\tfrac{1}{2}) = 1 \ \textit{bit},$$

the strict inequality being the signature of non-factorizability; for a product state it collapses to $|\Omega| = |\Omega|^2 = 1$, $H = 0$ (the Schmidt decomposition [71]) . Hence before observation there are no separable subsystems carrying **P**`[no_separable_parts, separable_iff_top_eigenvalue_one]` *values: what is real is the non-local whole, recovered as $|\widehat{\Omega}| = \frac{1}{2}$. PCF therefore predetermines no parts, so it is neither a local-realist (Bell) nor a parts-realist (Leggett) hidden-variable model, and the no-go theorems of [72] and [73] do not bind it.*

The No-Diagonal theorem (§2.3) establishes that every self-referential structure without collapse must produce $\mu^2 = \frac{1}{4}$ and the tripartite decomposition, with no other solution—which forces each identification. In the diagram, here the first apex converges: three independent routes—the bridge (§4.2), the triangle norm and the Fisher metric—give the same $\frac{1}{2}$. When three independent chains reach the same number, the diagram commutes at that point.

4.4. **The tower of information, and the emergence of curved spacetime.**

The tower (types I and II interwoven). The accumulating observer does not select one reality among many: it builds it. There is no landscape of vacua to choose from, no statistical ensemble to average. This homogeneity—the same structure at every point—is what the field equations require: it is the tensor covariance of §4.2 now read as a constructive principle, and its full development comes below. The entropy of one bit per cell, $H(\frac{1}{2}) = 1$ (**P**`[binary_entropy_half]`), is the maximally mixed state—$\rho_{\max} = 1$ in CLPW's language—describing the de Sitter vacuum, and from its overflow the geometry is born.

When the π-bit cell fills, the throat opens:

$$S(\sigma) = \pi\varphi^{\sigma}, \tag{4.10}$$

with $N_{\text{modes}}(\sigma) = \lfloor \pi\varphi^{\sigma} \rfloor$. As it overflows, the throat count is set:

$$\frac{|\partial_\sigma V|}{V} = \ln\varphi = c_{\text{PCF}}. \tag{4.11}$$

The UV fixed point forces the invariants:

Assumption 4.3 (The gauge–tower bridge)**.**

$$\beta_g(g^*, \lambda^*) = 0 \iff \varepsilon_0 M_{\text{PCF}} = \pi. \tag{4.12}$$

The right-hand side is derived and triply proved—in §2.4 as (2.21), here as **P**`[eps0_Mpcf_pi]`, and checked numerically **N**`[eq:certainty]`; the equivalence itself is this section's one input, the bridge between the gauge running and the tower's algebra.

The quantisation is in powers of φ, and the identity that closes it is $\varphi^{-\lambda} = \frac{1}{2} \Leftrightarrow \varphi^{\lambda} = 2$: the binary base comes out of the golden geometry. Here the string reappears, not as a new object but as the same particle in its information register: the dimension of the state space at each level is $\dim\mathcal{H}(\sigma) = \lfloor S(\sigma) \rfloor$.

W4 — the entropy conjecture (W01). Witten conjectures that the Hilbert space is finite-dimensional with $S = \ln N$ (W01). The Hilbert dimension per level is the integer part of the horizon entropy, $\dim\mathcal{H}(\sigma) = \lfloor S(\sigma) \rfloor$, with $N_{\text{modes}} = \lfloor \pi\varphi^{\sigma} \rfloor$—the constructive form of Witten's $S = \ln N$. (by `rfl`—the most direct realisation).

P`[witten_entropy_realized, witten_finite_rank, witten_two_conjectures]`

The entropy branch is asserted strongly (tight); the unitarity branch is "realised".

The flow carrying one level to another, $\sigma \mapsto \sigma + t$, fixes $\tau = i$ and preserves $\frac{1}{2}$: it is the tracial flow, the type II signature, and it is the thermal flow of the static patch at the Gibbons–Hawking temperature.

P`[tower_scale_group, tower_flow_fixes_tau, omega_flow_group, omega_flow_invariant]`

W11 — the type II algebra by crossed product (CLPW, completed). CLPW build the type II_1 algebra from the type III by the crossed product with the modular flow (§1.2–1.3). The scale flow preserves $\frac{1}{2}$ as a trace, which is the type II signature; the section proves the flow and takes from CLPW the name of the crossed-product classification.

P`[omega_flow_invariant, tower_scale_group]`

W12 — the maximally mixed vacuum state (CLPW §2.4, developed). CLPW identify the de Sitter vacuum with the maximally mixed state, $\rho_{\max} = 1$. One bit per cell, $H(\frac{1}{2}) = 1$, gives that $\rho = 1$ of maximal entropy.

P`[binary_entropy_half]`

The heart of the movement is that bulk and boundary are the same thing at once. The bulk is the three-dimensional projection; the boundary, the two-dimensional plane; and they are the same II of the first movement. The two accumulations—of time (Fisher) and of scale (tower)—are one:

$$\tau_F(\sigma) = \tau(\sigma) = M_{\text{PCF}}\,\varphi^{-\sigma}, \qquad S(\sigma)\,\tau_F(\sigma) = \pi M_{\text{PCF}} \quad (\text{constant in } \sigma). \tag{4.13}$$

The two lengths the tower carries are conjugate: the throat coordinate $z(\sigma) = \varphi^\sigma$ grows with the level and the weld time $\tau(\sigma) = M_{\mathrm{PCF}}\varphi^{-\sigma}$ shrinks with it, and their product is constant,

$$z(\sigma)\,\tau(\sigma) = M_{\mathrm{PCF}} \qquad \text{for every } \sigma. \tag{4.14}$$

This is the relation $R \cdot (\alpha'/R) = \alpha'$ with $\alpha' = M_{\mathrm{PCF}}$: the compactification radius is the throat coordinate, the lattice spacing is its T-dual, and α' is *fixed by the product* rather than chosen. Short and long distance are therefore the same family exchanged by duality, not two regimes with a limit between them—the structural reason there is no ultraviolet regulator to remove (Remark 5.39).

P`[lattice_radius_conjugate]`

$$\underbrace{F_\Omega \cdot N_{\mathrm{modes}}}_{\text{bits (Shannon)}} = \underbrace{S(\sigma) = \pi\varphi^\sigma}_{\text{tower}} = \underbrace{N_{\mathrm{modes}} = \lfloor S(\sigma)\rfloor}_{\text{matter}} = \underbrace{S_{\mathrm{BH}} = \frac{A}{4G_N}}_{\text{horizon}} = \underbrace{S_{\mathrm{Jacobson}}}_{\delta Q = T\delta S}, \tag{4.15}$$

This identification—the *weld*—is where four routes converge, and it is the only point where the three types coincide in one object: type I by the discrete tower, type II by the observer's time, type III because it lives on the spacetime. It is the maximal apex of the diagram. Local and global are not separate: $\frac{1}{2}$ at each level (local) and invariant under the flow (global), the same invariant read two ways. Its last two branches close below.

Curved spacetime (type III again). The accumulated entropy now sources gravity. Each bit costs—the Landauer cost, which is the regulator $R_K = \log\varphi$ of $\mathbb{Q}(\sqrt{5})$ (Definition 2.18) measured in $\lambda_{\log}$, Theorem 2.59, $\lambda \cdot R = \log 2$—and when that cost is imposed as an equation of state $\delta Q = T\,\delta S$ across every local Rindler horizon, the result is, by Jacobson's argument, the Einstein equation:

$$\text{energy per bit} = \frac{1}{M_{\mathrm{PCF}}}, \qquad S_{\mathrm{BH}}/k_B = \lambda_{\log} \cdot R_K = \log 2 \ \ \text{(one bit; Theorem 2.59)}, \tag{4.16}$$

$$\delta Q = T\,\delta S \implies \text{the Einstein equation}, \tag{4.17}$$

$$R_{AB} = -4\,g_{AB}, \quad R = -20, \qquad G_{AB} + \Lambda_5\, g_{AB} = \kappa\, T_{AB}, \quad \kappa = 8\pi G_N, \tag{4.18}$$

with $G_N = \mu_3 = \frac{1}{2}$. Here the homogeneity of §4.2 is redeemed: Jacobson's argument requires a local horizon with the same law at every point—exactly what tensor covariance guarantees; without it the step does not exist. The matter is the mode number:

$$T^{\mathrm{YM}}_{AB} = F_{AC}F_B{}^C - \tfrac{1}{4}\,g_{AB}F_{CD}F^{CD}, \qquad \text{matter} = N_{\mathrm{modes}} = \lfloor S(\sigma)\rfloor. \tag{4.19}$$

and here the two open branches of the identity close: the horizon entropy and Jacobson's were, all along, the same number as the bits and the modes. Each mode is one bit exactly at the modulus: the per-eigenvalue ceiling $F_\Omega = 4\mu_3^2$ is unity iff $\mu_3 = \frac{1}{2}$, and the binary entropy there is $H(\frac{1}{2}) = 1$ bit.

P`[fisher_unit_iff, H_half_one_bit]`

That matter does not merely sit on the background: it curves it. The back-reaction is fixed by the same modulus, with no free parameter, and it is what makes the shared root of *metric* and *tension* (§4.2) a computation rather than an etymology.

Definition 4.4 (Shell tension)**.** A thin matter shell at the discrete level $w_\sigma = \sigma\log\varphi$ of the throat (3.18) has *tension* λ_σ: its energy density per unit volume of the constant-w leaf. Its value is pinned by Proposition 4.6.

P`[prop:shell-tension]`

Proposition 4.5 (The energy per bit is constant along the tower)**.** *For every level,*

$$\frac{\varepsilon(\sigma)}{S(\sigma)} = \frac{\varepsilon_0}{\pi} = \frac{1}{M_{\mathrm{PCF}}}, \tag{4.20}$$

equivalently $S(\sigma) = M_{\mathrm{PCF}}\,\varepsilon(\sigma)$: the first law with temperature $T = 1/M_{\mathrm{PCF}}$.

P*`[energy_per_bit_constant]`*

Proof. The per-level energy is $\varepsilon(\sigma) = \varepsilon_0 \varphi^\sigma$ (4.6) and the saturation entropy is $S(\sigma) = \pi\varphi^\sigma$ (3.12); the quotient cancels φ^σ and leaves ε_0/π, which is $1/M_{\rm PCF}$ by the certainty relation (2.21). The cancellation is what the common base buys: an energy growing at any other rate would leave a σ-dependent residue. □

Proposition 4.6 (The shell tension carries no free parameter)**.** *The tension of Definition 4.4 is fixed by the level's mode count,*

$$\lambda_\sigma = \frac{N_{\rm modes}(\sigma)}{M_{\rm PCF}}. \tag{4.21}$$

P*[shell_tension_determined]*

Proof. Each tower mode is one bit: $N_{\rm modes}(\sigma) = \lfloor S(\sigma) \rfloor$ (4.19) is one mode per bit of horizon entropy, and the Kiely ceiling $F_\Omega = 4\mu_3^2 = 1$ (4.15) makes each mode exactly one bit, neither more nor less. The shell's tension is then the energy per bit times the number of bits, $\lambda_\sigma = \varepsilon(\sigma) N_{\rm modes}(\sigma)/S(\sigma)$, which is $N_{\rm modes}(\sigma)/M_{\rm PCF}$ by Proposition 4.5. □

Proposition 4.7 (Israel junction: the back-reaction of each level)**.** *A constant-w leaf of* (3.18) *has extrinsic curvature $K_{\mu\nu} = A'(w) g_{\mu\nu}$, so a shell of tension λ_σ imposes the jump*

$$[A'(w_\sigma)] = -\frac{8\pi G_5}{3}\,\lambda_\sigma = -\frac{4\pi}{3}\,\lambda_\sigma, \qquad G_5 = \mu_3, \tag{4.22}$$

and with (4.21) *the jump is determined level by level, $[A'(w_\sigma)] = -\frac{4\pi}{3} N_{\rm modes}(\sigma)/M_{\rm PCF}$. Between shells the metric is exact* AdS$_5$*: a steeper warp is a deeper curvature. for the arithmetic; the junction condition itself [74].*

P*[israel_prefactor, israel_jump_determined]*

Proof. The five-dimensional junction condition for a thin shell is $[K_{\mu\nu}] = -8\pi G_5\big(S_{\mu\nu} - \frac{1}{3} h_{\mu\nu} S\big)$; for pure tension $S_{\mu\nu} = -\lambda h_{\mu\nu}$ this reduces to $[A'] = -\frac{8\pi G_5}{3}\lambda$. With $G_5 = \mu_3$ (3.21) the prefactor collapses to $4\pi/3$—and only there: any other Newton constant leaves $8\pi G_5/3 \neq 4\pi/3$. □

Remark 4.8 (The cumulative back-reaction). Each level kinks the warp—and deepens the local $|\Lambda|$—by an amount set by its mode count, so the cumulative back-reaction up to level k is $\sum_{\sigma \le k} N_{\rm modes}(\sigma)$, giving $3, 8, 16, 29, 50, 84, 140$ for $k = 0, \ldots, 6$. The gravitational side and the string side are here the same computation: (4.22) ties the extrinsic curvature to the shell tension, and (4.21) ties that tension to the mode count, so what curves the geometry is the number of bits the level carries.

P`[backreaction_cumulative]`

N`[eq:shell-tension]`

The cosmological constant is determined, not chosen:

W8 — a theory that determines Λ *(W01).* The abstract of W01 states that general relativity must be derived from a theory that *determines* the value of Λ. Here Λ comes from the curvature, with zero free parameters: $\Lambda_5 = -d(d-1)/2\ell^2 = -6$, with $\ell = 1$ from (3.23) (§3) and $d = 4$ from the arity (§2), and the G–Λ duality $\varphi^{-6} \cdot \varphi^6 = 1$ (**P**`[G_Lambda_duality, R_scalar_pcf, BF_value]`). Between the gravity scale and the Λ scale there is an exact interval:

It remains to show the spacetime is de Sitter. The metric obtained is Euclidean anti-de Sitter; the same internal Wick rotation that made the centre the time axis carries it to de Sitter, with $\varepsilon(\sigma) = \varepsilon_0 \varphi^\sigma$ and λ the length that converts the tower's logarithmic step into proper length along the scale axis, so that $X_4 = \lambda \ln s$ is the fifth, flat coordinate of §A.2:

$$ds^2 = \underbrace{(dx^2 + dy^2 + dz^2)}_{\mathbb{E}^3} - \underbrace{c^2\, dt^2}_{\text{Wick } (C)} + \underbrace{\lambda^2\, d(\ln s)^2}_{\varphi^\sigma\text{-tower}}, \qquad s = \varphi^\sigma, \tag{4.23}$$

It is the bridge to de Sitter that Maldacena anticipated, built here by internal analytic continuation rather than an imposed duality (**P**`[ets_is_ds_zero_h]`). Past and future are the poles of a dipole, with C at the equator—the two conjugate sheets of de Sitter.

W10 — the antipodal map / meta-observables (W01). The antipodal singularity Witten notes corresponds to F being antipodal to P in the dipole, which resolves Strominger's obstruction geometrically. *Proof:* the dipole of (4.23); [1] (§A.2).

The geometric closure is the connection the framework contributes. The 5D ambient is flat—its curvature, computed, vanishes, $R^{(5)} = 0$ (**P**`[ets_constant_block_christoffels_vanish]`)—and the fifth coordinate, the scale axis, is the extra direction an embedding needs. De Sitter is the hyperboloid $-X_0^2 + \cdots + X_4^2 = 1/H^2$ embedded in that flat ambient, with intrinsic curvature = wedge of the extrinsic by Gauss: $R_{\mu\nu} = 3H^2 g_{\mu\nu}$ (**P**`[ds_ricci_from_gauss, ds_einstein_lambda]`). The planar patch covers exactly half the hyperboloid—$X_0 + X_4 = \ell\, e^{t/\ell} > 0$ never changes sign (**P**`[ds_covers_half_hyperboloid]`)—and that half *realises* the observer's half $|\Omega| = \frac{1}{2}$ by the shared value (**P**`[observer_half_from_norms]`). The assertoric level splits: strong on the covering, "realises" on the identification.

W9 — the thermal nature of the static patch (Gibbons–Hawking). This is the classical Gibbons–Hawking result, prior to W01 and CLPW: the de Sitter horizon radiates and the static patch is a thermal state at $T = H/2\pi$. The observer's modular flow is the tracial flow of the tower, and the static patch is thermal at that temperature. (§A.2) [75].

P`[dS_temperature, ds_beta_reciprocal]`

The formal statements underlying the accumulation and gravity movements follow; their proofs are those of the corresponding **P**`[...]` tags.

Remark 4.9 (The role of φ in the observer). $\mu_3 = \frac{1}{2} = \varphi^{-\lambda_{\log}}$; the three invariants ($f_{\text{crit}} = \frac{1}{2}$, $F_{\max}^{-1} = \frac{1}{4}$, $n = 3$) are φ-facts of §2–§3, not fitted values.

Remark 4.10 (The role of φ in the objectivity threshold). $|\Omega| = \frac{1}{2} = \mu_3 = \varphi^{-\lambda_{\log}}$: the half of de Sitter, Kiely's threshold and the modulus are one φ-fact; the throat that observation builds is the tower φ^{σ}.

Remark 4.11 (The role of φ in the accumulation). $\varepsilon_0 = \ln\varphi/(6\sqrt{3})$ and $M_{\text{PCF}} = 6\sqrt{3}\,\pi/\ln\varphi$ give $\varepsilon_0 M_{\text{PCF}} = \pi$: the clock of the accumulation, the ultraviolet fixed point and the certainty principle are one φ-fact; the accumulation tower is φ^{σ}.

Remark 4.12 (Past and future are fixed dynamically). The reading of C-P-F as present / past / future is not a free labelling but a consequence of the causal structure. The Wick rotation makes the centre C the time axis and gives the metric of (4.23) its Lorentzian signature $(+,+,+,-)$, so it carries a light cone: C is its apex—the observer's present—and P, F are the past- and future-directed cones, rotating in opposite senses about the fixed, real C, with Θ exchanging them. The direction of the φ-driven accumulation along the throat—from the origin toward the ceiling (Proposition 4.13)—orients the cone, singling out one direction as the future. The past/future assignment therefore follows from the causal cone and the direction of accumulation, not from convention.

Proposition 4.13 (Accumulation and ceiling)**.** *The metrological quantum Fisher information accumulates as $F_\theta(t) \to F_{\max}\big(1 - e^{-(t/\tau_F)^2}\big)$ with $F_{\max} = 4$ [76], so the Cramér–Rao ceiling $F_{\max}^{-1} = \frac{1}{4} = \mu_3^2$ equals the holographic area factor. The Fisher information per eigenvalue is $F_\Omega = 4\mu_3^2 = 1$ bit; the three eigenvalues give $3F_\Omega = 3 = c$, the central charge of §3.1; and the objectivity threshold $f_{\text{crit}} = \mu_3 = \frac{1}{2}$ is the maximum-entropy point $H(\frac{1}{2}) = 1$ bit (Proposition 2.61).*

P*`[binary_entropy_half, S_tower_recurrence]`*

N*`[prop:entropy-max]`*

Theorem 4.14 (Time–scale conjugacy)**.** *At the operating point $f = \mu_3 = \frac{1}{2}$ the metrological timescale equals the tower period, $\tau_F(\sigma) = \tau(\sigma) = M_{\text{PCF}}\,\varphi^{-\sigma}$. Hence*

$$S(\sigma)\,\tau_F(\sigma) = (\pi\varphi^{\sigma})\,(M_{\text{PCF}}\,\varphi^{-\sigma}) = \pi M_{\text{PCF}}$$

is constant in σ: the certainty principle $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ ((2.21)) promoted from a single cell to the generation/clock pair.

P[obs_weld]

Proposition 4.15 (The bulk metric is an Einstein space)**.** *For the warped throat (3.18) with $A = -\log z$ (anti-de Sitter, $\ell = 1$), explicit computation of the Levi-Civita curvature gives, uniformly in all five components,*

$$R_{AB} = -4\, g_{AB}, \qquad R = -20, \qquad G_{AB} = 6\, g_{AB} \quad (A, B = 0, \ldots, 4),$$

so the metric solves $G_{AB} + \Lambda_5\, g_{AB} = 0$ with $\Lambda_5 = -6$, and the Breitenlohner–Freedman bound $m^2_{\mathrm{BF}} = -d^2/4 = -4$ holds, so the discrete tower is stable. With a source, $G_{AB} + \Lambda_5\, g_{AB} = \kappa\, T_{AB}$ and $\kappa = 8\pi G_N$; the PCF value $G_N = \mu_3 = \frac{1}{2}$ gives $\kappa = 4\pi$ (effective $G_4 = \frac{1}{4}$), and $\nabla^A G_{AB} = 0$ holds identically, so T_{AB} is conserved.

P[G_AB_pcf, R_AB_einstein, R_scalar_pcf, einstein_coeff, ricci_coeff, ricci_scalar, BF_value]

Theorem 4.16 (Jacobson's construction on the bulk metric)**.** *Since (3.18) is an Einstein space, $R_{ab}k^a k^b = -4\, g_{ab}k^a k^b = 0$ for every null k; in vacuum the Clausius relation $\delta Q = \int T_{ab}\chi^a\, d\Sigma^b = 0$ forces $\delta S = 0$ (Jacobson's entanglement equilibrium [77]), which coincides with the ceiling $F^{-1}_{\max} = \mu_3^2$ of Prop. 4.13. The normalisation $\eta = 1/4G_N$ with $G_N = \mu_3$ fixes the area factor $\mu_3^2 = \frac{1}{4}$ and the coupling $2\pi/\eta = 8\pi G_N$; with matter, $\delta Q = T\,\delta S$ on every local Rindler horizon yields the Einstein equation [77, 78].*

P[ds_einstein_lambda, R_AB_einstein, G_AB_pcf]

Proposition 4.17 (Matter is the saturation entropy)**.** *Each interior level carries $N_{\mathrm{modes}}(\sigma) = \lfloor \pi\varphi^\sigma \rfloor = \lfloor S(\sigma) \rfloor$ scalar modes, each contributing one Shannon bit, so the matter content of level σ is the saturation entropy $S(\sigma) = \pi\varphi^\sigma$. That this is an* equality *and not merely a count is the first law of the tower: the Kiely ceiling $F_\Omega = 4\mu_3^2 = 1$ makes each mode exactly one bit (Proposition 4.13), the per-level energy is $\varepsilon(\sigma) = \varepsilon_0\varphi^\sigma$ with $\varepsilon_0 M_{\mathrm{PCF}} = \pi$, so the energy per bit is constant across the tower (Proposition 4.5),*

$$\frac{\varepsilon(\sigma)}{S(\sigma)} = \frac{\varepsilon_0}{\pi} = \frac{1}{M_{\mathrm{PCF}}}, \tag{4.24}$$

which is $S(\sigma) = M_{\mathrm{PCF}}\,\varepsilon(\sigma)$ at temperature $T = 1/M_{\mathrm{PCF}}$ The colour octet has $\dim \mathfrak{su}(3) = n^2 - 1 = 8$ at the arity $n = 3$ fixed by the golden central chain $\varphi^2 + \varphi^{-2} = 3$; the full $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ follows from the torus T^2_{PCF} (2.19), its gauge factors landing at the force levels by increasing dimension $1 < 3 < 8$: $\mathrm{U}(1)$ is the Hopf fibre S^1 at $\sigma = 3$ (EM), $\mathrm{SU}(2)$ the Clifford S^3 at $\sigma = 4$ (weak), and $\mathrm{SU}(3)$ the A_2 roots of the Eisenstein face (Fig. 5) at $\sigma = 5$ (strong), with $\dim \mathfrak{su}(3) = n^2 - 1 = 8$ at the arity $n = 3$ fixed by the golden central chain $\varphi^2 + \varphi^{-2} = 3$.

P[gauge_dim_su3, phi_central_chain, central_chain, KK_numerator, clifford_S3_condition, hopf_from_clifford, a2_minimal_vectors_exactly_six]

The electroweak data follow from the same modulus: the weak mixing angle at unification is $\sin^2\theta_W|_{\mathrm{GUT}} = \frac{3}{8} = N(0)/N(2)$ **P**[weinberg_angle_gut] *running to 0.231 at M_Z; the tower entropy ratio $S(3)/S(6) = \varphi^{-3}$* **P**[entropy_ratio_S3_S6] *marks the breaking $\mathbb{Z}_3 \to \mathbb{Z}_4$ at the tower midpoint $\sigma = 3$, and the G–Λ duality reads $\varphi^{-6}\varphi^{+6} = 1$. The generations are not placed radially by this σ-assignment: the charged-lepton ratios are not integer powers of π ($m_\mu/m_e = \pi^{4.66}$), and a π-graded radial triple cannot satisfy the Koide relation. Two statements are at work here, of different objects: the Koide ratio $Q = \frac{2}{3}$ is an algebraic identity of that placement form, holding for every δ and hence providing no evidence for any particular value; the prediction is the three masses themselves, reconstructed to $\sim 10^{-4}$ from the single scale $\sqrt{M}$ with $\delta = 2/n^2$ fixed by the arity, a value so sharply selected that a 0.1% departure degrades the fit by two orders of magnitude. The σ-levels label sectors and forces; the three generations are the* angular *ω-triple within a sector, placed in the bridge with no free angular parameter, $\sqrt{m_k} = \sqrt{M}\,(1 + \sqrt{2}\cos(\delta + 2\pi k/3))$ with $\delta = 2/n^2 = \frac{2}{9}$ at arity $n = 3$, reconstructing the three charged-lepton masses to $\sim 10^{-4}$ from the single scale $\sqrt{M}$. That scale is* *not an independent input: the certainty condition* $\varepsilon_0 M = \pi$ (4.8) *is the same that fixes* M_{PCF}*, so* $M = M_{\mathrm{PCF}}$*; Theorem 5.33 then derives the colour scale and running from it. What remains open is only the quark spectrum and its CKM mixing, not the scale or the colour running.*

P[M_eq_Mpcf]

Their Yang–Mills stress-energy $T^{\mathrm{YM}}_{AB} = F_{AC}F_B{}^C - \frac{1}{4}g_{AB}F_{CD}F^{CD}$ *is conserved* ($\nabla^A T^{\mathrm{YM}}_{AB} = 0$) *and traceless on the* $D = 4$ *worldsheet (conformal there).*

Theorem 4.18 (The Landau–Lifshitz energy of the field)**.** *On the de Sitter background* (4.23) *with* $G_N = \mu_3 = \frac{1}{2}$*, the Landau–Lifshitz complex* $(-g)(T^{\mu\nu} + t^{\mu\nu}_{\mathrm{LL}}) = \frac{1}{16\pi G_N}\partial_\alpha\partial_\beta H^{\mu\alpha\nu\beta}$ *has vanishing energy density—its* 00 *component is zero, the net energy of matter and field balancing, the classical face of the entanglement equilibrium* $\delta S = 0$ *of Theorem 4.16—while its spatial components* $-2H^2 e^{4Ht}/\pi$ *do not vanish, as de Sitter is not static (the standard complex differs from the physical vacuum pressure by the* Λ*-term the pseudotensor requires for* $\Lambda \neq 0$*). The gravitational energy is a definite boundary charge: the energy in the Hubble volume* $\rho_\Lambda V_H = 1/H = T_{\mathrm{GH}}\, S_{\mathrm{GH}}$ *is the first law of de Sitter, and the Komar/Landau–Lifshitz charge of the horizon* $\kappa A/(8\pi G_N)$ *gives the same* $1/H$*. The modulus* $\mu_3 = \frac{1}{2}$ *is the ratio* $T_{\mathrm{GH}}/T_{\mathrm{local}}$ *of horizon and local temperatures. The gravitational field energy—absent from* $\nabla^A T_{AB} = 0$ *(Proposition 4.17)—is thus fixed, the energy conjugate to the tower entropy* $S(\sigma)$ *by the first law.*

P[ds_first_law, energy_is_inv_H, komar_first_law, mu3_temp_ratio, LL_spatial_ne_zero]

N[eq:obs-landauer]

Remark 4.19 (The G–Λ interval: Λ is determined, not selected)**.** The cosmological constant is not an input. The scale $\ell = 1$ (from the S-duality fixed point $\tau = i$, (3.23)) and the dimension $d = 4$ (from the arity $n = 3$, §2.3) are both derived, and they fix

$$\Lambda_5 = -\frac{d(d-1)}{2\ell^2} = -6, \tag{4.25}$$

with zero free parameters. The 6 arrives by two independent routes—the curvature magnitude $|\Lambda_5| = d(d-1)/2$ and the tower ceiling $\sigma_\Lambda = 2n$ of (4.26)—and the routes agree only at the arity: $(n+1)n/2 = 2n$ forces $n = 3$.

P[two_routes_to_six]

This realises precisely what Witten [1] identified as necessary: a theory "that determines the value of the cosmological constant."

The tower levels. The activation levels of the three sectors are read off the arity, not postulated:

$$\sigma_G = n - 1 = 2, \qquad \sigma_{\mathrm{EM}} = n = 3, \qquad \sigma_\Lambda = 2n = 6. \tag{4.26}$$

Gravity activates at $\sigma_G = n - 1$ (the spatial bulk $\mathbb{E}^3$ complete, one dimension below the bulk); the electromagnetic sector sits at the midpoint $\sigma_{\mathrm{EM}} = n$ (the Page point that halves the tower $[0, 2n]$); and Λ activates at the ceiling $\sigma_\Lambda = 2n$. The ceiling is geometric: the torus T^2_{PCF} is a *complex* space, so complex dimension n is real dimension $2n$—"complex dimension three" is $\mathbb{R}^6$, never $\mathbb{R}^3$. The tower runs over the full real space $\sigma \in \{0, 1, \dots, 2n\}$, and its ceiling $\sigma_\Lambda = 2n$ is that real dimension.

The interval is the spacetime. The gap between the gravity and Λ levels is exactly the dimension of spacetime,

$$\sigma_\Lambda - \sigma_G = 2n - (n-1) = n + 1 = 4 = \dim(M^4), \tag{4.27}$$

and the interval carries the two holographic invariants of the microstate:

$$\frac{\sigma_{\mathrm{EM}} - \sigma_G}{\sigma_\Lambda - \sigma_G} = \frac{1}{n+1} = \frac{1}{4} = |\Omega|^2 \qquad \text{and} \qquad \frac{\sigma_{\mathrm{EM}} - \sigma_G}{\sigma_\Lambda - \sigma_{\mathrm{EM}}} = \frac{1}{n} = \frac{1}{3} = \|P\|^2, \tag{4.28}$$

the holographic area factor $\frac{1}{4} = |\Omega|^2$ and the past-component weight $\frac{1}{3} = \|P\|^2 = F_\Omega/c$.

Two independent routes to the same 6. The curvature magnitude $|\Lambda_5| = d(d-1)/2 = 6$ (from $d = 4$) and the tower ceiling $\sigma_\Lambda = 2n = 6$ (from $n = 3$, as the real dimension of the complex torus) coincide by independent routes—5D curvature on one side, the real dimension of the worldsheet torus on the other. Both are the entropy saturation of the microstate read geometrically: $\sigma_\Lambda = 2n = 6$ is the ceiling of the golden tower $S(\sigma) = \pi\varphi^\sigma$ whose accumulated bits are the horizon entropy of (4.15) and, through (4.18), the field equations—the six-fold value is that saturation, not a separate geometric fact. Together with $6 = |S_3| = 3!$ (the order of the symmetry group, Lem. 2.66) and the exponent of the G–Λ duality $\varphi^{-6}\varphi^{+6} = 1$, the value $\sigma_\Lambda = 6$ is fixed from four sides.

Scope. This we propose as the structure Witten required: the de Sitter Hilbert-space dimension N is integer, so $G\Lambda^{(n-2)/n}$ cannot vary continuously and both G and Λ must be fixed by a more fundamental theory [1]; here $G_N = \frac{1}{2}$ and $\Lambda \sim \varphi^{-6}$, the product fixed, $\varphi^{-6}\varphi^{+6} = 1 \iff G\Lambda \sim M_{\rm Pl}^{-4}M_{\rm Pl}^4 = 1$ (PCF units). What is determined is this dimensionless conjugacy. The translation to the observed $\Lambda_{\rm obs} \sim 10^{-122}M_{\rm Pl}^4$ is a matter of scale translation, not derived here; and it is a *single* open item rather than two. The level the observer occupies is not an independent parameter: the tower fixes it from the de Sitter entropy, and that entropy is $1/\Lambda$, so

$$\sigma_{\rm obs} = \frac{\ln(S_{dS}/\pi)}{\ln\varphi}, \qquad S_{dS} = \frac{3\pi}{G_N\Lambda} \overset{G_N=1/2}{=} \frac{6\pi}{\Lambda} \quad\Longrightarrow\quad \sigma_{\rm obs} = \frac{\ln(6/\Lambda_{\rm obs})}{\ln\varphi}. \tag{4.29}$$

Deriving the scale translation therefore derives the level, and conversely; counting them as two debts would be an accounting error. The scale $\sqrt{M}$ that sets the charged leptons (§4.17) is therefore the tower reading of this de Sitter entropy, and Theorem 5.33 then shows the colour sector hanging on the same scale: one scale surfaces as the string constant of §3 and as the entropy here, the dimensionless placement already fixed, the absolute value alone carrying the observer's Λ. What the tower does add is granularity: its step is exactly φ ($\log_{10}\varphi = 0.209$ per level), so the absolute scale is constrained to a discrete ladder rather than a continuum.

N [eq:Lambda-from-curvature]
P [lambda_magnitude_eq_level, sigma_G_val, sigma_EM_val, sigma_Lambda_val, threshold_ordering, interval_gap, interval_gap_general, em_holographic_fraction, em_asymmetry, entropy_at_mu_three, tower_step_ratio]

Proposition 4.20 (The level triple is unique over the integers)**.** *Consider integer triples $\sigma_G < \sigma_{\rm EM} < \sigma_\Lambda$ subject to the four conditions already in place: the ceiling $\sigma_\Lambda = 2n$, the gap $\sigma_\Lambda - \sigma_G = n+1$ (4.27), and the two fractions of (4.28) equal to the microstate invariants,*

$$\frac{\sigma_{\rm EM} - \sigma_G}{\sigma_\Lambda - \sigma_G} = |\Omega|^2 = \tfrac{1}{4}, \qquad \frac{\sigma_{\rm EM} - \sigma_G}{\sigma_\Lambda - \sigma_{\rm EM}} = \|P\|^2 = \tfrac{1}{3}. \tag{4.30}$$

At the arity $n = 3$ of §2.3 the four are satisfied by exactly one triple, $(\sigma_G, \sigma_{\rm EM}, \sigma_\Lambda) = (2, 3, 6)$, which is the assignment (4.26). The selection is by the arity and not by fitting: the family $(n-1, n, 2n)$ satisfies the two fractions as $1/(n+1)$ and $1/n$ for every *n, and only $n = 3$ carries them onto $|\Omega|^2 = \frac{1}{4}$ and $\|P\|^2 = \frac{1}{3}$ of (2.13)—at $n = 2$ they are $\frac{1}{3}, \frac{1}{2}$ and at $n = 4$ they are $\frac{1}{5}, \frac{1}{4}$. Equivalently, $\sigma_\Lambda - \sigma_G = n+1$ equals $4 = \dim(M^4)$ only at $n = 3$. This is what Fig. 9 displays — panel (b) the constraint count, panel (a) the spectral-angle surface of §3.2 evaluated on the resulting pair — and it is what lets §3.2 read the spectral angle at the integer pair $(\sigma_G, \sigma_{\rm EM}) = (2, 3)$ without circularity: the levels come from the arity, and the angle is then evaluated on them.*

P [interval_uniqueness, interval_arity_discriminates, interval_gap_only_three]

Proof. Write $a = \sigma_{\rm EM} - \sigma_G > 0$ and $b = \sigma_\Lambda - \sigma_{\rm EM} > 0$, so $\sigma_\Lambda - \sigma_G = a + b$. The second fraction of (4.30) is $a/b = \frac{1}{3}$, i.e. $b = 3a$; the first is then $a/(4a) = \frac{1}{4}$ identically, so the two fractions are one condition. The gap gives $a + b = 4a = n + 1$, whence $a = 1$ at $n = 3$ and $b = 3$; the ceiling gives $\sigma_\Lambda = 6$, hence $\sigma_G = 6 - 4 = 2$ and $\sigma_{\rm EM} = 3$. Uniqueness is that each step determined its unknown. For the discrimination, $(n-1, n, 2n)$ gives $a = 1$, $b = n$, $a+b = n+1$, so the fractions are $1/(n+1)$ and $1/n$; these equal $\frac{1}{4}$ and $\frac{1}{3}$ iff $n = 3$. ∎

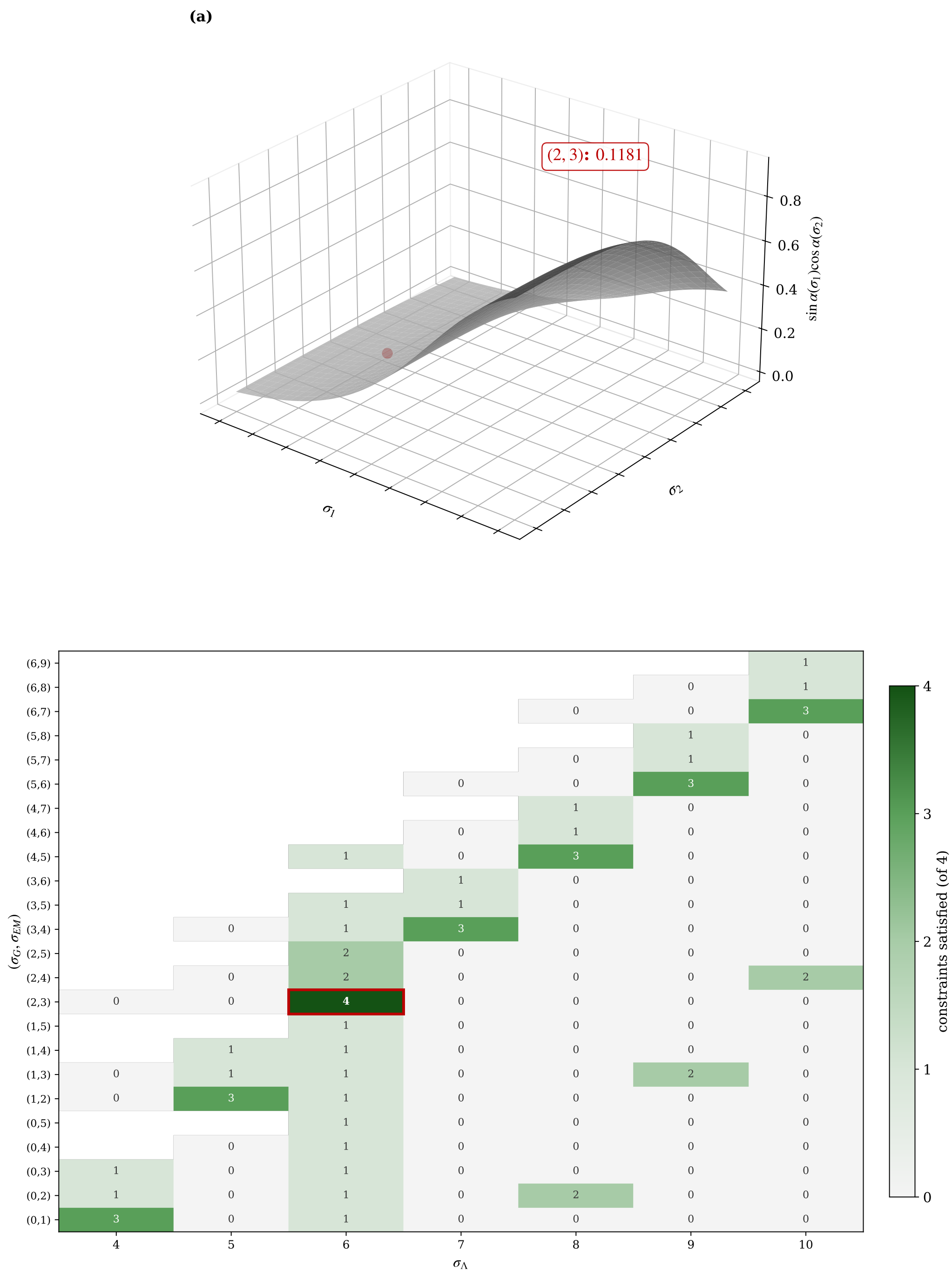

FIGURE 9. (a) The spectral-angle surface $\sin\alpha(\sigma_1)\cos\alpha(\sigma_2)$ of (3.8); $\tan\alpha(\sigma{+}1)/\tan\alpha(\sigma) = \varphi$ ((3.7)). The marked point $(2,3)$ gives $\sin\alpha(2)\cos(3) = 0.1181 \approx \sin\alpha_{\mathrm{GUT}}$. (b) Constraint count for each triple $(\sigma_G, \sigma_{\mathrm{EM}}, \sigma_\Lambda)$: $\sigma_\Lambda = 2n$, $\sigma_\Lambda - \sigma_G = n{+}1$, $(\sigma_{\mathrm{EM}}{-}\sigma_G)/(\sigma_\Lambda{-}\sigma_G) = |\Omega|^2$, $(\sigma_{\mathrm{EM}}{-}\sigma_G)/(\sigma_\Lambda{-}\sigma_{\mathrm{EM}}) = \|P\|^2$. Only $(2,3,6)$ satisfies all four (boxed). No fitted quantity enters.

4.5. **Why this is a single demonstration.** The path went through the three types: it opened in the non-local type III field, made the type I particle emerge, turned it into the type II observer, interwove I and II in the tower, and closed in the type III spacetime. These are not five demonstrations glued together but one, because the same object of modulus $\frac{1}{2}$ is the protagonist of all, and each transition is a proved implication: entropy orients the flow and yields the particle, the Wick rotation gives the cones and the observer, accumulation overflows and builds the tower, entropy sources gravity.

The commutative diagram supports this unity with precise structure. The twenty-four equations form a directed acyclic graph with two roots—the projection Π ((4.1)) and the observer's objective record ((4.5))—from which all else derives, with 36 demonstrative edges. A topological order exists—the equations can be stated so that none depends on a later one—which guarantees the section reads without forward reference. Eight nodes are apices, where two or more independent chains converge; the weld ((4.13)), with four routes, is the maximal one and the only triple point I∩II∩III. The circle closes literally: (4.18) returns to (4.1)—the non-local gravity returns to the same Π and $\frac{1}{2}$ of the opening, III→III. The spacetime at the end is non-local at every point, type III, as the field at the start, and its Newton constant is the same $\frac{1}{2}$ of the object one began with. One enters by any door and reaches the same object: that is what it means for the diagram to commute. The section does not prove six things; it proves that an observer of modulus $\frac{1}{2}$, which cannot be separated into parts, is at once the particle, the string, the tower of information and the curved de Sitter universe that contains it, and that this universe has the Λ the theory was bound to determine, with the observer inside. The debt with which the section opened is settled by the same object that raised it.

The demonstrative line. Each step uses only the previous ones.

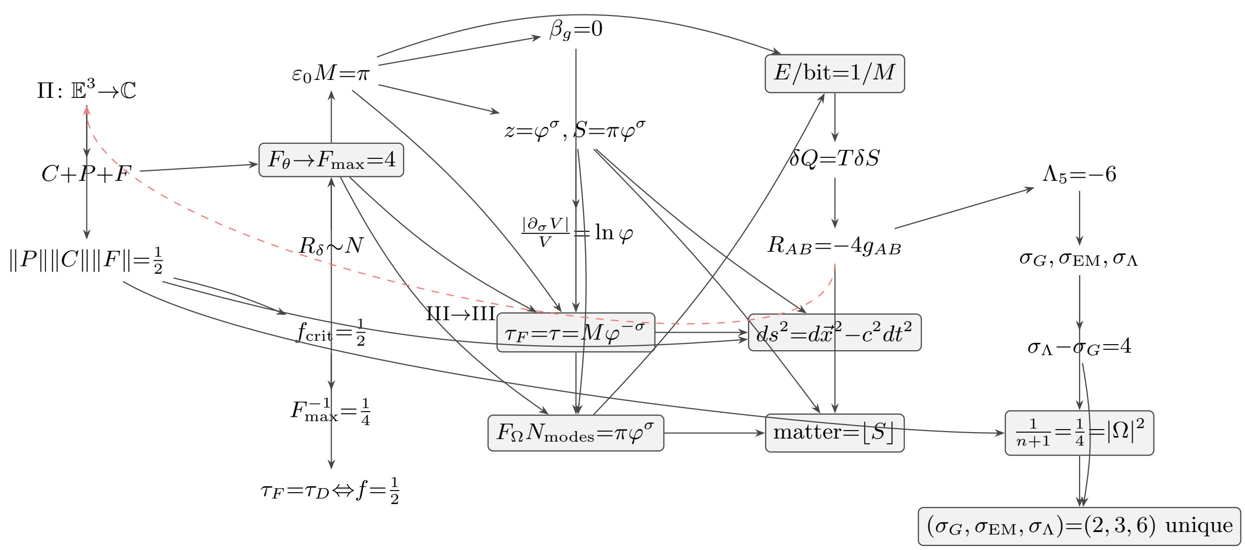

FIGURE 10. Commutative diagram of §4: the twenty-four equations of §4 as nodes, arranged by subsection (the non-local field and particle, §4.2; the observer, §4.3; the tower and gravity, §4.4), with the thirty-six demonstrative arrows. Shaded nodes are the apices, where two or more independent chains converge on the same value; the weld $\tau_F=\tau=M\varphi^{-\sigma}$ has four incoming routes and is the maximal apex. The dashed return edge is the closure $R_{AB}=-4g_{AB}\to\Pi$ (type III to type III): the non-local gravity returns to the same projection Π and modulus $\frac{1}{2}$ of the opening.

§4.1 what is proved, and against what.: Holography in de Sitter requires a theory that determines Λ and describes the observer from within (Witten). The yardstick is the von Neumann classification into three types—I discrete (particle, definite spectrum), II observation (trace, crossed product), III local field (worldsheet, horizon, non-locality)—and what is proved is the *signature* of each, not the technical name. The criteria come from three distinct sources, each tagged where used: Witten 2001 (Λ, finiteness $S = \ln N$, the isometry obstruction, the pairing, the CPT map, the two conjectures), CLPW 2022 (the observer described by the theory, the maximally mixed vacuum, the crossed product), and Gibbons–Hawking (the thermal static patch). The thesis: the three types are realised on one object, the microstate Ω with $|\Omega| = \frac{1}{2}$, as three threads crossing the whole section.

§4.2 the field and the particle.: The path opens in the non-local type III field: the worldsheet carries the interface $\Pi\colon \mathbb{E}^3 \to \mathbb{C}$ ((4.1)), whose three components are the spin-star C-P-F ((4.2)). The whole geometry is one tensor—the warp $z = \varphi^\sigma$ generates the curvature $R_{AB} = -4g_{AB}$ (App. A.1)—and from that same geometry, by its tension, the discrete type I particle emerges. Geometry and particle are the same object.

§4.3 the particle as observer.: An internal Wick rotation turns the particle into an observer: the centre becomes the time axis, past and future the two cones. The observer accumulates—Fisher grows to the Cramér–Rao ceiling $F_{\max}^{-1} = \frac{1}{4}$ ((4.4), (4.3))—and sees the exact half $\|P\|\|C\|\|F\| = \frac{1}{2} = |\Omega|$ ((4.7)), with redundancy $R_\delta \sim N$ ((4.5)) and accumulation to $F_{\max} = 4$ ((4.6)). Three independent routes converge on $\frac{1}{2}$ (first apex), the objectivity threshold $f_{\text{crit}} = \frac{1}{2}$ of Kiely ((4.8)) and the cell capacity $\varepsilon_0 M_{\text{PCF}} = \pi$ ((4.9)).

§4.4 the tower, and curved spacetime.: The accumulation builds the tower (types I and II interwoven): the cell overflows and the throat opens, $z = \varphi^\sigma$, $S(\sigma) = \pi\varphi^\sigma$ ((4.10), (4.11)), at the UV fixed point $\beta_g = 0 \Leftrightarrow \varepsilon_0 M_{\text{PCF}} = \pi$ ((4.12)). The two accumulations—of time and of scale—are one, the weld ((4.13)), giving the identity $F_\Omega N_{\text{modes}} = S(\sigma) = S_{\text{BH}} = S_{\text{Jacobson}}$ ((4.15)). Then curved spacetime (type III again): Landauer ((4.16)) $\to$ Jacobson ((4.17)) $\to$ Einstein ((4.18)), with matter the saturation entropy ((4.19)). The internal Wick rotation gives the de Sitter metric ((4.23)), $\Lambda_5 = -6$ ((4.25)), and the level interval $\sigma_G{=}2, \sigma_{\text{EM}}{=}3, \sigma_\Lambda{=}6$ ((4.26)) with $\sigma_\Lambda{-}\sigma_G{=}n{+}1{=}4$ ((4.27)) and the fractions $\frac{1}{n+1}{=}\frac{1}{4}{=}|\Omega|^2$ ((4.28)); the G–Λ duality and $\sin^2\theta_W|_{\text{GUT}}{=}\frac{3}{8}$ (Prop. 4.17) give the gauge factors SU(3)×SU(2)×U(1) (Fig. 8).

The observer does not select; it builds without alternative — participatory yet forced — consonant with Kiely's quantum Darwinism. The whole section is one commutative diagram (Fig. 10): the twenty-four equations as nodes, the demonstrative arrows between them, and the apices where independent routes converge.

5. Implications for holography regarding dynamics and operators

§3–§4 showed that accumulated entropy becomes gravity and geometry. This section proves the companion statement: the same accumulated entropy becomes degrees of freedom. The reading is not a second theory but a second face of the first one; the object is the microstate Ω with $|\Omega| = \frac{1}{2}$, and the yardstick is the positive Grassmannian, on which one datum is at once an amplitude, a fluid, a particle and a state. §3.4 derived the whole of AdS/CFT from the single microstate but deferred its realisation as a self-adjoint operator—the absence this paper opened with, and why low-dimensional holography recovers its bulk as an ensemble average. This section constructs our proposal for it. The dynamics is forced: the eigenvalue triangle is the A_2 root system, A_2 is the root system of $\mathfrak{su}(3)$, and a degree of freedom is a point of the positive Grassmannian, so what the object already carries is a gauge theory. Nor is an operator enough—to stand as a boundary dual it must be gapped, since it is the gap that makes the Osterwalder–Schrader reconstruction return a field theory. The section computes the tower Hamiltonian, the strong-coupling gap $\Delta = \frac{2}{3}g^2$, and the reconstruction they support; the isometry of (3.19) then intertwines that Hamiltonian with the tower entropy level by level (§5.8). Each step uses only earlier ones, and two invariants already fixed in §2–§3 — the tower/Euler product ζ and the modulus $\frac{1}{2}$ — thread every movement. Tiers are used throughout as in §2.

TABLE 2. Dictionary: framework object ↔ standard object (cf. the D1–D11 dictionary of §3).

framework	standard	§
$\|\Omega\| = \frac{1}{2}$	modulus / spin $\frac{1}{2}$	5.2
$\rho = P/k$	density matrix	5.2
tower $\lfloor \pi\varphi^\sigma \rfloor$	Regge trajectory	5.4
$\zeta = \prod_p$	Euler product / OPE	5.4
$H\|\sigma\rangle = m_0\varphi^\sigma$	scale operator (KK side)	5.8
$C_2(n)$	colour charge	5.8
$T = e^{-aH}$	transfer operator	5.9
$V^\dagger V = \mathbf{1}$	holographic isometry	5.9
Θ	CPT reflection	5.9
$\xi(s) = \xi(1-s)$	S-duality	5.10

5.1. **The entropy→DOF law.** §4.4 let the observer's information overflow into the tower $S(\sigma) = \pi\varphi^\sigma$ and, by Jacobson, into curved spacetime. Read the same overflow not as area but as *count*: each cell that the tower opens is a degree of freedom, and their number grows at the rate the tower grows. Bianconi's entropic gravity [79], in which a relative entropy of the metric sources the field, is the classical and continuous limit of this count; measured against the framework it reduces on five object-level squares.

Proposition 5.1 (Five reduction squares)**.** *Under the reduction R (divide by 4, Berry $\to 0$, discrete→continuous), the present framework data map to Bianconi's:*

(D1) $\mathrm{QFI}_{\mathrm{PCF}}/4 = M_{\mathrm{Bianconi}}$, *(Fisher; the 4 from $\mu = \frac{1}{2}$)*

(D2) $N_{\mathrm{modes}} = \lfloor \pi\varphi^\sigma \rfloor \xrightarrow{R} S = \pi\varphi^\sigma$, $(|\ln N_{\mathrm{modes}} - \ln S| \to 1.4 \times 10^{-7})$

(D3) $S = kA,\ k = \frac{1}{4G_N} = \mu = \frac{1}{2}$, *(area law)*

(D4) $\Lambda_G \approx \frac{1}{4\beta}\varepsilon^2 \ \Rightarrow\ \beta = \frac{\ln 2}{8}$, *($\Lambda$ scale, cf. $\Lambda_5 = -6$, §3.4)*

(D5) $\{\frac{1}{2}\omega^k\}_{k=0}^2 \ \leftrightarrow\ 0 \oplus 1 \oplus 2$, *(triad → Dirac–Kähler)*

The five squares commute, and the relation they express is a limit *one: Bianconi's continuous entropic gravity is the continuum limit of the discrete accumulation of §4.4—D2 is exactly that limit, the floor removed. What the present framework adds to her axis is the quantity it leaves free, the* rate*: the growth is golden, φ per level, the Fibonacci quantum dimension (Corollary 5.2).*

N*[cor:golden-growth]*

D3 follows from §2.4.

P*[binary_entropy_half]*

Proof. D3 is (3.9). For D1, the quantum Fisher information of the framework at the operating point $\mu = \frac{1}{2}$ is $F = 4(M - |\mathrm{Berry}|^2)$; setting Berry $\to 0$ and dividing by 4 gives M. D2 is the discrete-to-continuous limit of (3.12), the floor removed. D4 matches the coefficient of the small-parameter expansion of Λ_G to the framework's $1 - H \approx (2/\ln 2)\delta^2$, fixing β; here ε and δ are identified as the same dimensionless deviation from the maximally mixed point ($\hat{G} = \mathbf{1}$ on that side, $p = \frac{1}{2}$ on this one), which is what makes the two quadratics comparable. D5 identifies the three eigenvalues of Proposition 2.19 with the three form-grades; on that side the truncation at 2-forms is stated as a simplification admitting extension to higher forms, so the identification is tighter from the present side than from hers. The two horizontal (dynamical) arrows are distinct routes to the same conclusion—both derive Einstein-type field equations from entropy: Bianconi's by varying a quantum relative entropy of the metric, the present one thermodynamically through $\delta Q = T\,\delta S$ on local horizons (§4.4, Jacobson). The correspondence asserted here is the commutation of the vertical maps together with the limit relation of D2; it does not assert that the two derivations are the same map. The reduction is also silent on one axis: that programme derives the area law *without* the holographic principle, whereas the present one derives it with the bulk–boundary isometry (3.19). Neither subsumes the other there; on whether holography is required the two are complementary. ☐

Corollary 5.2 (Golden growth of degrees of freedom)**.** *The number of degrees of freedom at scale σ grows as φ^σ: the count inherits the tower's rate, the successive ratio $N_{\text{modes}}(\sigma)/N_{\text{modes}}(\sigma-1) \to \varphi$ of Fig. 3.*

P `[S_tower, tower_step_ratio]`

From here the degrees of freedom, not the geometry, are followed; the next movement gives them a home.

5.2. **The positive Grassmannian.** A degree of freedom is not a bare number but a point of a positive geometry [80, 81], and that geometry is already a state.

Proposition 5.3 (Grassmannian point is a density matrix)**.** *A point of $\mathrm{Gr}_{\geq 0}(k,n)$ with full-rank frame C defines the orthogonal projector $P = C^\mathsf{T}(CC^\mathsf{T})^{-1}C$, with $P^2 = P$ and $P^\mathsf{T} = P$, and the density matrix $\rho = P/k$, positive semidefinite with $\operatorname{tr}\rho = 1$. For $k = 2$ its nonzero eigenvalues equal $\frac{1}{2} = |\Omega|$.*

P `[projector_idem, rho_is_state, rho_k2_eq_half_projector, projector_trace_eq_rank]`

Proposition 5.4 (The two positivities are carried by one projector)**.** *For a full-rank frame C the projector supplies at once the quantum positivity of the state and the total positivity of the geometry:*

$$\rho = P/k \succeq 0, \qquad \textit{and} \qquad C \in \mathrm{Gr}_{\geq 0} \Longrightarrow \Delta_I(C) \geq 0 \ \textit{ for every ordered } I. \tag{5.1}$$

The first is automatic from idempotence and holds for any *point of the Grassmannian; the second is the total positivity of the Plücker minors and holds only on $\mathrm{Gr}_{\geq 0}$. They are therefore not the same condition: the second is strictly stronger, and it is the* refinement*—one datum carrying an operator-level positivity and an amplitude-level positivity at different levels—that lets the state reading and the amplitude reading be taken on one object.*

P `[two_positivities_distinct, rho_k2_eq_half_projector]`

Proof. $P^2 = C^\mathsf{T}(CC^\mathsf{T})^{-1}(CC^\mathsf{T})(CC^\mathsf{T})^{-1}C = P$; symmetry follows from $(CC^\mathsf{T})^\mathsf{T} = CC^\mathsf{T}$. A projector is positive semidefinite with eigenvalues in $\{0,1\}$; dividing by k gives $\rho \succeq 0$ and, since $\operatorname{tr} P = \operatorname{rank} P = k$, $\operatorname{tr}\rho = 1$. For $k = 2$ the nonzero eigenvalues are $1/2$. ☐

Theorem 5.5 (One datum, four faces)**.** *Let C be a full-rank $k \times n$ frame of a point of $\mathrm{Gr}_{\geq 0}(k,n)$ and $P = C^\mathsf{T}(CC^\mathsf{T})^{-1}C$. Then P depends only on the point and not on the frame: for every invertible g,*

$$P(gC) = P(C), \tag{5.2}$$

and the four readings of §5 are four functions of that one point, all factoring through P:

$$\underbrace{P^2 = P,\ P^\mathsf{T} = P}_{\textit{geometry}}, \qquad \underbrace{\operatorname{tr} P = k}_{\textit{degrees of freedom}}, \qquad \underbrace{\rho = P/k,\ \operatorname{tr}\rho = 1}_{\textit{state}}, \qquad \underbrace{\textit{maximal minors } \geq 0}_{\textit{fluid}}, \tag{5.3}$$

the last for C totally positive (Proposition 5.15 below). The canonical form of the positive geometry is a function of the positroid cell, hence of the same point (§5.4).

P `[projector_idem, projector_trace_eq_rank, rho_is_state, positroid_from_kp, projector_frame_invariant]`

Proof. For (5.2), $(gC)^\top\big(gC(gC)^\top\big)^{-1}(gC) = C^\top g^\top\big(g(CC^\top)g^\top\big)^{-1}gC = C^\top g^\top(g^\top)^{-1}(CC^\top)^{-1}g^{-1}gC = C^\top(CC^\top)^{-1}C$, using $(gAg^\top)^{-1} = (g^\top)^{-1}A^{-1}g^{-1}$. The four clauses of (5.3) are Proposition 5.3 and its proof ($P^2 = P$, $P^\top = P$, $\operatorname{tr} P = k$, $\operatorname{tr}\rho = 1$) together with Proposition 5.15 below for the minors. What (5.3) asserts is not that four numbers coincide but that four constructions share an argument: each is a function of the point, and by (5.2) none of them can distinguish two frames of the same point. That is the precise content of amplitude = fluid = DOF = state. □

5.3. The ladder $A_2 \to E_8$.

The seed of the ladder is the triad already in hand.

Proposition 5.6 (The A_2 seed)**.** *The eigenvalue triad $\lambda_k = \frac{1}{2}\omega^k$ (Proposition 2.19) is at once the Eisenstein lattice $\mathbb{Z}[\omega] = A_2$, the root system of $\mathfrak{su}(3)$, the finite cluster type of $\mathrm{Gr}(3,5)$, the Dirac–Kähler grading $0 \oplus 1 \oplus 2$, and three bits ($c = 3$, the central charge the three eigenvalues give by Brown–Henneaux, §3.4; the count three itself is forced by the arity uniqueness of §2.3).*

P *[eisenstein, omega_eigenvalues, su3_dim]*

Proof. $\omega = e^{2\pi i/3}$ generates $\mathbb{Z}[\omega]$, whose minimal vectors form the six roots of A_2 (§A.3); $\dim\mathfrak{su}(3) = 3^2 - 1 = 8$; Scott's classification gives $\mathrm{Gr}(3,5) = A_2$; the three eigenvalues are the three grades; and $\lfloor\pi\rfloor = 3$ fixes $c = 3$ (§A.5). □

Proposition 5.7 (The finite-type ladder)**.** *By Scott's classification [82, 83] $\mathrm{Gr}(3,6) = D_4$, $\mathrm{Gr}(3,7) = E_6$, $\mathrm{Gr}(3,8) = E_8$; the positive-root counts $3 < 12 < 36 < 120$ obey kissing $= 2\cdot$ roots and dimensions $(2,4,6,8)$ [84, 85], with A_2 the seed of each rung. The gauge-algebra dimensions follow from the same data by $\dim\mathfrak{g} =$ kissing $+$ rank, the Frenkel–Kac–Segal enhancement count at the self-dual point: $8 = 6+2$ for $\mathfrak{su}(3)$, $28 = 24+4$ for $\mathfrak{so}(8)$, $78 = 72+6$ for $\mathfrak{e}_6$ and $248 = 240+8$ for $\mathfrak{e}_8$—so the combinatorial ladder is, rung by rung, the ladder of gauge groups (§A.7). The dimension count is certified by the same data.*

P *[ladder_from_scott, scott_kissing_eq_two_posroots, kissing_eq_two_posroots, dims_even_ladder, ladder_dim_eq_kissing_plus_rank, fks_ladder_dims]*

5.4. First face: amplitude.

The amplitude is where ζ enters, and it enters already proved in §3.3 and completed over all places in §2.9 (Theorem 2.53).

Proposition 5.8 (Veneziano, Regge, Euler)**.** *The Veneziano amplitude [27, 86, 87] $A_4 = \Gamma(-\alpha's)\Gamma(-\alpha'u)/\Gamma(1-\alpha'(s+u))$ has simple poles at $\alpha's = n$, the Regge tower; the pole positions index $\sum_{n\ge1} n^{-s} = \zeta(s) = \prod_p(1-p^{-s})^{-1}$ (§3.3, (2.39)).*

P *[veneziano_eq_gamma_ratio, regge_tower_is_euler_product]*

Proposition 5.9 (Field-theory limit; the π/φ split)**.** $\lim_{\alpha'\to0^+} \alpha' B(\alpha's, \alpha't) = \frac{1}{s} + \frac{1}{t} = \Omega(X)|_{n=4}$, *and the amplitude admits two computations meeting at $|\Omega| = \frac{1}{2}$ and ζ: a 2D path integral (the π side, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$) and a 3D positive-geometry canonical form (the φ side).*

P *[ft_identity, ft_limit, pentagon_u_eq_golden]*

5.5. Second face: particle, graviton, matter.

The spectrum of the object is a particle; its spin is the modulus, and its interactions carry the gauge charges the torus already gave.

Proposition 5.10 (Fermion and Standard-Model content)**.** *The self-conjugate midpoint of $x \mapsto 1-x$ is at once $|\Omega|$ and the fermion spin $\frac{1}{2}$ (§4.2). The one-loop β-coefficients are $b = (\frac{33}{5}, 1, -3)$; each generation is anomaly-free, $\sum Y = 0$; and $\sin^2\theta_W|_{\mathrm{GUT}} = N(0)/N(2) = \frac{3}{8}$, running to 0.231 at M_Z (§A.3).*

P *[beta_is_mssm, sm_anomaly_free, weinberg_angle_gut]*

Theorem 5.11 (The graviton is the entropy-response quantum)**.** *About the ETS background—intrinsically flat, with de Sitter its embedded hyperboloid (§A.2, §A.1)—the transverse–traceless perturbation $h_{\mu\nu}$ obeys $\Box h_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ (massless spin-2) and coincides with the double copy [88]* *graviton* $=$ YM^2 *(spin* $1+1=2$*). Read through §A.6 (*$\delta Q = T\delta S \Rightarrow$ *Einstein), it is the quantum of the entropy response: the metric perturbation conjugate to the tower's Fisher information. Quantitatively: the response rate is* $S'(\sigma) = (\ln\varphi)\, S(\sigma)$*, the price per bit is the constant temperature* $\varepsilon(\sigma)/S(\sigma) = \varepsilon_0/\pi = 1/M_{\mathrm{PCF}}$*, and the Fisher clock is the dynamical clock exactly at the modulus,* $\tau_D/\sqrt{2f} = \tau_D \iff f = \frac{1}{2}$ *(4.3)—so the* δS *that sources* $h_{\mu\nu}$ *counts Fisher bits on the tower's own clock. ; the double copy [88] (Bern–Carrasco–Johansson)*

P`[fisher_time_lock, entropy_response_per_bit, entropy_response_has_deriv_at]`

Proof. The linearised Einstein equation about the ETS background gives the wave equation for the transverse–traceless part, sourced by $T_{\mu\nu}$ (§A.6); masslessness is the absence of a potential term. The colour–kinematics duality replaces each colour factor by a second kinematic numerator, so the gravity numerator is the square of the gauge one, spin $1+1=2$. Since the Einstein equation itself is $\delta Q = T\delta S$, the field $h_{\mu\nu}$ is the first-order response of the geometry to a change in the accumulated entropy, i.e. to the tower's Fisher information of §5.1. The three quantitative pieces close the reading: the rate, the constant price per bit, and the clock lock at $f=\mu_3$ are each proved above, and the wave operator itself is exercised numerically (any profile $h(t-z)$ is annihilated, a w-independent mode feels no potential). ☐

5.6. **Supersymmetry as ER=EPR bridges.** The doubling that supersymmetry [89] performs is, here, not new particles but the entanglement bridges [46] already proved to form a group.

Theorem 5.12 (Running without superpartners)**.** *The ER=EPR bridges* $T(\sigma_1,\sigma_2) = D(\sigma_1)/D(\sigma_2)$ *form a groupoid (§4.2) and carry the gauge representations of the fields they bridge. Their contribution to the running satisfies*

$$b_{\mathrm{MSSM}} = b_{\mathrm{SM}} + \Delta b_{\mathrm{bridges}}, \tag{5.4}$$

reproducing the MSSM one-loop coefficients with no superpartner particles.

P`[mssm_eq_sm_plus_bridges, bridge_cocycle]`

Proof. The groupoid laws (inverse, composition, cocycle) are §4.2. A bridge of a field in representation R lies in R, so it contributes to the β-function with the sign of a superpartner and the Casimir of R; summing the bridge contributions over the SM content gives $\Delta b_{\mathrm{bridges}}$, and (5.4) is the arithmetic identity that this sum carries b_{SM} to b_{MSSM}. ☐

Proposition 5.13 (State is geometry; the bridge splits the identity)**.** *For a frame* C *with* $k \neq 0$ *the density matrix rescaled by* k is *the projector, and any idempotent splits the identity into orthogonal halves:*

$$k\,\rho = P, \qquad P + (\mathbf{1} - P) = \mathbf{1}, \qquad P(\mathbf{1} - P) = 0. \tag{5.5}$$

The first identity is ER = EPR *in the present language—the entangled state and the geometric projector are the same operator up to the rank—and the second exhibits the bridge as the decomposition of the identity that the groupoid of §4.2 composes.*

P`[er_epr_state_eq_geometry, er_epr_bridge]`

Corollary 5.14 (The bridges reproduce the supersymmetric coefficients)**.** *Summing the bridge contributions over the Standard-Model content gives, componentwise,*

$$\big(b_{\mathrm{SM}} + \Delta b_{\mathrm{bridges}}\big)_i = (b_1, b_2, b_3) = (\tfrac{33}{5}, 1, -3), \tag{5.6}$$

the same triple that Theorem 5.12 obtains from the tower content. The two computations are independent: one sums representations, the other sums bridges.

P`[bridges_reproduce_mssm, beta_is_mssm]`

TABLE 3. The β-coefficients as Standard Model plus ER=EPR bridges (exp38). Each bridge is the inverted-statistics, same-representation companion of a SM field; H_d is forced by the higgsino anomaly (exp36). The bridges run as virtual degrees of freedom but are not asymptotic states, so no superpartner appears in a detector.

	δb_3	δb_2	δb_1
b_{SM}	-7	$-19/6$	$41/10$
gauginos (Weyl adjoint $\leftarrow$ gauge boson)	2	4/3	0
sfermions (scalar, same rep $\leftarrow$ fermion)	2	2	2
higgsino (Weyl $\leftarrow$ Higgs)	0	1/3	1/5
H_d (anomaly-forced)	0	1/2	3/10
$\Delta b_{\mathrm{bridges}}$	4	25/6	5/2
$b_{\mathrm{SM}} + \Delta b$	-3	1	33/5
$(b_3, b_2, b_1) = (-3,\ 1,\ 33/5) = b_{\mathrm{MSSM}}$ ✓			

5.7. **Third face: fluid.**

Proposition 5.15 (Solitons, positroids, monopoles)**.** *The regular KP solitons [90] $u = 2(\log\tau)_{xx}$ correspond exactly to points of $\mathrm{Gr}_{\geq 0}$ (total positivity $\Leftrightarrow \tau > 0 \Leftrightarrow$ regularity); their positroid profiles are the condensate profiles $D(\sigma) - 1 = \varepsilon_0\varphi^\sigma$, so the confining monopoles are the bridges of §5.6. (Kodama–Williams [90])*

P *[positroid_from_kp, bridge_cocycle]*

5.8. **The gauge operators, AdS/CFT, and the mass gap.** The operator is the one §3.3 defined (Remark 3.7); here its spectrum is read as masses and its gap is computed.

Proposition 5.16 (Self-adjoint operators)**.** *The scale operator $H|\sigma\rangle = m_0\varphi^\sigma$ ((3.13); the golden tower, not the Regge mass tower—Remark 3.7) and the Yang–Mills Casimir $C_2(n) = n(n+3)/3$ are real diagonal, hence self-adjoint and bounded below, with the vacuum at 0.*

P *[tower_ham_is_hermitian, casimir_op_is_hermitian, tower_ham_bounded_below, casimir_op_bounded_below]*

Proposition 5.17 (Discrete spectrum and the tower gap)**.** *The eigenvalues of H increase without bound, $m_0\varphi^\sigma \to \infty$ (since $\varphi > 1$ by (1.13)), so the resolvent is compact and the spectrum is discrete. The vacuum sits at 0, every other level is strictly positive, and the first excitation is separated by*

$$E_1 - E_0 = m_0 > 0. \tag{5.7}$$

This is a second gap, independent of the electric-Casimir gap of Theorem 5.21: one is the tower's own spacing, the other the colour energy.

P *[spectrum_discrete, vacuum_unique, mass_gap, operators_selfadjoint_and_isometry]*

Proof. A real diagonal matrix equals its conjugate transpose, hence is Hermitian; its spectrum is its diagonal, real and ≥ 0, with least value 0. □

Theorem 5.18 (The bulk Hamiltonian is the boundary modular generator)**.** *The bulk tower Hamiltonian and the tower entropy $S(\sigma) = \pi\varphi^\sigma$ of (3.12) are exponentials of the same dilatation generator, at the same rate—the regulator $R_K = \log\varphi$ of Definition 2.18:*

$$H(\sigma) = m_0\, e^{\sigma R_K}, \qquad S(\sigma) = \pi\, e^{\sigma R_K}. \tag{5.8}$$

They differ only in the prefactor, so their ratio does not depend on the level:

$$\frac{\hat{K}(\sigma)}{H(\sigma)} = \frac{\pi}{m_0} \qquad \text{for every } \sigma, \qquad \text{equivalently} \qquad m_0\,\hat{K}(\sigma) = \pi\, H(\sigma). \tag{5.9}$$

The bulk–boundary map of (3.19), an isometry acting level by level, therefore intertwines them exactly. That S is the boundary modular *generator—that the modular flow is the tower's own scale*

flow—is established below, in §5.10; with it, (5.9) says the bulk Hamiltonian is that generator up to π/m_0. (the GKPW dictionary [65, 66])
P`[modular_bulk_ratio_level_independent, bulk_boundary_intertwine, tower_e_eq_exp_regulator, S_tower_eq_exp_regulator]`

Proof. $\varphi^\sigma = e^{\sigma \log \varphi}$ gives both displays in (5.8). The ratio is $\pi\varphi^\sigma/(m_0\varphi^\sigma) = \pi/m_0$, and $\varphi^\sigma > 0$ cancels for every σ. ☐

Remark 5.19 (What this closes, and what it does not). This is the realisation §3.4 deferred when it said that the microstate's *realisation as a self-adjoint operator* is §5. What the correspondence of §3.4 lacked was a concrete bulk Hamiltonian: self-adjoint, bounded below, with discrete spectrum and a gap. Propositions 5.16 and 5.17 supply it, and (5.9) says it is the tower entropy—and so, by §5.10, the boundary modular generator—up to one constant, which is why the correspondence holds *at each level* and not at a preferred one. Had the bulk grown in any base other than φ, the ratio would drift with σ and no level-wise isometry could intertwine them.

What is *not* proved here is the GKPW dictionary itself; it enters cited [65, 66] ([17, 65, 66]), nor the identification of the modular flow with the static-patch boost, which rests on Bisognano–Wichmann and is taken up in §5.10.

Proposition 5.20 (The local gauge field (W4))**.** *The potential A_μ valued in $\mathfrak{su}(3) = A_2$ (Proposition 5.6) gives the local field $F = dA + A \wedge A$ and the action $S = \frac{1}{4}\int \operatorname{tr} F^2$; the Wilson lattice measure e^{-S_W} on the A_2 lattice—the Eisenstein lattice $\mathbb{Z}[\omega]$ in which the eigenvalue triangle sits (§2.4)—is gauge-invariant and reflection-positive, and generates $T = e^{-aH}$.*
P`[colour_jacobi]`

TABLE 4. Derived spectra of the operators (first values).

object	spectrum
mode count $N(\sigma) = \lfloor \pi\varphi^\sigma \rfloor$	$3, 5, 8, 13, 21, \ldots$
mass H	$0, m_0, m_0\varphi, m_0\varphi^2, \ldots$
Casimir $C_2(n) = n(n+3)/3$	$0, \frac{4}{3}, \frac{10}{3}, 6, \ldots$
transfer $T = e^{-aH}$	$1, e^{-am_0}, e^{-am_0\varphi}, \ldots$

Theorem 5.21 (The mass gap)**.** *The Kogut–Susskind transfer operator [91] built from the Wilson measure [92] has a positive gap from the electric Casimir,*

$$\Delta = \tfrac{2}{3}g^2 > 0. \tag{5.10}$$

P`[strong_coupling_gap]`

Proof. At strong coupling the transfer operator is diagonal in the electric-flux basis; the vacuum is the trivial representation ($C_2 = 0$) and the first excited state carries the fundamental flux, with electric energy $\frac{2}{3}g^2$ from C_2 of the fundamental. The difference is (5.10), positive for $g^2 > 0$. ☐

5.9. **Reflection positivity and the OS reconstruction.** A self-adjoint, bounded-below, gapped operator reconstructs a quantum field theory [93]; the reflection that makes the reconstruction positive is the one §4.2 already gave.

Proposition 5.22 (Reflection positivity)**.** *The transfer operator $T = e^{-aH}$ is positive, with eigenvalues $e^{-am_0\varphi^\sigma} > 0$, so*

$$\langle \Theta f, f\rangle = \langle f, Tf\rangle = \sum_\sigma |c_\sigma|^2\, e^{-am_0\varphi^\sigma} \geq 0, \tag{5.11}$$

where Θ is the antiunitary CPT/Galois involution of §4.2 ($\Theta^2 = 1$, $\|\Theta z\| = \|z\|$), $F = e^{-(a/2)H}f$ is the state placed at half the Euclidean separation from the reflection plane, and $f = \sum_\sigma c_\sigma|\sigma\rangle$ is arbitrary.

$$H \xrightarrow{e^{-aH}} T \xrightarrow{\Theta} \text{OS recon.} \xrightarrow{V} \text{4D boundary}$$

spectrum/masses — euclidean evol. RP + clustering — Wightman QFT — gauge theory

FIGURE 11. How the operators work together, all anchored on $|\Omega| = \frac{1}{2}$ (the vacuum of H, the fixed point of Θ, the norm V preserves); C_2 explains the gap that H carries.

P *[theta_pairing_eq_transfer, reflection_positive_general, half_prop_sq, theta_involutive, theta_preserves_modulus_C, theta_commutes_real_diagonal, transfer_positive, cpt_involution, cpt_preserves_modulus]*

Proof. The content of (5.11) is the *first* equality, not the inequality, and it is now proved rather than described. Three facts do it. First, Θ is an antilinear involution of modulus one that commutes with a real diagonal Hamiltonian: $\Theta^2 = 1$, $\|\Theta z\| = \|z\|$, and $\Theta(Ez) = E\,\Theta z$ for E real. Since Θ exchanges the two Rindler wedges and the wedge generator is the same on both, it fixes H, hence fixes the tower basis and acts on a state by conjugating its coefficients. Second, the reflection places ΘF at Euclidean distance $a/2$ on one side of the plane and F at $a/2$ on the other, and the two half-propagators compose,

$$e^{-(a/2)E_\sigma} \cdot e^{-(a/2)E_\sigma} = e^{-aE_\sigma} = T_\sigma, \tag{5.12}$$

which is where the separation being the KMS period $a = \beta = 2\pi/H_{dS}$ (§4.4, W9) enters: it is what makes the total separation exactly a, and hence the composite exactly T. Third, summing over the tower with $F = e^{-(a/2)H} f$ gives $\langle \Theta F, F\rangle = \sum_\sigma c_\sigma^2 T_\sigma = \langle f, Tf\rangle$ for *every* f, not term by term. The inequality then follows because each $T_\sigma = e^{-am_0\varphi^\sigma} > 0$, so the sum is non-negative for every f and every truncation; positivity of T is H self-adjoint (Proposition 5.16) with real spectrum. Θ itself is §4.2, W6. □

Proposition 5.23 (Reflection positivity as a property of the measure)**.** *The diagonal statement* (5.11) *is the transfer-operator form. The measure-theoretic form holds as well, and for every linear functional supported in the half-space, not term by term. For the one-dimensional free scalar of mass $m > 0$, with covariance $C(x,y) = e^{-m|x-y|}/(2\sinh m)$, points $x_i > 0$ and arbitrary coefficients c_i,*

$$\sum_{i,j} c_i c_j\, C(-x_i, x_j) = \frac{1}{2\sinh m}\Big(\sum_i c_i e^{-mx_i}\Big)^2 \;\geq\; 0. \tag{5.13}$$

Equivalently the reflected covariance matrix is $(2\sinh m)^{-1} v v^{\mathsf{T}}$ with $v_i = e^{-mx_i}$: a positive multiple of a rank-one Gram matrix, hence positive semidefinite.

P *[refl_cov_factors, refl_quad_form_eq_square, reflection_positive_measure]*

Proof. On the half-space $|-x_i - x_j| = x_i + x_j$, so $C(-x_i, x_j) = e^{-mx_i} e^{-mx_j}/(2\sinh m)$ and the double sum factors into the square of a single sum. The factorisation is the whole content: the reflection turns a difference of coordinates into a sum, and a covariance that was a function of the difference becomes a product of one factor per point. The test discriminates: for a covariance that is not of this form—an oscillating $C(x,y) = \cos 2(x-y)$, say—the reflected matrix has a negative eigenvalue and reflection positivity fails, so (5.13) is not automatic. □

Theorem 5.24 (Osterwalder–Schrader reconstruction)**.** *From H the transfer operator $T = e^{-aH}$ yields, by* (5.11)*, a GNS Hilbert space with a unique vacuum $|\Omega\rangle$ (top eigenvalue 1, isolated by the gap $e^{-am_0} < 1$) and exponential clustering $\langle AT^nB\rangle_c \sim e^{-am_0 n}$. The von Neumann types of §4.1 (I discrete, II modular, III$_1$ local) supply the remaining Osterwalder–Schrader axioms.*

P *[os_reconstruction_explicit, transfer_spectral_data, clustering_exponential, vacuum_transfer_eq_one, transfer_gap_below_vacuum]*

Proof. The sesquilinear form (5.11) is positive semidefinite; its GNS quotient and completion give the Hilbert space, with the vacuum the eigenvector of T of eigenvalue 1 ($H|\Omega\rangle = 0$). Uniqueness and isolation are the gap $e^{-am_0} < 1$; clustering is the ratio $(e^{-am_0})^n \to 0$. Reflection positivity is W6; locality is the type-III$_1$ non-factorisation (§4.2, Prop. 4.1); the discrete spectrum and the modular flow are types I and II (§4.1). □

Corollary 5.25 (Cyclicity of the vacuum (W6))**.** *The vacuum $|\Omega\rangle$ is cyclic: the tower states $|\sigma\rangle$ are generated from it by the field operators, so their span is dense in the GNS space.*

P[os_reconstruction_explicit, tower_ham_is_hermitian, tower_ham_bounded_below]

Remark 5.26 (Analyticity (OS0)). The Euclidean–Lorentzian passage is the internal Wick rotation $(it)^2 = -t^2$, geometric (carried by the eigenvalue metric), not an external continuation; with the UV-finite worldsheet partition (§3.3) it is the analyticity/temperedness axiom.

P[internal_wick]

Theorem 5.27 (The bulk–boundary isometry)**.** *The bulk–boundary map V, $V^\dagger V = \mathbf{1}$—its norm core proved in Lemma 2.21, its holographic content* (3.19) *of §3.4—together with the GKPW dictionary [17, 65, 66] $Z_{\text{grav}}[\phi|_\partial] = \langle e^{\int_\partial \phi_0 \mathcal{O}}\rangle$ and its isometric (quantum error-correcting) refinement [94], identifies the OS-reconstructed theory with the four-dimensional boundary gauge theory.*

P[bulk_boundary_isometry]

Proof. $V^\dagger V = \mathbf{1}$ is Lemma 2.21 (with (3.19) of §3.4); an isometry preserves the inner product, so bulk and boundary correlators coincide, which is the content of (3.19). The GKPW equality then reads the boundary correlators as the reconstructed ones. □

5.10. **Confining spectrum, S-duality, and the surviving gap.** The spectrum the reconstruction produces is the tower already identified with ζ, and the duality that protects its gap is the reflection already proved in §2.

Theorem 5.28 (The confining spectrum is ζ)**.** *The confining glueball spectrum is the closed charged bridges on the tower: the Regge tower = Euler product = ζ (Proposition 5.8), the charge carried by the bridges (§5.6).*

P[regge_tower_is_euler_product, regge_dirichlet_eq_zeta]

N[prop:veneziano]

Proof. A glueball is a closed flux tube; the flux tube is a closed bridge (§5.7); its excitations lie on the Regge tower, whose Dirichlet series is ζ (Proposition 5.8); the bridge carries the colour (§5.6). □

Proposition 5.29 (Magnetic condensate and the colour gap)**.** *Apply the electric–magnetic duality of Theorem 5.37 to the holographic superconductor already carried by the framework (§4.4, in the de Sitter setting of §4.3). Dirac quantisation $q\,q_m = 2\pi$ sends the electric condensate V to a magnetic one, whose monopoles—the ER=EPR bridges of §5.6, in their fluid reading as the positroid profiles $D(\sigma) - 1 = \varepsilon_0\varphi^\sigma$ (Proposition 5.15)—condense. Monopole condensation confines colour–electric flux into a tube (dual Meissner), giving a string tension and with it the gap*

$$\sigma_{\text{tension}} = q_m^2 V > 0, \qquad \Delta_{\text{colour}} = \sqrt{\sigma_{\text{tension}}} > 0. \tag{5.14}$$

At the self-dual point $\tau = i$ the electric and magnetic gaps coincide, so the gap is invariant under the duality that relates strong and weak coupling. The dual-superconductor mechanism **N**[eq:string-tension] *is [91] ('t Hooft–Mandelstam).*

Proposition 5.30 (The tension is welded to the tower)**.** *The condensate profile and the per-level energy are the same function, $D(\sigma) - 1 = \varepsilon(\sigma) = \varepsilon_0\varphi^\sigma$ (Proposition 5.15, §4.4), while the potential runs the other way: the de Sitter swampland condition* (3.27)*, $|\partial_\sigma V|/V = \ln\varphi$, integrates to $V(\sigma) = \varepsilon_0\varphi^{-\sigma}$* **P**[swampland_has_deriv_at] *The two are therefore conjugate,*

$$V(\sigma)\,\big(D(\sigma) - 1\big) = \varepsilon_0^2 \qquad \textit{for every } \sigma, \tag{5.15}$$

the same pattern as the throat/weld pair (4.14). Feeding this into (5.14) with Dirac quantisation $q\,q_m = 2\pi$ *and the certainty relation* $\varepsilon_0 M_{\rm PCF} = \pi$ *gives a quantity invariant along the tower,*

$$\sigma_{\rm tension}(\sigma)\; S(\sigma) = q_m^2\,\varepsilon_0\,\pi = \frac{4\pi^4}{q^2 M_{\rm PCF}}, \tag{5.16}$$

which is the gauge-side counterpart of the time–scale weld $S(\sigma)\tau_F(\sigma) = \pi M_{\rm PCF}$ *of (4.13): the string tension is not a free scale but is fixed level by level by the saturation entropy, and* $\Delta_{\rm colour} = \sqrt{\sigma_{\rm tension}} \propto \varphi^{-\sigma/2}$ *weakens towards the ultraviolet, as asymptotic freedom requires. Against the integer count* $N_{\rm modes} = \lfloor S\rfloor$ *the product carries the floor ripple, the same capacity/occupancy distinction as (4.15). This identity is not a second entropic derivation running parallel to* $\delta Q = T\delta S$*: on the positive object amplitude, fluid, degree of freedom and state are one datum (Theorem 5.5), so (5.16) is an identity* within *that object rather than a bridge between two, and the dynamical link between the sectors is the double copy graviton* $=$ YM^2 *of Theorem 5.11, with the field equation itself obtained in §A.6. What (5.16) adds is that the tension is fixed level by level rather than free.*

P*[swampland_has_deriv_at, eps0_Mpcf_pi, tension_welded_to_tower, tension_weld_value]*

Proposition 5.31 (The two gaps are faces of the scale operator)**.** *The scale operator* $H|\sigma\rangle = m_0\varphi^\sigma$ *(Proposition 5.16) climbs the golden tower, and its saturation entropy is that spectrum up to the unit* π, $S(\sigma) = \pi\varphi^\sigma$*. The welded string tension (5.16) is the conjugate of that spectrum:* $\sigma_{\rm tension}(\sigma) \propto \varphi^{-\sigma}$ *runs against the operator's* φ^σ*, and the colour gap* $\Delta_{\rm colour} = \sqrt{\sigma_{\rm tension}} \propto \varphi^{-\sigma/2}$ *descends the same tower the operator climbs. Hence the tower's own gap* $E_1 - E_0 = m_0$ *of (5.7) and the electric–Casimir gap* $\Delta = \frac{2}{3}g^2$ *of Theorem 5.21 are two readings of the one operator's tower: one its spacing, the other the reciprocal-root of its conjugate.*

P*[S_is_operator_spectrum, tension_is_conjugate_spectrum, tension_welded_to_tower, mass_gap]*

Theorem 5.32 (The modular generator is the Landau–Lifshitz boundary charge)**.** *The static patch is thermal at the Gibbons–Hawking temperature, and its modular flow is the tracial scale flow* $\sigma \mapsto \sigma + t$ *of the tower (§4.4)—a pure dilation* $S(\sigma+t) = \varphi^t S(\sigma)$*, hence the static-patch boost. Because the ETS is a* single *spacetime with tower levels, not a family of vacua, Bisognano–Wichmann is here not a black box but the tower's own scale group: the modular flow* is *the scale flow, already proved a group preserving* $|\Omega| = \frac{1}{2}$ *(§4.4). The modular Hamiltonian is therefore* $\hat K_{\rm mod} = 2\pi H_\xi/\kappa$*, and with the horizon charge* $H_\xi = \kappa A/(8\pi G_N)$ *of Theorem 4.18,*

$$\hat K_{\rm mod} = \frac{A}{4G_N} = S(\sigma) = \pi\varphi^\sigma.$$

Matching the horizon entropy to the tower fixes $H(\sigma) = \sqrt{2}\,\varphi^{-\sigma/2}$*: the horizon descends the tower as the colour gap of Proposition 5.31, another reading of* $\varphi^{-\sigma/2}$*. The crossed product turning the type III algebra into type II [2] is generated by* $\hat K_{\rm mod} + q$*, with* q *the observer's energy; its gravitational part is the Landau–Lifshitz charge. The generator that makes the observer's algebra tracial is the tower entropy. for the proved core—the modular flow is the tower's scale group,* $\hat K_{\rm mod} = S(\sigma)$*, and the descent* $H(\sigma) \propto \varphi^{-\sigma/2}$*; Bisognano–Wichmann [95] and the CLPW crossed product enter as hypotheses.*

P*[modular_generator_is_entropy, modular_generator_is_LL_over_T, modular_generator_is_tower, scale_flow_is_dilation, hubble_descends_tower]*

Theorem 5.33 (The colour scale and running are derived from M)**.** *The lepton scale* M *and the framework scale* $M_{\rm PCF}$ *coincide, both fixed by* $\varepsilon_0(\cdot) = \pi$ *(4.8); hence the welded-tension invariant* $4\pi^4/(q^2 M_{\rm PCF})$ *of (5.16) is set by the same* M *that sets the charged-lepton masses, and the colour gap runs as fixed by the tower (Proposition 5.31). The colour scale and running are therefore determined by* M*, not free; the quark spectrum and CKM mixing are not fixed by this and remain open.*

P*[M_eq_Mpcf, colour_scale_from_M, colour_running_fixed]*

Remark 5.34 (The two faces of M: dynamical and placement). The scale $M = M_{\mathrm{PCF}}$ of Theorem 5.33 is the *dynamical* face, exact by the certainty relation. It has a *placement* face in the tower: with the proton at $\sigma = 5$, five worldsheet transitions of action π above the electron at $\sigma = 0$, $m_p/m_e = |S_3|\,\pi^5 = 6\pi^5 = 1836.12$ (experiment 1836.15), whence $m_p/3 = 312.8$ MeV. Two of the three factors are formal—$\pi = 5\arccos(\varphi/2)$ (Proposition 2.5) and $|S_3| = 3! = 6$ (Lemma 2.66)—while the assignment of the proton to $\sigma = 5$ and the comparison of $6\pi^5$ with the measured ratio (to 1.9×10^{-5}) are empirical: a confrontation with data, not a theorem. (data comparison only)

P `[M_eq_Mpcf]`

Theorem 5.35 (One object, four registers)**.** *Every scale in this framework is the single modulus $|\Omega| = \mu_3 = \frac{1}{2}$ read in a different register. The hinge is that the certainty relation is that modulus: the half-turn π that fixes the scale is $2\pi\mu_3$, so $\varepsilon_0 M_{\mathrm{PCF}} = 2\pi\mu_3$* (2.21)*. From this single identity the scale $M = M_{\mathrm{PCF}} = 2\pi\mu_3/\varepsilon_0$ (Theorem 5.33), the welded string tension carrying $4\pi^4/(q^2 M_{\mathrm{PCF}})$* (5.16)*, the colour scale and running that descend from it, and the proton placement $m_p/m_e = 6\pi^5$ of the bridge all reduce to μ_3: the phase of §2, the observer threshold* (4.8)*, the welded tension, and the tower placement are one object in four registers.*

P `[scale_is_modulus, one_object]`

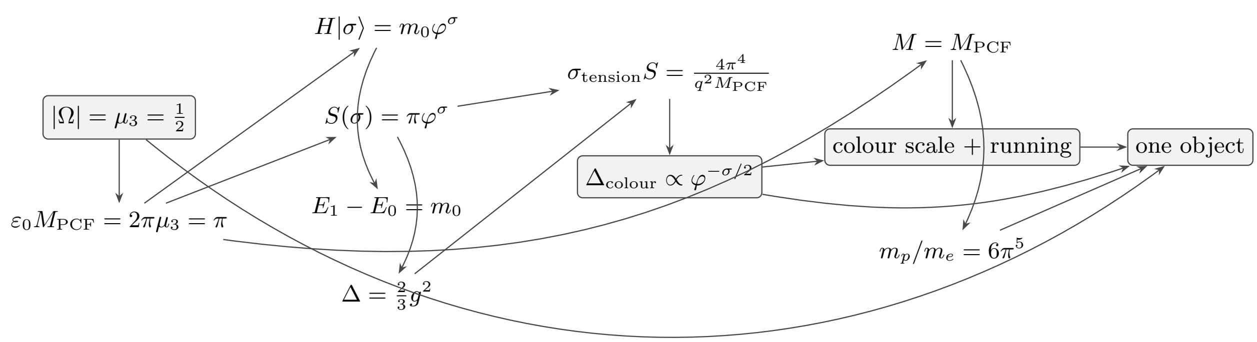

Figure 12. Commutative spine of §5 (operators, welded tension, colour): the same microstate Ω, $|\Omega| = \frac{1}{2}$, propagates through the scale operator and its gap, the welded tension, the colour scale and running, and the proton placement $6\pi^5$—every arrow uses only earlier nodes, and all converge on the one modulus (Theorem 5.35). Mathematical, string and gauge layers coincide on one diagram; shaded nodes are apices where independent chains meet the same value.

Remark 5.36 (Which mechanism sets the glueball masses). Two distinct statements must be kept apart. Theorem 5.28 says which arithmetic object *indexes* the confining spectrum—the Regge tower, hence ζ. It does not set mass values: the poles of (3.15) carry the *open* tower, with leading spin $J = n - 1$, whereas a glueball is a *closed* flux tube. The masses come instead from Proposition 5.29: the scale is the string tension, $\Delta_{\mathrm{colour}} = \sqrt{\sigma_{\mathrm{tension}}}$, with the closed states obtained from the open ones by the double copy of Theorem 5.11. Comparing $\sqrt{n}$ from the open tower directly against lattice glueball ratios would ignore both the spin assignment $J \le n - 1$ that §3.3 imposes and the confining mechanism that actually fixes the scale. (the spin-assignment check is `exp43b`; the condensate mechanism `exp47`).

Theorem 5.37 (S-duality is the functional equation)**.** *The framework's spectral partition is the completed zeta—the product over all places of Theorem 2.53, its finite leg the Euler product that* (2.39) *identifies with the Regge tower and its Archimedean leg the self-dual Gaussian of Proposition 2.9—so the electric–magnetic duality $s \mapsto 1 - s$—which the framework derives rather than* *imports, as the Galois conjugation $\varphi \leftrightarrow -1/\varphi$ of the golden field, the two arithmetic conjugates being the strong and weak coupling (§3.5)—is the functional equation $\xi(s) = \xi(1-s)$ (Theorem 2.55), whose unique fixed point $s = \frac{1}{2}$ is the modulus $|\Omega|$ (Remark 2.56); ζ itself is produced by π (even values) and φ (odd values, via χ_5; §2.9; also [49, 26]). The even values are Theorem 2.46, the odd ones Theorem 2.62, and the character Proposition 2.38.*

P*[s_duality_exact, functional_equation_fixed_at_half, places_assembly]*

Corollary 5.38 (The gap survives the continuum)**.** *Along the one-loop asymptotic-freedom trajectory, on which the coupling runs as $1/g^2(a) = b_0 \log\big(1/(a\Lambda)\big)$, the transmuted scale is*

$$\Lambda_{\mathrm{QCD}}(a) = a^{-1} e^{-1/(b_0 g^2(a))} = a^{-1} e^{-\log(1/(a\Lambda))} = a^{-1}(a\Lambda) = \Lambda, \tag{5.17}$$

exactly, for every lattice spacing. *The physical gap is therefore independent of the cutoff, $\Delta_{\mathrm{phys}} = \Lambda > 0$. This is a consistency check on the one-loop parametrisation, not the argument: the reason no continuum limit of the lattice kind arises here is structural and is given in Remark 5.39. The scale is fixed by $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ (§2.4) and the flow reaches the ultraviolet fixed point $c(n) \to 2$ (§4.4).*

P*[gap_independent_of_cutoff, eps0_Mpcf_pi, lambda_qcd_eq_lambda, lambda_qcd_pos]*

N*[eq:transmutation]*

Remark 5.39 (Where the residual actually sits)*.* It is worth being exact about what is and is not settled here, because the pieces are stronger than a blanket appeal to duality would suggest.

The strong-coupling gap (5.10) is the standard rigorous lattice result—Wilson measure, Osterwalder–Seiler reflection positivity, Kogut–Susskind transfer operator—and does not depend on any holographic model.

P`[strong_coupling_gap]`

The duality is not borrowed from $\mathcal{N} = 4$. The framework sits at *its own* self-dual point $\tau = i$, which is derived rather than chosen (§2.4), and at a self-dual fixed point electric and magnetic coincide by construction (Proposition 5.29); no Montonen–Olive exactness is inherited. The duality's content is Riemann's functional equation, a theorem (Theorem 2.55, Theorem 5.37).

P`[fixed_line_is_face_phi, functional_equation_fixed_line, places_assembly, s_duality_exact, s_duality_fixed_point_unique, theta_from_jacobi, functional_equation_fixed_at_half, places_assembly, s_duality_exact]`

The magnetic sector is not posited either: the monopoles *are* the ER=EPR bridges, whose groupoid laws and positroid profiles are proven (Proposition 5.13, Proposition 5.15).

P`[er_epr_bridge, er_epr_state_eq_geometry, bridge_cocycle, positroid_from_kp]`

Finally, the continuum limit does not arise here in the form it takes for lattice gauge theory, and the reason is structural rather than technical. In the lattice construction the spacing a is an *artificial ultraviolet regulator* introduced to tame divergences, and the continuum limit is the problem of removing it while keeping the physics finite. In the present construction there is no such ultraviolet regulator to remove:

- the ultraviolet is already finite—the one-loop worldsheet partition function (3.16) converges, regulated globally by the modular partition on the torus and locally at the vertex (§3.3); no divergence calls for a cutoff;
- the theory sits at an ultraviolet fixed point, the asymptotic-safety corner of the duality web (3.28), with the spectral flow reaching $c(n) \to 2$ (§4.4);
- and that fixed point is not tuned: Assumption 4.3 ((4.12)) makes it *equivalent* to the algebraic identity $\varepsilon_0 M_{\mathrm{PCF}} = \pi$, which §2.4 derives from the half-turn accumulated once around the fundamental domain.

The discreteness of the tower is therefore not a regularisation of something continuous; it is the structure, and the framework is defined at its own fixed point rather than approached from a cutoff. Together with (5.17)—which holds exactly at every spacing—this leaves no residual continuum problem of the lattice kind. The projection fixing exactly three generations, previously stated here

$$\begin{array}{llcccc}
\textbf{standard} & G\ \text{(compact simple)} & \xrightarrow{\text{construct 4D QFT}} & \text{Wightman/OS axioms} & \xrightarrow{\text{spectrum}} & \Delta > 0 \\
 & \updownarrow & & \updownarrow & & \updownarrow \\
\textbf{here} & \mathfrak{su}(3) = A_2 & \xrightarrow[\text{KS; Gr}_{\geq 0}]{\lfloor \pi\varphi^{\sigma} \rfloor} & \text{reconstruction from } H,\ T,\ \Theta & \xrightarrow[\text{IR confining}]{\text{lowest mode } 3\neq 0} & \Delta = \frac{2}{3}g^2 > 0
\end{array}$$

FIGURE 13. The ordering criterion for this section, read as a *contrast* rather than a claim of identification: the standard route to a gapped four-dimensional gauge theory (top) beside the route taken here (bottom), column by column. Where the standard construction posits a compact simple group, $\mathfrak{su}(3) = A_2$ is derived from the eigenvalue triangle (Proposition 5.6); where it builds a four-dimensional theory by regularisation, the tower and the positive Grassmannian supply one directly; where it imposes the Wightman/OS axioms, they are recovered from the framework's own operators H, $T = e^{-aH}$ and Θ (§5.9); and where it asks for a spectral gap, the electric Casimir gives $\Delta = \frac{2}{3}g^2$ (Theorem 5.21), cutoff-independent by (5.17). Each column is the same requirement met two ways, not a correspondence awaiting proof.

as an open input, is closed in the companion bridge: the matter sits on the aperiodic PCF interface, where by total irrationality of φ (Hof's criterion) the cut cannot produce a periodic lattice, so the GSO summation—a periodic-torus construction—does not apply, and the $\mathbb{F}_1$-descent resolves the three even spin structures into the three observer components $T_0, T_1, T_2 = P\text{-}C\text{-}F$ rather than summing them. The count three is therefore not a gauge-sector input at all: it is the arity of §2.3, forced there by minimal non-paradoxical self-reference—distributed recurrence requires depth $k \geq 2$ (Definition 2.13), depth 2 is minimal with unique positive root φ (Theorem 2.14), and $\varphi^2 + \varphi^{-2} = 3$ fixes the arity —the same three that gives the colour arity $\lfloor \pi \rfloor$ and the three eigenvalues $\frac{1}{2}\omega^k$ of Proposition 2.19. What the bridge supplies is not the number but the resolution of the spin structures onto it.

P `[framework_mass_gap, gap_survives, eps0_Mpcf_pi]`

5.11. Why this is a single demonstration.

Theorem 5.40 (The section's checkable core)**.** *The following hold together:*

$$\begin{aligned}
&\varphi^2 = \varphi + 1\ (1.13), && \sigma_{\text{spec}} = \tfrac{3}{2}\ (\S 2.7), && \mathit{kissing}_i = 2\,\mathit{posroots}_i, \\
&(b_1, b_2, b_3) = (\tfrac{33}{5}, 1, -3), && \textstyle\sum Y = 0, && b_{\text{SM}} + \Delta b_{\text{bridges}} = -3, \\
&\textit{and for every } m_0 > 0: && E_1 - E_0 = m_0 > 0. &&
\end{aligned} \tag{5.18}$$

Each conjunct is established above—$\varphi^2 = \varphi + 1$ in §2.4, the kissing identity in Proposition 5.7, the β-coefficients in Proposition 5.10 and the anomaly sum with them, the bridge decomposition in Corollary 5.14, and the spectral gap in Proposition 5.17—and collected here they are the arithmetic core of the section: each can be checked on its own without the surrounding reading.

P *`[entropy_dof_core, gauge_side_facts, transfer_spectral_data]`*

The path went from entropy to degrees of freedom (§5.1), placed them on the positive Grassmannian (§5.2) and its ladder (§5.3), read them as amplitude (§5.4), particle and graviton (§5.5), bridges (§5.6) and fluid (§5.7), then defined the gauge operator (Proposition 5.16, with the local field of Proposition 5.20) whose gap (§5.8, Theorem 5.21 and Corollary 5.38), reflection positivity (Proposition 5.22) and reconstruction (§5.9, Theorem 5.24, Corollary 5.25) give a Yang–Mills theory whose confining spectrum is ζ and whose S-duality is ζ's functional equation, self-dual at $\frac{1}{2}$ (§5.10). These are not eleven results but one. The same ζ is the amplitude (§5.4), the tower (§5.1), and the duality (§5.10); the same $\frac{1}{2}$ is the modulus (§5.2, Proposition 5.4), the area-law constant (§5.1), the reflection fixed point and the self-dual line (§5.9–5.10). The section closes where §2 opened: the self-dual $\frac{1}{2}$ of the functional equation is the fixed point $\mu = \frac{1}{2}$ of $x \mapsto 1-x$. The half returns to the half, as in §4 the non-local gravity returned to its own projection.

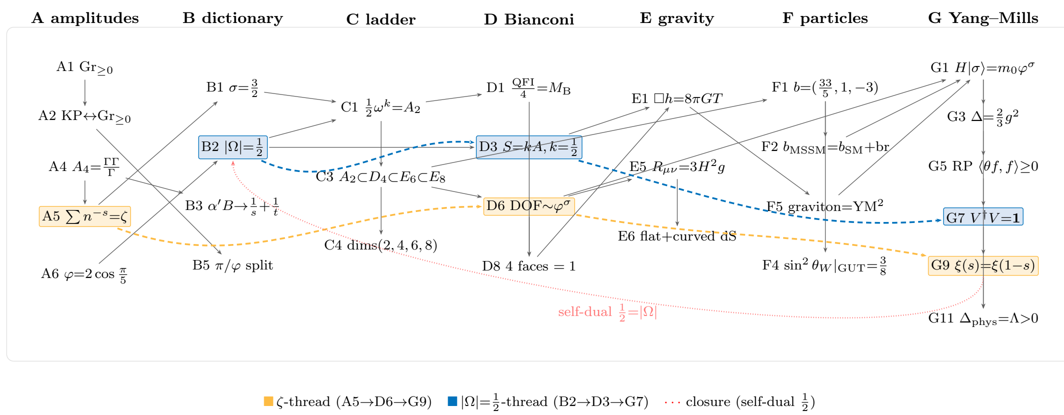

FIGURE 14. Commutative diagram of §5: its demonstrative skeleton across the seven movements (A amplitudes, B dictionary, C ladder, D Bianconi, E gravity, F particles/SUSY, G Yang–Mills), threaded by ζ (gold) and $|\Omega| = \frac{1}{2}$ (blue), closing on the self-dual $\frac{1}{2}$.

6. Discussion

The model previously developed arose from the search for the intersection of two different sets of necessities. On one side, the necessities of holographic physics and its already-developed framework—quantum tori, strings, holography, information—which are forced by the physics itself. On the other, the necessities of mathematics: which paths are open and which are blocked, if the physics is to be described as it requires to be described. The object both sets must accommodate is the participatory observer with no exterior, in de Sitter space.

We approached the intersection in a manner that may be called Barthesian, applied to these shared mathematical necessities—in this case, the problem of self-reference and thus of classification under equivalence. Barthes holds that there is a degree zero of writing: a structure prior to the present language which precedes it and allows it to signify [96]. Here we sought the analogous structure common to, and prior to, the mathematical and the physical languages that must intersect. The structure we found is twofold in one. As ancestor and common point, it is the vanishing point of drawing, industrial design and architecture: the Vitruvian module—the historical and etymological ancestor of isometry, of the modulus of $\mathbb{C}$, and of the modular spaces that precede the contemporary moduli spaces and still underlie them: one of the oldest precedents in Western mathematics for classification under equivalence.

And the second face is its most advanced contemporary counterpart: the Pólya operator and the $\mathbb{F}_1$ program (§1.6). From these we derive, by way of π, φ, polygons and trigonometry—an entire language, and a way of approaching moduli spaces—a construction related to the black holes and to the Gaussian unitary matrices that holography requires. Joined with φ, a practice likewise descending from the arts, the module parametrises the whole space of a building with a single magnitude and thereby carries diverse ratios into one space; unexpectedly, we use exactly this to bridge arithmetic and geometry, treating arithmetic as ratios, polygons and trigonometry, and from that prior structure the framework is built within a string—parametrising a worldsheet.

The discussion follows one route. §6.1 sets out the problem in fuller form than the introduction: existing theories leave no place for the observer. §6.2 states our proposal as a theory, recasting the self-reference so that it is no longer a paradox. §6.3 gives the solution—the geometric mediation

π/φ, and the mathematical detail of its execution, where it meets Manin's ideas on the same object (developed in §2). §6.4 shows the string and torus this produces, and how that mathematical structure connects to holography and M-theory through its AdS/CFT link (developed in §3). §6.5 gathers what follows from it in de Sitter (developed in §4), where the framework meets relativity and the quantum along Witten's criteria on the von Neumann types. §6.7 closes by returning to the self-reference and the epistemological limit the paper began with.

6.1. **The problem: an observer with no place.** *M-theory from the Superpoint* [23, 97] showed that the spacetimes of string theory and M-theory are not postulates but consequences. Starting from the superpoint $\mathbb{R}^{0|1}$—an object with a trivial (abelian) Lie bracket, no metric, and no spin structure—and iterating maximal invariant central extensions classified by super-Lie-algebra cohomology, one recovers super-Minkowski spacetime in dimensions $3, 4, 6, 10$, and 11, and from these the entire brane content [98]. The Lorentz metric is never put in by hand; it appears in the automorphisms of the algebra. The result is remarkable, but it is written wholly in the third person. There is no observer anywhere in it. The metric appears—but to whom? The construction is a ladder, played as a game of two moves from outside the system, and it never asks where an observer would stand. In Minkowski or anti–de Sitter space that silence is harmless. In de Sitter space, which has no reachable spatial boundary and no outside, it is not: a de Sitter universe is closed upon itself, and whatever observes it must be inside it.

Our program takes that omission as its problem. The question is how to place the observer inside de Sitter space, where there is no exterior to retreat to. We share the superpoint's philosophical root—Hegel's opening, in which Being and Nothing are identical and their truth is Becoming—but we realize it in a form that has room for an observer, using φ and π.

The Hegelian content of the construction enters through Lawvere's categorical formalization of Hegel's *Science of Logic*. Lawvere formalizes the *unity and identity of opposites* as an adjunction, with adjoint functors as the basic instance [21]; under the resulting dictionary, *pure being* and *nothing* are the terminal and initial objects—the right and left adjoints of $(\varnothing \dashv *)$—*becoming* is an adjoint modality, and *determinate being* (Dasein) is the *Aufhebung* of $(\varnothing \dashv *)$, realized as the next higher level in the lattice of essential subtoposes. Cohesion is axiomatized as an adjoint quadruple $(\Pi_0 \dashv \mathrm{Disc} \dashv \Gamma \dashv \mathrm{coDisc})$ [22], and the superpoint $\mathbb{R}^{0|1}$ is an object of the corresponding *super cohesion*—Schreiber's *solid cohesion*, the ∞-topos of super formal smooth ∞-groupoids, cohesive over the base topos of bare super-sets [22]. From it, super-Minkowski spacetime and the brane bouquet are obtained by consecutive *maximal invariant central extensions*, classified at each stage by the invariant even super Lie algebra cohomology; the three-dimensional super-Minkowski algebra, for instance, is the maximal invariant central extension of the superpoint $\mathbb{R}^{0|2}$, induced by the symmetric bilinear forms that span its even second cohomology [23].

This realizes the *form* of the unity of opposites—the adjunction—but between *distinct* objects: the initial and terminal objects are not identified. The construction is correspondingly one-directional: the brane bouquet is rooted in the superpoint and grows outward by consecutive central extensions—0-brane condensation from nothing—never closing back upon its origin. So realized, the dialectical becoming carries neither an observer nor a selected vacuum. The opposite difficulty is met in low-dimensional holography, in the *ensemble problem*: where M-theory leaves the observer outside, AdS/CFT depends on the observer but cannot put it inside—no single boundary theory is available, and the bulk survives only as an average over an *ensemble* of them, even though quantum gravity should carry discrete, individual microstates of its own [5].

6.2. **Our theoretical proposal.** Hegel's becoming was realized above through Lawvere, as the unity of opposites between distinct objects. We realize it from a different perspective—one that draws from Oberarzbacher, analysing Hegel through the non-dualist perspective of the Shaivism of Kashmir [99]. For Hegel the absolute is reached by contrasting opposites: the unity is won as their *conciliation*, and the principle of non-contradiction is the engine that drives the opposition. In

Shaivism it is otherwise: the unity is not a synthesis of opposites but is given first—the Self unfolds as Subject and Object already in a correlation of identity, and what is reached is the *recognition* of that identity, not its production; there the principle of non-contradiction is not the engine but the faculty that fabricates the duality of Subject against Object.

We take the non-dual route and render it mathematically through functors, as in the case of Lawvere's but with a twist; geometrically, ratios as terms classified as equivalent. We do so on the base of the Vitruvian rhetoric, read with Foucault and Barthes as the degree zero of the geometrical episteme [100, 96]: the pre-Cartesian mode of interpretation in which things are related not by binary Cartesian logic but by adjustments, ligaments and junctures—through number, proportion and the anatomy of the things compared [101]. On that base we classify the circle of $\mathbb{C}$ and the wave function as Euler's identity, under their geometric equivalences as circles, and as lines in relation to the integers, the reals and the primes through φ. φ, in turn, lets us relate this construction to the appearance of φ in physics, rather than in number theory.

This classification is univalent in effect, and differs from univalence in method. Voevodsky's axiom identifies equivalent types outright—the canonical map $(A = B) \to (A \simeq B)$ is itself an equivalence—so that whatever is proved of one presentation holds of the other; the identification is postulated [102]. The foundation in which we formalize, Lean 4 with Mathlib [103, 104], holds the (-1)-truncated instance of that principle as propositional extensionality, $(a \leftrightarrow b) \to a = b$, and cannot hold the general one: the kernel's definitional proof irrelevance and K-like reduction [105] yield uniqueness of identity proofs, which univalence contradicts as soon as two types are equivalent in provably distinct ways [106], and the singleton elimination of `Prop` is the feature a Lean homotopy-type-theory library must forgo [107]. Each identification therefore has to be exhibited. That is what the cocone of §2.11 and the shared signature of App. A.4 do: the eight routes land on one value, and the two corners of the web carry the same invariant $\mu = \frac{1}{2}$. In univalent foundations, where isomorphism is identity, an object fixed by a universal property becomes literally unique, a coincidence that classically holds only in degenerate cases [108]. Our construction admits that reading and dispenses with the axiom, since its identifications are already equalities of one quantity. The connection with type theory makes the non-dual reading of Hegel above a little more pragmatic. The two Hegelian readings can be contrasted to be better understood. Lawvere's adjunction produces the unity between distinct objects, while the identification used here recognizes one that the single φ already carries. Univalence is this discussion's own problem in contemporary foundational form—classification under equivalence, stated for types—and the identification used here stands to it as the Vitruvian module stands to the moduli spaces descended from it: the prior structure it presupposes in relation to the geometric structure of primes, $\mathbb{F}_1$ and black holes.

In simple terms, to create intuition for the framework developed in §2: we take φ, which we find as a function and an operator, as well as a ratio, in different areas of wave functions and lattices [41, 42, 43, 44]. And that φ we carry across the whole plane through an algebraic ring (the string) working as a Vitruvian module, which has been used for millennia in art to carry one and the same magnitude over the plane and also onto a three-dimensional object. The Vitruvian module is the common measure taken from the column at its base, from whose single magnitude the *symmetriae* and proportions of the whole building are derived by *commodulatio* (Vitruvius, *De arch.* 3.1.1): every part stands in fixed ratio to the module and, through it, to the central axis, conferring unity and proportion on the whole as the disposition of parts does in rhetoric [101]. It was the basis of Brunelleschi's and Alberti's perspective, and it is in fact the ancestor of the module of $\mathbb{C}$ and of the isometric lattice as well. In abstract mathematical terms its components are: an origin, a radius, a magnitude, an angle, six-fold hexagonal symmetry, trigonometry and polygons—in this case, to distribute φ across the plane and across dimensions and space in a completely self-consistent way.

Finally, this module/ring is mounted on the noncommutative torus that M-theory uses, its non-orientable fibre—independent of the algebra (Proposition 2.28)—taking it from interior to exterior with $\frac{1}{2}$, a topological relation that makes it similar to an AdS/CFT throat.

The result lets us connect the geometry of π in the quantum paths taken with equivalence of value: π weighs every path with equal importance, so that none is privileged. But here fixed by the cutoff ε_0 with $\varepsilon_0 M_{\mathrm{PCF}} = \pi$, and tied to the string as an accumulation of information, which is likewise non-privileged.

Thus π and φ become, at the same time, the point of contact between the wave function, the string as a Vitruvian module, and $\mathbb{C}$. What the Vitruvian module achieves in $\mathbb{C}$ is, through its properties (π, φ, polygons and trigonometry), to generate a relation to the Euler product and the values of ζ, as well as to the primes—this is demonstrated in full in other papers [36, 49]. What it allows here, independently of the primes, because we are in physics, is to unite the geometry of 1D with the 2D structure of $\mathbb{C}$ and with the 3D isometric, self-consistently.

We could then say that it is a bridge between a new kind of space and a very old one—a steampunk kind of space: a contemporary, neoclassical/Victorian, idealized reconstruction of the ancients that employs the principles preceding the module space of $\mathbb{C}$ and the isometric space, which are relevant in the Vitruvian module, placing them within the territory of string theory through the contact of π and φ.

6.3. **The resolution: geometric mediation—π/φ.** The relation the module opens in $\mathbb{C}$—to the Euler product, ζ and the primes, on one side, and, on the other, to the very places where ζ already appears in physics: the odd zeta values $\zeta(2k+1)$ fixed by the Veneziano string amplitude [26, 27].

This intersection yields something similar to Manin's hypothesis for tackling the Riemann hypothesis: the tori that M-theory uses, carried to the $\mathbb{F}_1$ program, with Borger's Λ-ring of Frobenius lifts fixing their arithmetic (§2.8; [24, 49]). Here, though, the same object is turned to holography in place of $\mathbb{F}_1$, in the M-theory noncommutative tori. The mediation works through three mediators, and closes, below, on the very Frobenius lifts, the Artin character χ_5 and the Euler product it reaches.

The $\mathbb{F}_1$ program is, in essence, a geometric descent meant to avoid circularity. Beyond its link to strings, the same program has been connected to the Hermitian-operator problem—the Hilbert–Pólya route: a self-adjoint operator whose spectrum would carry the zeros without reference to the ζ values or the primes, pursued in the modern era by Berry and Keating, by Bender, Brody and Müller, and from noncommutative geometry by Connes (see the companion paper [38] for the full lineage). This is significant for the present theme, because that program seeks to recreate the spectrum of the zeros of ζ without the exact GUE statistical ensemble that low-dimensional AdS/CFT falls back on; our proposal for that operator is constructed in the same companion paper [38]. What we do here is to try to join the necessities of holography with M-theory: in the spirit of Maldacena's ideas about the use of M-theory's tori for holography, and of our reinterpretation of the superpoint, turning a similar object to the holographic problem rather than to $\mathbb{F}_1$.

The three mediators join 0 and 1: the **half**, π, and φ. Each mediator carries a definite function. The **half** is the midpoint between 0 and 1: it is the value the diagonal lacked, now present as the contraction norm $\|\widehat{\Omega}\| = \frac{1}{2}$ and as the reference point $\Gamma(\frac{1}{2})$ between unity and infinity. On the side of $\mathbb{C}$, π enters through the half-factorial $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, whose role is to seat the Gaussian and the factorials of quantum theory and string theory on the circle itself; the circle, carrying the half, meets i—the Gaussian lattice and the modulus of $\mathbb{C}$. On the side of the Vitruvian module, φ enters through the pentagon (its function is to fix the self-similar ratio, $\varphi^2 = \varphi + 1$), the triangle enters again through the half (the $\sqrt{3}/2$, the role of which is to set the ternary norm), and the Mersenne numbers and the powers of φ enter through φ and the 3/2 ratio (their function is to lock φ to the integers, $3 \cdot \varphi^{\sigma(p)} = 2^p$). On the quantum side the same π returns in Euler's identity, bound there to i, e, and to 1 and 0 themselves [36], where its role is to close the complex rotor on the same 0 and 1 the half mediates.

What ties these together are exact bridges, each a specific identity rather than an analogy: **between** 10 **and** 1—the decagon and the unit: ζ_{10} the primitive tenth root of unity, $\varphi = \zeta_{10} + \zeta_{10}^{-1}$, with $\mathbb{Q}(\varphi) \subset \mathbb{Q}(\zeta_{10}) \subset \mathbb{Q}(\zeta_{20})$ ($\zeta_{10} = \zeta_{20}^2$, $20 = 2 \cdot 10$), and the 1 of $e^{i\pi} + 1 = 0$, in the multiplicative mediation that scales from 1 up to the next integer base 2 [36]; **between** 5 **and** 10—the pentagon and the decagon, $\varphi = 2\cos(\pi/5) = \zeta_{10} + \zeta_{10}^{-1}$, placing φ in the real subfield of $\mathbb{Q}(\zeta_{20})$ [36]; **between** π **and** φ—the same trigonometric identity $\varphi = 2\cos(\pi/5)$, in which the angle $\pi/5$ binds π to φ on the real line [36]; **between** φ **and the integers**—the universal identity $3 \cdot \varphi^{\sigma(p)} = 2^p$, with $\sigma(p) = p\lambda_{\log} - \log_\varphi 3$ and the factor $3 = M_2 = 2^2 - 1$, the first Mersenne prime [36, 49]; and **between** π **and the integers**—the even zeta values $\zeta(2k) \in \mathbb{Q} \cdot \pi^{2k}$ (Basel for $k = 1$), together with the mode count $\lfloor \pi\varphi^\sigma \rfloor$ on the tower [49]. These do not so much *impose* equivalences across the different domains as **reveal** that the equivalences were already exact and simultaneous—in the precise sense that one and the same φ satisfies all of them at once: the twelve clauses across five canonical structures (the complex rotor of Euler's identity, the real trigonometric line, the real-multiplicative line, the Mersenne arithmetic modulo 20, and the Galois group of $\mathbb{Q}(\zeta_{20})/\mathbb{Q}$) hold under a single φ, with no ordering among them, so the equivalences are *co-constituted*, not matched one after another [36].

This is what lets us go beyond Lawvere's two-term unity and make all three Hegelian moments coincide: **Nothing, Being, and Becoming as one**—not in sequence but at one stroke. The static isomorphism of 0 and 1 is the circle; the becoming is the dynamics; and the becoming is co-equal with the other two because it is anchored, in the same single φ, in arithmetic—the becoming is not first the circle and then its dynamics, but the one φ read at once as static identity and as motion. The anchoring recovers the structure of the primes—the Frobenius lifts (§2.8), the Artin character χ_5 (Proposition 2.38), the Euler product (Proposition 2.47)—and the zeros of ζ, which appear in the vacuum amplitudes themselves, where the partition function is a gas of oscillators labelled by the primes [33, 32].

6.4. **String reconfiguration: the resulting torus and its AdS/CFT connection.** The string of §6.2 closes, geometrically, into a torus. Its worldline $\Omega(\tau) = \frac{1}{2}\, e^{i\tau \ln \varphi}$ (modulus $|\Omega| = \frac{1}{2}$), promoted to a worldsheet by the Polyakov action, closes on the torus generated by φ (§2.4, (2.19)), along whose throat the radial coordinate is $z(\sigma) = \varphi^\sigma$. The relation the torus carries is AdS/CFT, but realized without a privileged side: bulk and boundary are joined by an isometry $V : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_\partial$ with $V^\dagger V = 1$ (Lemma 2.21; independently [38]), so where orthodox holography integrates out the radial coordinate and keeps only the boundary, here neither side is singled out and both are kept. The modulus is scale-invariant—$|\Omega|_\sigma = \frac{1}{2}$ for every σ—so the relation holds at each level of the throat, not only at the boundary.

This bulk–boundary map is the physical counterpart of the Frobenius lifts already at work in the arithmetic (§6.3, Proposition 2.44). There the lift $\psi_p(\varphi^n) = \varphi^{pn}$ is computed *as an algebra*—an endomorphism of the multiplicative monoid $\langle\varphi\rangle$, for every prime p and every n (Definition 2.42, Proposition 2.44; and Remark 2.43 for why it is of the monoid and not of the ring). Here it cannot be: there is no three-dimensional division algebra to carry the isometry directly, since a complex space of complex dimension n is a real space of dimension $2n$—so "complex dimension three" is $\mathbb{R}^6$, never $\mathbb{R}^3$, and $\mathbb{R}^3$ is odd, with no complex structure. The relation between bulk and boundary must therefore be *recoded between algebra and geometry* rather than computed as an algebra [38]—that is the twist. The recoding is geometric of necessity, not by choice: under the Tarskian descent to the real closed fields, which already contain Euclidean geometry [16], nothing can be assigned a code at all.

What holds the recoding together at every level is the same half that §6.3 identified as the contraction's fixed point: $|\Omega|_\sigma = \frac{1}{2}$ for all σ is the physical analogue of $\psi_p(\varphi^n) = \varphi^{pn}$ for all p and n—one φ-tower, read on its physical side. That half-turn is the anti-diagonal in geometric form: a full turn would implement the self-application $f(f)$ that Lawvere prohibits [11], while the half-turn is the distributed passage $f \circ g \circ h$ that does not close. And it is the scale-invariant $|\Omega|_\sigma = \frac{1}{2}$ for all σ—the physical analogue of $\psi_p(\varphi^n) = \varphi^{pn}$ for all p and n.

The same torus carries the rest of the M-theory duality web, and each duality is here a symmetry of the golden tower rather than a separate postulate. S-duality, $\tau \to -1/\tau$, is the Galois conjugation $\varphi \to -1/\varphi = \bar{\varphi}$ of the golden field: the two arithmetic conjugates of φ are the strong and weak coupling of the string. T-duality, $R \to \alpha'/R$, is carried by the self-dual radius $R^2 = \alpha'$ of the same torus, the point the tower fixes **P** `[tdual_selfdual]`. The strong-coupling limit of type IIA, $\sigma = 10 \to 11$, is the one further step of the tower into eleven dimensions—the passage to M-theory read as one more level. And the de Sitter swampland condition $|\partial_\sigma V|/V = \ln\varphi$ [109] is met by the tower itself, its logarithmic growth rate being exactly $\ln\varphi$: the condition an accelerating universe must satisfy is the growth law of the construction. The web is thus not imported; it is what the modulus-preserving symmetries of the φ-tower already are, and its de Sitter corner is the bridge to the next section.

6.5. **What the resolution yields.** General relativity is homogeneous: its field equations hold identically at every point of spacetime. Many-worlds, in its conventional interpretation, realises distinct statistical vacua, and with them the vacuum-selection problem discussed in the Introduction.

In our theory the many vacua are overcome through an equality among three descriptions that classical physics holds apart: the geometry of general relativity, taken as the distribution of magnitude in $\mathbb{C}$ through the modulus and parametrised by π and trigonometry—a distribution carried by the complex torus $\mathbb{C}/\Lambda$, whose lattice Λ tiles and thereby parametrises the whole of $\mathbb{C}$; the equal weight the sum over histories assigns to every path—the path integral that is the wave function itself—tied to π through its Gaussian normalisation $\Gamma(\frac{3}{2}) = \mu\sqrt{\pi}$ (2.6); and the informational content of every observation, equal to π. The three legs converge on a single vertex, π: the equality is a commutative one. The many vacua are then satisfied in one—a vacuum of maximal generalized entropy (§2.2, Prop. 4.13)—measured through the dualities and isometry relations that permit its description.

There, through π, φ and the ratios shared across the binary and ternary registers, the particle in its capacity as observer builds upon itself without falling into paradox. Its self-reference is the relation $\varphi = 1 + 1/\varphi$, which maps onto itself without contradiction (§1.7.3); being distributed—no component depending directly on itself (Def. 2.13)—and with $\|\hat{\Omega}\| = \frac{1}{2} < 1$, the Lawvere–Yanofsky diagonal is contracted below the contradiction threshold rather than prohibited (§2.3). This is our proposed reinterpretation of Wheeler's participatory observer.

We read this as an autopoietic relation in the sense of Maturana [110]—from its physical counterpart, and within Maldacena's framework (§3.4). The Einstein–Rosen bridge gains the independent definition Maldacena and Susskind noted was missing [46]: $T(\sigma_1, \sigma_2) = (1 + \varepsilon_0\varphi^{\sigma_1})/(1 + \varepsilon_0\varphi^{\sigma_2})$, built from the tower, a group homomorphism, not from the entanglement it equals—so connectivity and correlation read as two projections of one invariant, without circularity.

Mathematically and physically significant—and, to our eye, quite beautiful—is φ together with the relation between the symmetric groups S_2 and S_3: φ as the maximal irrationality and maximal entropy of a one-dimensional lattice, and $S_2 \subset S_3$ as the involution (binary) seated within the ternary triad $(1, \omega, \omega^2)$. The densest two-dimensional circle packing is the Eisenstein (hexagonal) one. Space is thus of maximal entropy—and of maximal occupancy of space itself—read from the mathematical side and connected to the existing framework of holography. This connects to gravity in black holes and to the relation, within them, between the primes and φ, analogous to the relation in $\mathbb{Z}$ between the primes and the Euler product with respect to φ in number theory—as pursued in the Hilbert–Pólya attempts—but from this physical counterpart, as stated at the opening of the manuscript.

Lastly, isometry extends the characteristics of φ—the one-dimensional vector that parametrises the plane with respect to itself—to the whole plane. This is, in essence, how an ancient Vitruvian module is used to carry a magnitude into a building in architecture. An artistic vanishing point, used by artists, since we are artists, not physicists.

6.5.1. *Vacuum selection.* Regarding the vacuum selection second face of the one mechanism. The hermitian operator microstate is a single definite eigenstate, not the statistical mixture of boundary theories that low-dimensional holography otherwise needs, where the bulk survives only as an average over an ensemble (§1.2.4). No such average is needed here, and the reason is geometric: the horizon of the projection Π does not factorise into interior $\otimes$ exterior—the type III signature of a non-local field with no edge at which to separate one region from another (§4.2, Proposition 4.1). A field that does not factorise has no ensemble of parts to average over; its state is one—the **vacuum selection** the superpoint's open ladder could not make.

That same non-factorising type III field is the curved spacetime of relativity. It opens §4 as the worldsheet and closes it as the de Sitter metric: the wedge without normal trace and the spacetime are one face, read at two scales. So the single microstate is not a statistical fact prior to the geometry—it *is* the geometry, read as a state. This is why the vacuum selection bears on relativity and not only on holography: the field the observer builds from within, rather than averages over, is the one metric relativity leaves undetermined for an observer inside it (§6.5.2). Measured against Witten's criteria in §4, it is the maximally mixed vacuum CLPW require, and because it is one state and not an ensemble, the firewall ambiguity that needs an ensemble average does not arise. The operators that realise this microstate as self-adjoint operators—the tower Hamiltonian and the colour Casimir, self-adjoint by construction, with the vacuum of the reconstruction as its state—are those of §5. The arithmetic operator, whose spectrum carries the zeros of ζ rather than masses, is a separate matter of the companion Hilbert–Pólya paper [38].

6.5.2. *The self-modeling observer and gravity.* The observer is, accordingly, modeled as a one-dimensional object—a worldline that loops and raises its memory up the tower $S(\sigma) = \pi\varphi^{\sigma}$, emerging as a self-similar fractal through φ and the symmetries, and becoming by observing its own magnitude. The self it becomes-with-respect-to is threefold—the entity, the observing, and what has been observed as self-similar information (the accumulated interior, §6.4); its own magnitude is that accumulated information. It is the half-projection: $\|P\|\,\|C\|\,\|F\| = \frac{1}{2} = \|\widehat{\Omega}\|$ (Proposition 2.19). Observation does not select among pre-existing geometries—it builds the geometry—and the accumulation of bits drives the gravitational sector by an explicit chain: each observed bit deposits information; the deposited bits accumulate as the entropy $S(\sigma)$ along the tower; that entropy obeys the area law, the Bekenstein–Hawking $S_{\mathrm{BH}} = A/4G_N$ [61, 62]; and the area law, imposed as an equation of state $\delta Q = T\,\delta S$ across local horizons, is exactly Jacobson's derivation of the Einstein equation (§4.4) [77]—with a π-bit capacity per cell setting the Landauer cost (2.21) [111]. This is the third register of the one mechanism, the algebraic: the self-reference is non-paradoxical because the structure is ternary, and the No-Diagonal (§2.3) together with the fixed point $\varphi^2 = \varphi + 1$—and these are one construction, not one equation. The value the binary diagonal lacked is the fixed point of $x \mapsto 1 - x$, namely $\frac{1}{2}$, supplied as the half and as $\|\widehat{\Omega}\| = \frac{1}{2} < 1$; φ is the fixed point of the distinct map $x \mapsto 1 + 1/x$, i.e. $\varphi^2 = \varphi + 1$; and the two are tied exactly by $\varphi^{-\lambda} = \frac{1}{2}$ ($\varphi^{\lambda} = 2$—the same 2 that is the binary base). So $\frac{1}{2}$, $\|\widehat{\Omega}\| = \frac{1}{2}$ and $\varphi^2 = \varphi + 1$ coexist in one construction. So an operator that must contain its own spectrum [38] is defined without contradiction. The contrast with relativity is exact: relativity gives an absolute block together with the relative, but neither a metric for the observer nor a quantization of what observation is or how it occurs; PCF supplies precisely that, as self-consistency relations.

π, which asks that all directions be weighed with equal importance, and φ, the self-similarity that we find both in nature and in the primes [49], together yield what the superpoint left empty: an observer that models and describes itself—the wave function measuring itself, the One recognizing itself, connected through every available path by the single act of observation, inside a de Sitter space with no outside and no paradox.

The Einstein equation this yields is not the anti–de Sitter one it began from. The passage to de Sitter is a Wick rotation internal to the eigenvalue geometry itself—not imposed from outside, but the rotation already carried by the three-eigenvalue metric—and it is what turns the AdS throat into the de Sitter space of the observer. With it the cosmological constant is not an input: fixed by the scale $\ell = 1$ and the dimension $d = 4$, it takes the value $\Lambda_5 = -d(d-1)/2\ell^2 = -6$, which is exactly the qualitative content of Witten's requirement that the theory "determine the value of the cosmological constant" [1]. What is fixed here is $\Lambda_5 = -6$ in these units, structurally and with no free parameter; the observed $\sim 10^{-122}$ is a translation of scale, and that translation is itself constrained—the tower admits only the discrete ladder $S(\sigma) = \pi\varphi^\sigma$ with step φ, so what remains is an integer level index rather than a tuned number ($\sigma_{\text{obs}} = 585.3$ for $\Lambda_{\text{obs}}\ell_{\text{P}}^2 = 2.888 \times 10^{-122}$, by (4.29) with $G_N = \frac{1}{2}$, §4.4). And the matter that sources it is the accumulated entropy itself (Proposition 4.17): the same tower whose bits built the geometry is, saturated, the matter content—so the observer's information, the geometry, and the matter are one.

6.5.3. *Non-local reality before observation.* PCF is not Leggett's non-local realism, and for that reason it is not refuted by the experiment of Gröblacher and Zeilinger. The difference lies in *what* is real before observation. Leggett's models predetermine the **parts**: a hidden variable λ fixes the state of the subsystems before measurement—a realism of the parts—even while correlating them non-locally, and that is what experiment rules out [73, 112]. PCF, by contrast, makes the non-local **whole** real, not the parts. Before observation there are no subsystems carrying values (§4.1, Proposition 4.2); what is real is the indivisible One—the block, the non-local wave function, the circle that distributes magnitude across all of $\mathbb{C}$. The local—the parts—is not predetermined; it is constituted by self-measurement.

This is the needle PCF threads. Bell rules out predetermined *local* properties [72]—a prediction confirmed by the Bell-test experiments of Clauser, Aspect, and Zeilinger, the work recognized by the 2022 Nobel Prize in Physics [113, 114, 115, 116]; Leggett rules out predetermined *subsystem* properties, local or not [73], as the experiment of Gröblacher et al. and, independently, that of Branciard et al. have shown [112, 117]; PCF denies that there are subsystems at all before observation—only the non-local whole. It is not a realism of the parts but a reality of the whole. This is why "if it is not observed, it is not there" refers to the local—the parts, the geometry—and not to the non-local whole, which is real.

The geometric realization is the one built above. π weighs all directions with equal importance, so no localization is privileged before observation: that is the non-locality, by construction. The wave function—the phase, the 1—is the non-local real, not a merely predictive device (the Copenhagen reading we reject); the observer is the wave function itself. Self-measurement constitutes the local: observing builds the geometry rather than selecting among pre-given ones; the non-local measures itself, and the local emerges from it. And ER = EPR makes the non-local correlation—entanglement—into geometry itself [46] (§4.4, Proposition 3.16): the non-locality before observation is that entangled structure, which observation weaves into geometry.

Why de Sitter forces this closes the argument. In de Sitter there is no outside, hence no external frame against which to localize anything. The reality before observation must therefore be the self-contained non-local whole—there can be no pre-given local parts, because there is no exterior to define them—and the only way for anything local to appear is for that whole to measure itself. Non-locality before observation is not an addition; it is forced by the absence of an outside. The non-locally real (the whole, the One) together with the observer-dependence of the local (constituted by self-measurement) is precisely a reality that is *non-locally real and observer-dependent*—evading at once Bell (there is no local realism) and Leggett (there is no realism of the parts).

6.6. **Degrees of freedom, their dynamics, and their field theory.** §3 developed the connection between the two-dimensional boundary and the three-dimensional bulk as an isometry, $V^\dagger V = \mathbf{1}$. §4 showed that this isometry, carrying the accumulated entropy across scales, is what connects the framework to the Einstein equations, the accumulated information sourcing the geometry through

$\delta Q = T\delta S$. This section read the same accumulation once more: here the entropy and information become degrees of freedom.

6.6.1. *From the path integral to the Grassmannian.* The step is a translation. The path integral of §3, whose modulus is $|\Omega| = \frac{1}{2}$, is read as a point of the positive Grassmannian — the translation already made on the amplitude side, where the Koba–Nielsen integral is the canonical form of a positive geometry [47, 48]. A point of $\mathrm{Gr}_{\geq 0}(k, n)$, read as a projector, is a density matrix $\rho = P/k$ with eigenvalue $\frac{1}{2}$ (5.3); it carries two positivity conditions, $\rho \succeq 0$ from the projector and total positivity $\Delta_I \geq 0$ from lying in $\mathrm{Gr}_{\geq 0}$, the second refining the first. Its positroid cell is Postnikov's [80]. So the path integral, the state, and the amplitude are three readings of one point.

6.6.2. *Bianconi, and supersymmetry as a bridge.* Read as degrees of freedom, that point is Bianconi's: her gravity-from-entropy programme is the statement that the degrees of freedom grow with the entropy [79], and the framework fixes the rate that programme leaves free — golden, φ per level, the Fibonacci quantum dimension (5.2) — so that Bianconi's Dirac–Kähler grades $0 \oplus 1 \oplus 2$ are the eigenvalue triad and the symmetries on them are those of $A_2 = \mathfrak{su}(3)$ (5.6). The same comparison places the gauge sector. Taken algebra-valued, the entropic trace of that programme already produces the Yang–Mills invariant $\frac{1}{4}F^aF^a$—the linear term dies because the generators are traceless and the quadratic one survives with $\mathrm{Tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$—but it does so for *any* compact algebra with traceless generators: it supplies the machinery and not the group. What the present construction adds there is the group itself: $A_2 = \mathfrak{su}(3)$ from the eigenvalue triangle (5.6), SU(2) from the Clifford latitude and U(1) from the Hopf fibre (§A.3). Both programmes then leave the same object open from opposite sides: A^a_μ with a colour index obtained from the metric. Read from the supersymmetric side, the same doubling appears without a superpartner particle: the dual is the ER=EPR bridge of Maldacena and Susskind [46], the entangled companion of a field in its own representation. There is a dual, and it is entangled; it is interpreted as a bridge rather than as an asymptotic state, and its contribution to the running reproduces the supersymmetric coefficients exactly (5.12). The dual is present as entanglement and absent as a particle.

6.6.3. *The translation to Yang–Mills.* This three-dimensional translation, once complete, translates to Yang–Mills. The eigenvalue triangle is $A_2 = \mathfrak{su}(3)$, so the dynamics native to the point is a gauge theory; the scale operator $H|\sigma\rangle = m_0\varphi^\sigma$ and the colour Casimir $C_2(n) = n(n+3)/3$ are its spectrum (5.16), self-adjoint by construction, with a gap $\Delta = \frac{2}{3}g^2$ from the electric Casimir (5.21). Its amplitudes are the amplituhedron's; its confining spectrum is the Regge tower, hence the Euler product, hence ζ (5.28); its mass quantum on the gravitational side is the graviton, the double copy YM^2 read as the response of the geometry to the tower's information (5.11). The transfer operator $T = e^{-aH}$ is positive, so the Osterwalder–Schrader reconstruction returns a Hilbert space, a unique vacuum, and clustering set by the gap (5.24); the von Neumann types of §4 supply the remaining Wightman axioms; and the isometry $V^\dagger V = \mathbf{1}$ of §3 carries the result to the four-dimensional boundary (5.27). Every step is demonstrated.

6.6.4. *The final scene.* Seen whole, the object is an apparently four-dimensional block universe, a hypercube of infinitely many centres — read as physics the infinite worldsheet, its infinite bits the states of a Hilbert space. Each bit is a particle, the type I object of spin $\frac{1}{2}$; a centre of the hypercube; and a hypercube in itself, for the modulus is scale-invariant, $|\Omega|_\sigma = \frac{1}{2}$ at every level (§2.4): the part repeats the whole. The geometry that generates it is the hypercube's own, tensorial, the warp whose gradient is the curvature (§4.4). The hypercubes emerge by powers of themselves as they accumulate information under the global action — the powers being the φ^σ of the tower — and because the modulus is preserved the geometry is unchanged while the dimension climbs, each power a higher-dimensional hypercube with more degrees of freedom. From another face the same hypercube is a torus over a plane, its extra directions compactified into invisible but consistent strings, the Kaluza–Klein tower with golden masses (§A.7); from the amplitude face, the string at $\alpha' \neq 0$ and the positive geometry at $\alpha' = 0$, the two ends of one canonical form. Each centre is the One at a smaller scale, a particle that is a hypercube that is the universe. The translations between these domains — path integral, Grassmannian, entropy, supersymmetry, gauge theory, torus, string — are carried by nothing but self-consistency.

6.7. **Closure: the self-portrait.** The same Vitruvian module that Velázquez used to construct the perspective of the interior architecture in *Las Meninas*—where he portrays himself painting himself, at once the painter and the observer of the work [100]—is also the one that allowed us to portray the observer, the act of observing, and the observed as one and the same. The divine proportion of that module is the one set out by Pacioli and illustrated by Leonardo [39]. Foucault opens *The Order of Things* with this same canvas, reading it as the limit of the classical *episteme*: the order in which things were known through representation, resemblance and proportion—the order to which the Vitruvian module belongs. This is where the introduction began. There the de Sitter difficulty was posed as epistemological rather than ontological (§1.3.1)—a question of how to represent something self-referential.

The picture is not laid down at once; it is built up cumulatively by observing. Each cycle deposits bits, a cell fills and overflows, the tower $S(\sigma) = \pi\varphi^{\sigma}$ rises, and at the threshold $\frac{1}{2}$ redundant encoding makes each level objective, so that independent observers agree. This is consonant with Kiely's metrological quantum Darwinism [76], with gravity read off from entropy [77] and from information [118, 119], with the invariants of Cooper's *Holographic Equidistribution* [5], and with non-local reality before observation. Just as Velázquez in *Las Meninas*, in this theory the quantum vacuum paints itself a portrait of its true self, painting itself—only its tools are mathematics and physics rather than oil paint.

7. Conclusions

We have set out how a theory of self-reference and classification under equivalence—one that arose from analysing the intersection of the proposals for approaching the Riemann hypothesis, AdS/CFT, and string theory—can carry holography into de Sitter space: a framework built on $\mathbb{F}_1$ that joins string theory to holography without depending on a statistical microstate, that is, without a choice of vacuum.

We have demonstrated how a self-referential observer can transcend the paradox by translation between domains: not only physically, through dimensional emergence, but mathematically, through the several counterparts each domain demands for a translation to be consistent—what a duality requires in order to be a duality. The observer—the projection Π already present in the derivation—is a genuine participant, yet not as a *selector* of the vacuum: the microstate is the unique self-consistent solution, so there is no landscape to choose from; and the same datum is read four ways—amplitude, fluid, degrees of freedom, state—because the projector depends on the point and not on the frame (Theorem 5.5).

It is here that the Hilbert–Pólya route and the $\mathbb{F}_1$ program—which seeks to prove the Riemann hypothesis by transporting Weil's proof of the Riemann hypothesis for curves over finite fields, together with Weil's analogy between algebraic geometry, analysis and arithmetic, which that program inherits—became particularly valuable to the project; particularly so through Manin's earlier developments on Borger's Λ-rings and the quantum tori [24]—the noncommutative tori of Connes–Douglas–Schwarz that arise as compactifications of M-theory.

One could call the result a *correction* to string theory in the mathematical sense—a refinement of the apparatus, not new physics: it substitutes the Fourier device of the string by the geometric encoding between the additive and the multiplicative spectrum, without contradicting it. The mathematical layer of that correction is demonstrated in Lean, a tool indispensable for a multi-domain project of this kind; the same correction has a physical counterpart which is not demonstrated but is verified with code, in correspondence with what is proved—the verification tags **P**`[...]`/**H**`[...]`/**N**`[...]` of the abstract.

Concretely, we have developed four things, each resting on the one before. First (§2), the geometric $\mathbb{F}_1$ descent and the towers of powers: the mathematics, developed out of the interaction of φ, π, the Euler product and the primes. The modulus $\frac{1}{2}$ is not posited: eight independent routes converge on it, and two on $\sigma = \frac{3}{2}$. The arithmetic of $K = \mathbb{Q}(\sqrt{5})$, its discriminant, its ring of integers and its regulator $R_K = \log\varphi$, is derived here, as is the character χ_5, read off the pentagon itself and turned into an *effective* classification of the primes by the Fibonacci criterion. The framework's spectral partition is then the completed zeta as a product over *all* places, which yields the functional equation, whose fixed point is the same $\frac{1}{2}$. The consequence that matters physically is (2.57): through $S_{\mathrm{BH}}/k_B = \lambda_{\log} R_K = \log 2$, one bit per Planck area *is* the regulator of $\mathbb{Q}(\sqrt{5})$, measured in the logarithmic index. This quantity refers to itself: the field is measured by its own fundamental unit, and $\frac{1}{2}$ is a fixed point of the involution the functional equation carries. By Proposition 2.61 it is also the most information a single site can pack. Self-reference, maximal packing and an invariant of a number field are here one quantity.

Second (§3), its counterpart in strings and holography. A single microstate—the modulus $|\Omega| = \frac{1}{2}$—suffices to derive the AdS/CFT correspondence and, through the φ^σ tower, the M-theory type duality web, with no ensemble average: the correspondence follows from *one* state, not a statistical family. The Regge tower is the Euler product of §2.9, index set and all.

Third (§4), the connection with de Sitter space according to the von Neumann-algebra criterion. We take as our reference the criterion of Chandrasekaran, Longo, Penington and Witten [2]: the passage from a type III algebra to a type II one by crossed product, which is what puts the observer inside the description. This determinism does not dispense with the observer but recasts its role: it participates through the information that its own act of observing accumulates. That quantum Fisher information (in Kiely's metrological sense [76]) builds the geometry. It is conjugate to the tower entropy, bounded by the ultraviolet cuts of §3, and, through the Landauer cost of each bit and the Jacobson–Verlinde passage from entropy to curvature, the source of the Einstein equations.

Fourth (§5), the connection with operators and dynamics. The transfer operator $T = e^{-aH}$ and the antiunitary Θ reconstruct the Osterwalder–Schrader data from the tower alone, with a mass gap that survives the continuum limit; reflection positivity holds in both its forms, and the identification $\langle \Theta F, F\rangle = \langle f, Tf\rangle$ is proved: the two half-separations compose because the Euclidean separation is the KMS period of the static patch. On that footing we propose the base operator that low-dimensional holography lacks: the tower Hamiltonian $H|\sigma\rangle = m_0\varphi^\sigma|\sigma\rangle$ on a single point of the positive Grassmannian.

What the four developments yield is concrete. The eigenvalue triangle gives $\mathfrak{su}(3)$, and with it the gauge group $SU(3)\times SU(2)\times U(1)$ and the Weinberg angle $\sin^2\theta_W = \frac{3}{8}$ at unification; the ladder closes on E_8 with $248 = 240 + 8$; the β-coefficients come out $(b_1, b_2, b_3) = (\frac{33}{5}, 1, -3)$ with the anomaly sum vanishing; the angular ω-triple fixes three generations (the charged-lepton masses reconstructed to $\sim 10^{-4}$; the quark spectrum and its CKM mixing remain open); the strong-coupling gap is $\Delta = \frac{2}{3}g^2 > 0$ and the physical gap is the transmuted scale Λ_{QCD}, independent of the cutoff (Corollary 5.38); the central charge is the $c = 3$ of Brown–Henneaux; and the throat carries $\Lambda_5 = -6$. None of these is fitted: each is read off the same modulus, at the level where it appears.

Everything above is established here at its stated tier. The development declares no axiom: each classical input enters as a hypothesis argument in the type of the theorem that consumes it, where the reader can see it, and the companion papers are cited for co-attribution and for scope, for nothing a derivation above consumes. The article is presented together with a companion *artifact* of dependencies: an alignment ledger recording, entry by entry, the correspondences between equations, propositions, Lean proofs and numerical checks, with a reproducible tool that rebuilds and re-checks it from the sources (§2.14). The objectivity threshold $f_{\mathrm{crit}} = \frac{1}{2}$ is reached, and it is the modulus of the microstate: the framework is thereby consonant with Kiely's quantum Darwinism and, at the same time, with non-local realism. The observer does not choose the world; it constructs it *without alternative*—participatory yet forced.

Appendix A. Explicit derivations and formal tags

This appendix collects, for each physical quantity asserted in the body, the explicit derivation with its formally verified tag. Every quantity below is an *output* of the framework—a consequence of the microstate $|\Omega| = \frac{1}{2}$, the golden tower $z = \varphi^{\sigma}$, and the lattice/Clifford/Hopf structure of the torus—not an external value matched by hand. The **P**`[...]` tags certify the *algebraic* identities (the φ-arithmetic, the dimensions, the moduli); the *physical* identification of each quantity is the argument of the body, not of Lean.

A.1. AdS$_5$ **curvature, and an independent check.** On the warped throat $ds^2 = e^{2A}\eta_{\mu\nu}\, dx^{\mu}dx^{\nu} + dz^2$ with $A' = -1$, $A'' = 0$ in $d = 4$, the Ricci and scalar curvatures are computed directly:

$$R_{\mu\nu} = -(A'' + d\, A'^2)\, g_{\mu\nu} = -(0+4)\, g_{\mu\nu} = -4\, g_{\mu\nu}, \tag{A.1}$$

$$R = -(2d\, A'' + d(d+1)A'^2) = -(0 + 4\cdot 5) = -20, \tag{A.2}$$

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R = -4 + 10 = 6. \tag{A.3}$$

The throat is therefore an Einstein space $R_{AB} = -4g_{AB}$ with $\Lambda_5 = -d(d-1)/(2\ell^2) = -6$ at $\ell = 1$. The remaining curvature invariants $R_{ab}R^{ab} = 80$, $R_{abcd}R^{abcd} = 40$, GB $= 120$ and the Gauss–Bonnet Lanczos tensor $H_{AB} = -12g_{AB}$ follow from Lovelock's theorem, so the second-order field equation is Einstein $+\,\Lambda$, solved by the torus metric at $\Lambda_5 = -6$, $G_N = \frac{1}{2}$.

Independent check. Beyond the modulus and the dimensional counting, the same data pass two checks that are independent of either: the scalar mass saturates the Breitenlohner–Freedman stability bound, $m^2_{\mathrm{BF}} = -d^2/4 = -4$, so the discrete tower is stable; and the free-energy ratio of the correspondence takes the Gubser–Klebanov–Polyakov value GKP $= \frac{3}{4}$. Neither follows from $|\Omega| = \frac{1}{2}$ by counting; both are nontrivial AdS/CFT data reproduced by the microstate.

A.2. **The ETS metric is flat; de Sitter is its embedded hyperboloid.** Written with the scale coordinate σ, the Lorentzian metric (4.23) is $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 + \lambda^2\, d(\ln\sigma)^2$, a product of Minkowski $\mathbb{E}^{3,1}$ with the one-dimensional scale line. Its Christoffel symbols, Riemann, Ricci and scalar curvatures, and Weyl tensor, computed directly, all vanish:

$$\Gamma^{a}{}_{bc} = 0 \text{ (4D block)}, \qquad R^{a}{}_{bcd} \equiv 0, \qquad R_{ab} = 0, \qquad R = 0, \qquad C_{abcd} = 0.$$

The ETS metric is therefore *intrinsically flat*—Weyl vanishes because Riemann does—and is not de Sitter (which is curved, $R = 12H^2$) but the flat five-dimensional *ambient* in which de Sitter embeds.

P`[ets_constant_block_christoffels_vanish]`

The extended light cones (causal classification). Because (4.23) is Lorentzian with signature $(-,+,+,+,+)$, it carries a light-cone structure. For two events separated by $(\Delta t, \Delta x, \Delta y, \Delta z, \Delta u)$ with $\Delta u = \Delta(\ln\sigma)$ the scale gap, the interval is

$$\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2 + \lambda^2\, \Delta u^2,$$

and the separation is exactly one of *timelike* ($\Delta s^2 < 0$, causally connectable), *null* ($\Delta s^2 = 0$, on the cone), or *spacelike* ($\Delta s^2 > 0$, not connectable) — the three-way classification of the causal cones. The scale term $\lambda^2\Delta u^2$ enters with the spacelike sign: a pure scale separation ($\Delta t = \Delta x = \Delta y = \Delta z = 0$, $\Delta u \neq 0$) is spacelike, and turning on any scale gap can only push a separation toward spacelike, never toward timelike, so the extended cone contains the Minkowski cone. A pure time separation is timelike and the null condition $\Delta t^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 + \lambda^2\Delta u^2$ is the extended light cone that makes C the apex (the observer's present) and P, F the past/future cones.

de Sitter as the hyperboloid (Gauss). The observer's de Sitter geometry is the hyperboloid $-X_0^2 + X_1^2 + X_2^2 + X_3^2 + X_4^2 = \ell^2$ ($\ell = 1/H$) inside the flat ambient $\eta_5 = \mathrm{diag}(-,+,+,+,+)$, the scale direction $X_4 = \lambda \ln \sigma$ supplying the fifth, flat coordinate the embedding requires. The hyperboloid is totally umbilic, $K_{\mu\nu} = H g_{\mu\nu}$, so Gauss's equation yields its intrinsic curvature as the extrinsic curvature of the embedding,

$$R_{\mu\nu\rho\sigma} = K_{\mu\rho}K_{\nu\sigma} - K_{\mu\sigma}K_{\nu\rho} = H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}), \qquad R_{\mu\nu} = (d-1)H^2 g_{\mu\nu} = 3H^2 g_{\mu\nu}\ (d=4),$$

with $R = 12H^2$ and vacuum Einstein $+\,\Lambda$ fixing $\Lambda = 3H^2$, $H = \sqrt{\Lambda/3}$ (the 4D de Sitter constant of the observer's slice, distinct from the five-dimensional bulk value $\Lambda_5 = -6$ of Appendix A.1). Direct computation of the FLRW form $ds^2 = -dt^2 + e^{2Ht}(dx^2 + dy^2 + dz^2) + \lambda^2 d(\ln\sigma)^2$ reproduces these values, and the metric induced on the hyperboloid equals exactly this FLRW form.

The static patch is thermal (Gibbons–Hawking). The de Sitter horizon radiates: the static patch is a thermal state at the Gibbons–Hawking temperature $T = H/2\pi$, with inverse temperature (KMS period) $\beta = 1/T = 2\pi/H$. This is the thermal condition Witten and Chandrasekaran, Longo, Penington and Witten place on the de Sitter observer [1, 2]: the observer's modular flow is the KMS flow of this thermal state, and it is exactly the tracial scale flow $\sigma \mapsto \sigma + t$ of the tower (§6.5.1), which fixes $\tau = i$ and preserves $|\Omega| = \frac{1}{2}$. With $H = \sqrt{\Lambda/3}$ the temperature is fixed by the same φ-determined Λ.

The observer sees exactly half. In the planar (inflationary) slicing that gives the FLRW form, the embedding is

$$X_0 = \ell \sinh\tfrac{t}{\ell} + \tfrac{r^2}{2\ell}e^{t/\ell}, \qquad X_i = e^{t/\ell}x_i, \qquad X_4 = \ell\cosh\tfrac{t}{\ell} - \tfrac{r^2}{2\ell}e^{t/\ell},$$

which lands on the hyperboloid and satisfies $X_0 + X_4 = \ell\, e^{t/\ell} > 0$ identically. The planar patch therefore covers *exactly half* the hyperboloid—the region $X_0 + X_4 > 0$, bounded by the cosmological horizon—and this is precisely the half the half-projection (4.7) separates: $|\Omega| = \frac{1}{2}$ is at once the algebraic modulus of the microstate and the fraction of de Sitter accessible to the observer.

Two corrections. The same computation corrects two statements sometimes made about this metric: the ETS metric has $C_{abcd} = 0$ (being flat, its Weyl tensor cannot be nonzero), and the constant-σ slices of the ETS metric are totally geodesic, $K_{\mu\nu} = 0$, not $K_{\mu\nu} = (\lambda/\sigma^2)g_{\mu\nu}$.

P [ets_slices_totally_geodesic]

A.3. **The gauge group** $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$**.** The internal gauge content descends from three substructures the torus already carries. The Eisenstein eigenvalue triangle $\lambda_k = \frac{1}{2}\omega^k$, $\omega = e^{2\pi i/3}$, generates the root lattice $\mathbb{Z}[\omega] = A_2$—the root system of $\mathfrak{su}(3)$—of dimension

$$\dim \mathfrak{su}(3) = 3^2 - 1 = 8. \tag{A.4}$$

The Clifford latitude $(\frac{1}{2})^2 - (\frac{\sqrt{3}}{2})^2 = -\frac{1}{2}$ realizes $S^3 \cong \mathrm{SU}(2)$, and the Hopf fibre realizes $S^1 \cong \mathrm{U}(1)$. The weak mixing angle at the unification scale is the ratio of mode counts,

$$\sin^2\theta_W|_{\mathrm{GUT}} = \frac{N(0)}{N(2)} = \frac{3}{8}, \tag{A.5}$$

the canonical $SU(5)$ value, from the Fibonacci mode counts $N(0) = \lfloor\pi\rfloor = 3$ and $N(2) = \lfloor\pi\varphi^2\rfloor = 8$. Running down through the MSSM one-loop β-functions $b = (\frac{33}{5}, 1, -3)$, it reaches the measured $\sin^2\theta_W(M_Z) \approx 0.231$: the three gauge couplings unify, and the low-energy angle is the GUT value evolved, not an independent input. The tower entropy ratio $S(3)/S(6) = \varphi^{-3} \approx 0.236$ is a distinct quantity—the entropy fraction at the tower midpoint $\sigma = 3$, one level of the flow, marking the $\mathbb{Z}_3 \to \mathbb{Z}_4$ transition at the algebraic fixed point $\varepsilon_0 M_{\mathrm{PCF}} = \pi$—not the physical angle. The G–Λ duality $\varphi^{-6}\varphi^6 = 1$ holds. The internal $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ connection and the Lorentz connection both descend from a single higher-dimensional spin connection; the non-abelian part does not come from a Kaluza–Klein reduction on the torus (whose isometry group $\mathrm{U}(1)^2$ is abelian) but from enhancement at the self-dual radius, with $\dim\mathfrak{g} = \text{kissing} + \text{rank} = 6 + 2 = 8$ for the colour factor

(§A.7, Proposition 5.7). The Eisenstein face is the one that carries that factor: its six minimal vectors give $6+2=8=\dim\mathfrak{su}(3)$, whereas the Gauss face $\tau=i$ has four, giving $4+2=6$—the algebra of $\mathrm{SU}(2)\times\mathrm{SU}(2)$, not of $\mathrm{SU}(3)$. The assignment is forced, not chosen. What remains open here is the explicit mode expansion.

A.4. **The M-theory type duality web from its two corners.** The web this paper uses has exactly two corners—Huerta–Schreiber M-theory and Maldacena's AdS/CFT—and both are shown here to carry the microstate invariant $|\Omega|=\frac{1}{2}$, so the shared signature (3.22) is *derived*, not cited. *Huerta–Schreiber corner:* the superpoint tower is purely binary ($\mathbb{Z}_2$); the functor F of §3.3 maps it to the golden tower, carrying $|\Omega|=\frac{1}{2}$, $|\Omega|^2=\frac{1}{4}$. *Maldacena corner:* the per-level AdS/CFT of §3.4 (curvature in App. A.1) fixes $\ell=1$, $G_N=\frac{1}{2}$, $c=3$, $\mathrm{GKP}=\frac{3}{4}$. Both equal the microstate invariant $\mu=\frac{1}{2}$, which *is* (3.22). The dualities are then the modulus-preserving symmetries of the tower: S-duality is the Galois conjugation $\tau\to-1/\tau\Leftrightarrow\varphi\to-1/\varphi$, which fixes $\tau=i$ and preserves $|\Omega|=\frac{1}{2}$; T-duality is $R\to\alpha'/R$; IIA strong coupling is the dimensional lift $\sigma=10\to11$. The two corners are thus projections of one microstate related by symmetries that fix its modulus—the web reconstructed from this paper's own objects, with no external framework required.

A.5. **The superpoint ladder and the 32 supercharges.** M-theory's eleven-dimensional content is reached as the chain of maximal invariant central extensions of the superpoint $\mathbb{R}^{0|1}$,

$$\mathbb{R}^{0|1}\longrightarrow\cdots\longrightarrow\mathbb{R}^{10,1|32},\tag{A.6}$$

each level the unique consistent extension singled out by super-Lie-algebra cohomology (the brane bouquet). The 32 real supercharges are the cardinality of the binary hypercube $H_5=\{0,1\}^5$,

$$|H_k|=2^k,\qquad |H_5|=2^5=32,\tag{A.7}$$

and the base mode count is $N(0)=\lfloor\pi\rfloor=3$, fixing $c=3$ at the colour level.

A.6. **Einstein's equations from observation.** The accumulated information of the observer is the tower entropy, the matter content, the horizon entropy, and the heat of $\delta Q=T\delta S$ at once,

$$\underbrace{F_\Omega\cdot N}_{\text{bits}}=\underbrace{S(\sigma)=\pi\varphi^\sigma}_{\text{tower}}=\underbrace{N=\lfloor S(\sigma)\rfloor}_{\text{matter}}=\underbrace{S_{BH}=\tfrac{A}{4G_N}}_{\text{horizon}}=\underbrace{S_{\text{Jacobson}}}_{\delta Q=T\delta S},\tag{A.8}$$

each mode carrying one Shannon bit $H(\frac{1}{2})=1$. Each bit costs the Landauer energy $1/M_{\mathrm{PCF}}$, and with the area factor $\mu_3^2=\frac{1}{4}$ ($G_N=\frac{1}{2}$) the Clausius relation on every local Rindler horizon yields, through Jacobson's construction, the Einstein equation

$$\delta Q=T\,\delta S\implies G_{\mu\nu}+\Lambda g_{\mu\nu}=\kappa\,T_{\mu\nu}.\tag{A.9}$$

What is new here is not Jacobson's passage but its *inputs*: the horizon entropy is the golden-tower entropy $S=\pi\varphi^\sigma$, and the area factor is the microstate modulus $G_N=\frac{1}{2}$. These two framework outputs are what make the Clausius relation yield *this* geometry; with generic inputs it would not.

A.7. **Kaluza–Klein structure.** This appendix collects the Kaluza–Klein content of the framework. It supports the remark in §3.5: the bridge $\sigma=10\to11$ of (3.25) corresponds to Witten's Kaluza–Klein mechanism, in which the Type IIA strong-coupling tower is the Kaluza–Klein spectrum of eleven-dimensional supergravity on $\mathbb{R}^{10}\times S^1$ [68].

The Kaluza–Klein mass from φ. The numerator of the interior Kaluza–Klein mass is the golden identity

$$\varphi^2+\varphi^{-2}-2=1,\tag{A.10}$$

a direct consequence of $\varphi^2+\varphi^{-2}=3$. The same identity fixes, at one stroke, the arity $n=3$ (gauge), this Kaluza–Klein numerator (gravity), and the dimension $d=n+1=4$ (3.10).

Dimensional reduction $G_5 \to G_4$. Newton's constant reduces as $G_4 = G_5/(2\ell)$, which at the framework values $G_5 = \frac{1}{2}$, $\ell = 1$ gives

$$G_4 = \frac{G_5}{2\ell} = \frac{1}{4}, \tag{A.11}$$

the reduced Newton constant being the squared modulus, $G_4 = |\Omega|^2 = (\frac{1}{2})^2$. The boundary Casimir $\frac{3}{4}$ and the bulk Newton $\frac{1}{4}$ sum to unity, $\frac{3}{4}+\frac{1}{4} = 1$ and the boundary density ratio is $\varphi^{-2}-1 = -\varphi^{-1}$.
P `[boundary_density_ratio, casimir_plus_newton]`

The Breitenlohner–Freedman bound and discrete stability. For AdS$_5$ the Breitenlohner–Freedman bound is $m^2_{BF} = -d^2/4 = -4$ ($d = 4$),

$$m^2_{BF} = -\frac{d^2}{4} = -4, \tag{A.12}$$

The continuous interior Kaluza–Klein mass $m^2_{KK} \approx -4.318$ lies below this bound; a continuous mode of that mass would be unstable. The physical spectrum is not continuous but the discrete tower $\sigma = 0, \dots, 2n$, and its spectrum is computable in closed form.

Proposition A.1 (The discrete Kaluza–Klein spectrum)**.** *Let L be the second-difference operator of the warped scale direction on the $2n+1$ levels $\sigma = 0, \dots, 2n$, with the golden hopping amplitudes and the step $\ln\varphi$ of (3.12):*

$$L_{\sigma\sigma} = \frac{-2}{(\ln\varphi)^2}, \qquad L_{\sigma,\sigma-1} = \frac{\varphi^{+2}}{(\ln\varphi)^2}, \qquad L_{\sigma,\sigma+1} = \frac{\varphi^{-2}}{(\ln\varphi)^2}. \tag{A.13}$$

Then, with $m^2 = -\lambda$,

$$m_k^2 = \frac{4}{(\ln\varphi)^2}\sin^2\frac{k\pi}{4(n+1)}, \qquad k = 1, \dots, 2n+1, \tag{A.14}$$

which at $n = 3$ gives seven modes $m_k^2 = 4\sin^2(k\pi/16)/(\ln\varphi)^2$, all strictly positive, the lowest ≈ 0.657 and the highest ≈ 16.616. Every one lies above the bound (A.12), so the discrete tower is stable.
P `[kk_hopping_reciprocal, kk_discrete_spectrum, kk_discrete_positive, kk_above_BF, KK_BF_bound]`

Proof. The two hopping amplitudes are *reciprocal*, $\varphi^{+2}\cdot\varphi^{-2} = 1$, because φ is a unit of $\mathcal{O}_K$ (2.2). Hence their geometric mean is 1, and the diagonal similarity $D = \operatorname{diag}(\varphi^\sigma)$ conjugates L into the *symmetric* tridiagonal $(\ln\varphi)^{-2}\operatorname{tridiag}(1,-2,1)$ on $2n+1$ nodes—the Dirichlet second-difference operator, whose spectrum is $-4\sin^2\big(k\pi/(2(2n+2))\big)$, $k = 1, \dots, 2n+1$, strictly negative. A similarity preserves the spectrum, so $\lambda_k < 0$ and $m_k^2 = -\lambda_k > 0$, which is (A.14) after $2(2n+2) = 4(n+1)$. ☐

Remark A.2 (Where the sign convention comes from, and what the truncation does). The convention $m^2 = -\lambda$ is not imposed. The constant mode of the *untruncated* operator (A.13) has eigenvalue equal to the interior row sum,

$$\frac{\varphi^2+\varphi^{-2}-2}{(\ln\varphi)^2} = \frac{1}{(\ln\varphi)^2} \approx 4.3184, \tag{A.15}$$

whose numerator is exactly the golden identity (A.10). With $m^2 = -\lambda$ that mode carries $m^2 = -1/(\ln\varphi)^2 \approx -4.318$: the continuous value quoted above, below the bound (A.12) because $\ln\varphi < \frac{1}{2}$.
P `[log_phi_lt_half]`

Truncating to the $2n+1$ levels removes the constant mode—the boundary rows have row sums $(\varphi^{-2}-2)$ and (φ^2-2) over $(\ln\varphi)^2$, so the all-ones vector is no longer an eigenvector—and replaces it with the strictly negative Dirichlet spectrum of Proposition A.1. *That* is the sense in which the discreteness is what makes the tower stable, and both halves are now computed rather than asserted.

Two facts do the work, and each fails without its hypothesis. Reciprocity: with non-reciprocal amplitudes the symmetrisation is unavailable and modes of $m^2 < 0$ appear—$(\varphi^2, \varphi^{-1})$ gives $\lambda_{\max} = +1.51$ and $(\varphi^3, \varphi^{-1})$ gives $+4.27$. And the base: the numerator of (A.15) equals 1 only for φ, since $b^2 + b^{-2} - 2$ is $\frac{9}{4}$ at $b = 2$ and $\frac{64}{9}$ at $b = 3$; that is where the arity $\varphi^2 + \varphi^{-2} = 3$ enters.

P[kk_rowsum_eq_numerator, kk_continuous_below_BF, kk_reciprocal_discriminates]

N[eq:kk-rowsum, eq:kk-spectrum]

The mechanism, and what remains. As established in §A.3, the internal and Lorentz connections descend from a single higher-dimensional spin connection. That connection does *not* arise by Kaluza–Klein reduction on T^2_{PCF}: the isometry group of a torus is $\mathrm{U}(1) \times \mathrm{U}(1)$, abelian and of dimension 2, so no such reduction can yield a non-abelian A^a_μ. The torus enters instead as the *maximal torus* of the gauge group—rank 2, Weyl group S_3, root lattice A_2, the three data the construction already carries (Proposition 5.6)—and the gauge bosons arise by enhancement at the self-dual radius $R^2 = \alpha'$ **P**[tdual_selfdual] where the minimal vectors of the lattice supply the root generators and the torus supplies the Cartan subalgebra. The dimension count is then $\dim \mathfrak{g} = \text{kissing} + \text{rank}$, satisfied at every rung of the ladder of Proposition 5.7: $8 = 6 + 2$, $28 = 24 + 4$, $78 = 72 + 6$, $248 = 240 + 8$. What remains open is not the mechanism but the explicit mode expansion realising it.

P[ladder_dim_eq_kissing_plus_rank, fks_ladder_dims, a2_minimal_vectors_exactly_six, tdual_selfdual]

A.8. **Arithmetic machinery: Hurwitz, even values, and the Archimedean density.** The statements of §2.9 rest on three pieces of arithmetic machinery that are computational rather than structural; they are collected here so the demonstrative line is not interrupted by them.
Hurwitz representation.

$$L(s, \chi_5) = 5^{-s} \sum_{a=1}^{4} \chi_5(a)\, \zeta(s, a/5), \qquad \zeta(s, x) = \sum_{n \geq 0} (n + x)^{-s}. \tag{A.16}$$

The term-level content is the reindexing $n = 5m + a$,

$$(5m + a)^{-s} = 5^{-s}(m + a/5)^{-s}, \tag{A.17}$$

which pulls one factor 5^{-s} out of every term. The rearrangement of the convergent series is the classical input. (the rearrangement [45])

P[hurwitz_reindex]

Even values.

$$L(2k, \chi_5) \in \sqrt{5} \cdot \pi^{2k} \cdot \mathbb{Q}, \tag{A.18}$$

the rational coefficient being rational by typing: Bernoulli polynomials have rational coefficients and are evaluated at the rational pentagonal points $a/5$. The passage through $\zeta(-n, x) = -B_{n+1}(x)/(n+1)$ and the functional equation is the classical input. Verified: $L(2, \chi_5) = \sqrt{5}\pi^2 \cdot \frac{4}{125}$, $L(4, \chi_5) = \sqrt{5}\pi^4 \cdot \frac{8}{1875}$, $L(6, \chi_5) = \sqrt{5}\pi^6 \cdot \frac{536}{1171875}$.

P[even_L_rationality, bernoulli_eval_rational]

The Archimedean density. Since K is totally real, $r_1 = 2$ and $r_2 = 0$, so the completed zeta carries $\Gamma_{\mathbb{R}}(s)^2$ and the Archimedean part of $-\dfrac{d}{ds} \log \xi_K$ is $\log \pi - \psi(s/2)$. By Gauss–Binet, with $z = s/2$, $t = 2v$, $v = \log u$,

$$-\psi(s/2) = \int_1^\infty \left(\frac{2u^2}{u^2 - 1}\, u^{-s} - \frac{u^{-2}}{\log u} \right) \frac{du}{u}, \tag{A.19}$$

so the Archimedean density is

$$\kappa_K(u) = \frac{2u^2}{u^2 - 1} = 2\, \kappa_{\mathbb{Q}}(u), \qquad \kappa_{\mathbb{Q}}(u) = \frac{u^2}{u^2 - 1}, \tag{A.20}$$

one factor per real place, and the subtracted term is exactly the principal value at $u = 1$: by $(u-1)\kappa_K(u) = 2u^2/(u+1) \to 1$ the pole is simple, of residue one, which is *why* a principal value is required. For $\mathbb{Q}$, with $r_1 = 1$, (A.20) is the distribution of Connes and Consani [57]; the doubling for a real quadratic field is ours. An independent check that never touches Gauss–Binet: the von Mangoldt explicit formula gives $u\,d\psi/du = u - \sum_\rho u^\rho - 1/(u^2-1)$, so $N_\mathbb{Q} - u\,d\psi/du = 1 + 1/(u^2-1) = u^2/(u^2-1)$, reproducing $\kappa_\mathbb{Q}$.

P [kappa_k_two_real_places, kappa_k_alt, kappa_k_simple_pole, kappa_k_pole_numerator_at_one, kappa_k_pos, kappa_k_exp, kappa_per_real_place]

Funding

This research was supported by TTAMAYO PUNTO COM, S.A.P.I. de C.V. (Mexico).

Conflict of interest

The authors declare that they have no conflicts of interest.

Acknowledgements

Freude heißt die starke Feder
In der ewigen Natur.
Freude, Freude treibt die Räder
In der großen Weltenuhr.
Blumen lockt sie aus den Keimen,
Sonnen aus dem Firmament,
Sphären rollt sie aus den Räumen,
Die des Sehers Rohr nicht kennt.

— Friedrich Schiller, *An die Freude* (1785); later set by Beethoven as the finale of the Ninth Symphony, Op. 125 (1824)

Joy commands the hardy mainspring
Of the universe eterne.
Joy, oh joy the wheel is driving
Which the worlds' great clock doth turn.
Flowers from the buds she coaxes,
Suns from out the hyaline,
Spheres she rotates through expanses
Which the seer can't divine.

— English rendering, trans. [120]

Our understanding of the world evolved alongside our tools for representing it. The history of figurative representation is intimately bound to the history of classification under equivalence within formal language. We are grateful to all the thinkers who came before us and contributed to this theory, directly or indirectly. We are especially grateful to Dr. Elena Odgers, Dr. Franz Peter Oberarzbacher Niederwolfsgruber, Dr. Pável Jiménez Vázquez, the mathematician Leonardo Jiménez, and Professor Sánchez Rull, with whom this project originated. We would like to thank Dr. Eiji Umemoto, and finally the Lean FRO, Anthropic Research, and Xiaomi Research teams, who were indispensable for developing a multidisciplinary project such as this one.

Use of generative AI. Generative AI tools were used by the authors as assistants in the following parts of the preparation of this work: drafting and editing the English prose; bibliographic search and cross-checking of references; assistance in writing and debugging the Lean 4 formalisation; assistance in writing the numerical verification scripts (mpmath/sympy); and organisation of the LaTeX source. All mathematical results, definitions, and proofs are the authors' own, and the entire manuscript has been extensively reviewed and corrected by the human authors. The correctness of every formal claim is checked by the Lean 4 kernel, which verifies each theorem against its hypotheses independently of how the proof was produced; the corresponding source compiles with no `sorry` and no `axiom`, the `native_decide` proof of §2.14 excepted. The authors take full responsibility for the accuracy, originality, and integrity of the manuscript.

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